An Introduction to Time Series Analysis

With a Focus on Macroeconomic & Financial Applications

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Outline

1. Introduction
2. Data Prep
3. Explore
4. Explain
5. Forecast

Learning Objectives

- Understand what is "special" about time series.

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- Learn how to adapt our typical workflow for time series data
- Introduce new techniques that are not feasible within non time series settings.

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- The applications will be cast within a macro and financial context, but the tools and concepts are widely applicable.

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Taxonomy of Time Series Models

- **Linear / Econometric Models**
  - AR, MA, ARIMA, ADL, ECM
- **State Space Models**
  - Kalman Filter, Structural TS, Deep State Space
- **Machine Learning Models**
  - Tree-based, Neural Networks (LSTM, N-BEATS)
- **Global / Panel Models**
  - DeepAR, hierarchical forecasting
- **Foundation Models**
  - TimeGPT, Chronos

Types of Data

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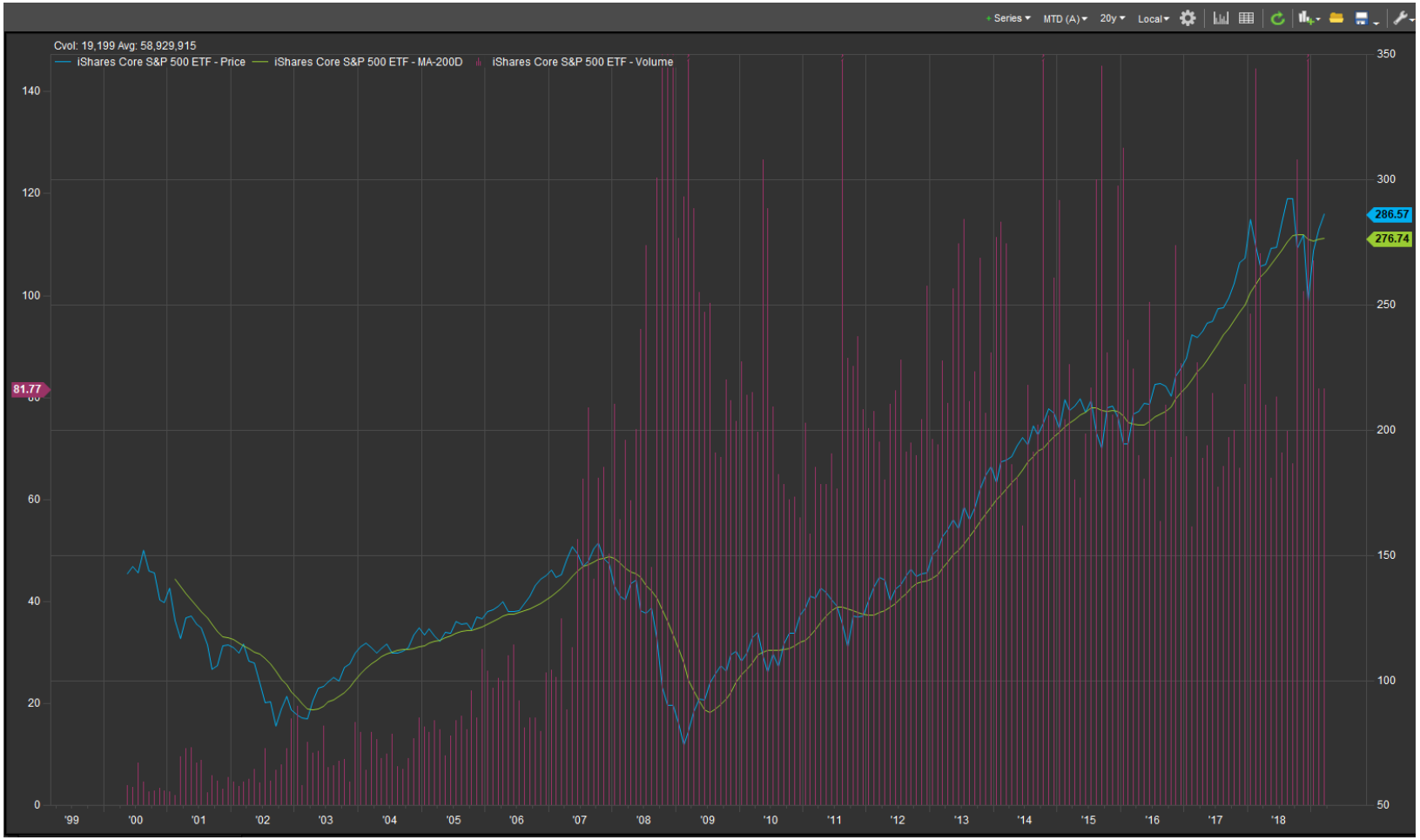
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What is so special about "time series" data?



Certain features and stylized facts appear with time series data that are not apparent in cross sectional. (e.g. momentum)

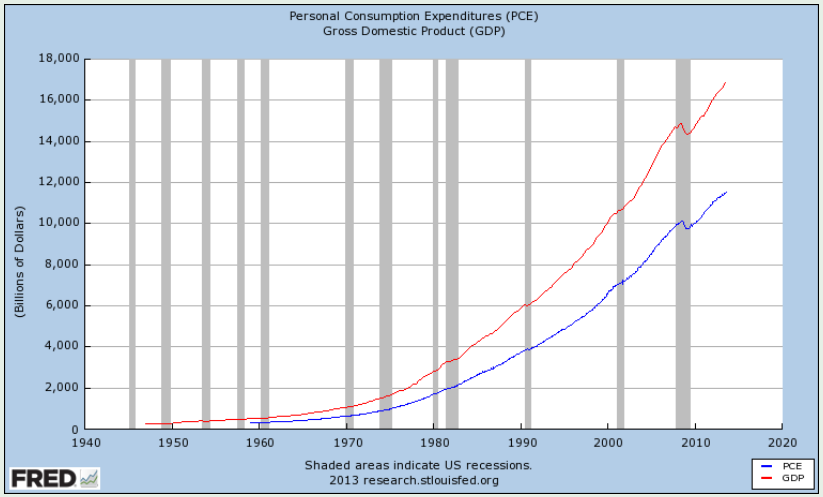
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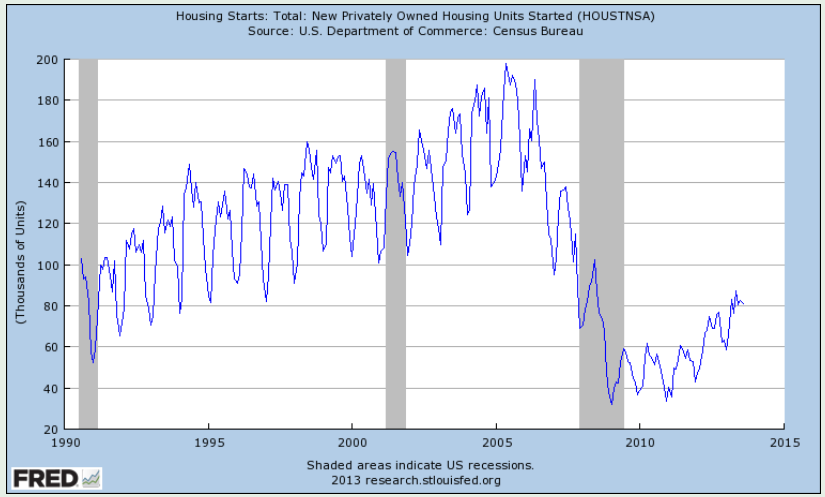
Example (Misleading Time Series Regressions)

Consider the time series models below. In each, discuss why running such a regression by OLS might be misleading.

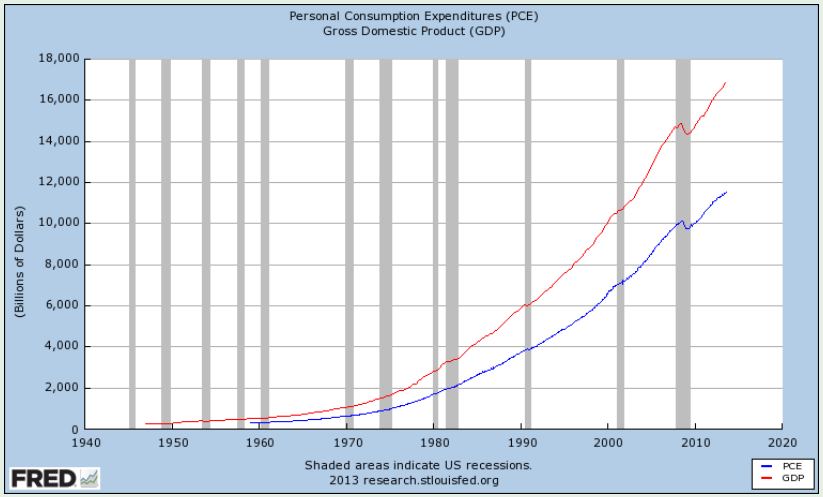
1. $C_t = \beta_0 + \beta_1 Y_t + u_t$
2. $HsngStarts_t = \beta_0 + \beta_1 MrtgRates_t + u_t$

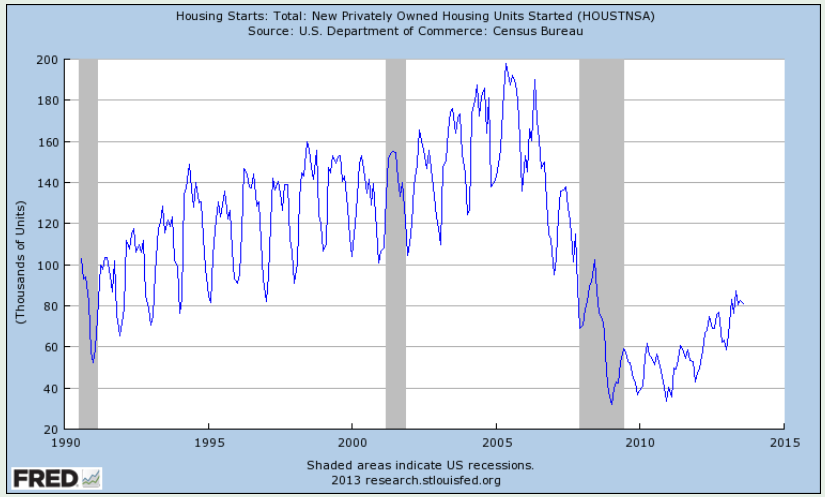
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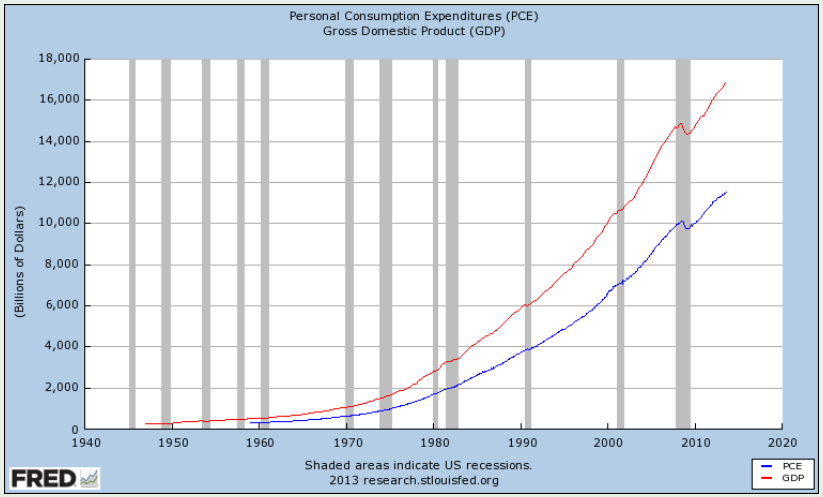
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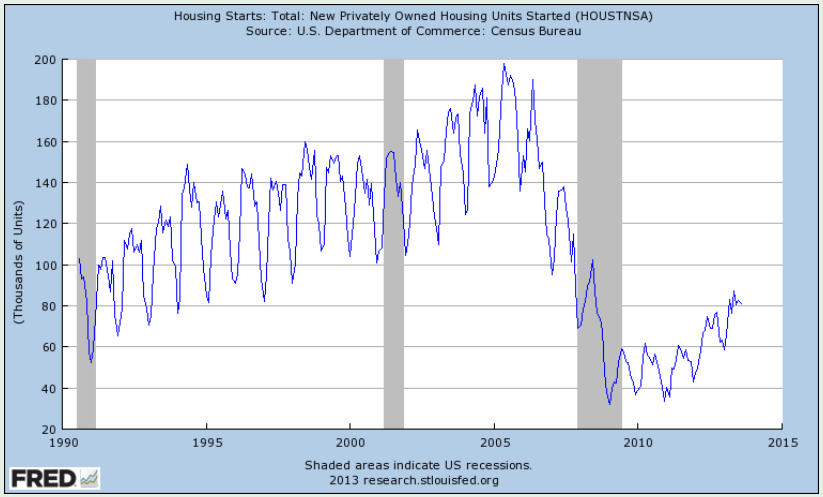




1. (Left) Spurious correlation due to omitted time trend

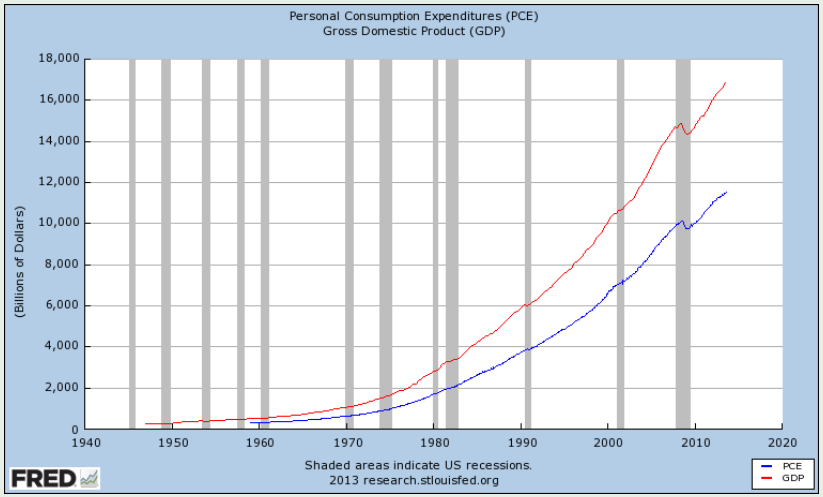
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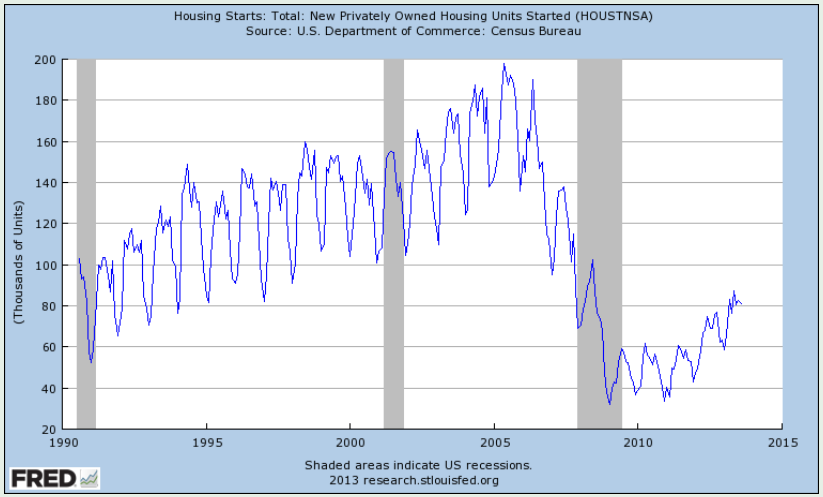




1. (Left) Spurious correlation due to omitted time trend
2. (Right) Seasonal patterns

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We need to accommodate for these common features of time series.

From Intuition to Modeling

Time series data exhibit features that challenge standard approaches:

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These features imply:

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Next: How does the time series workflow adapt to these challenges?

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Why Data Preparation Matters in Time Series

Time series data requires special care before modeling:

- **Ordering matters**
  - Incorrect ordering destroys temporal dependence
- **Missing data is not innocuous**
  - Dropping or imputing values can distort dynamics
- **Timing must be aligned**
  - Misaligned data introduces lookahead bias

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Poor data preparation leads to misleading inference and invalid forecasts

Sequencing Doesn't Matter for Cross Sectional Data

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Company	# Distribution Facilities	Order Fullfillment Time
ABC	10	18
DEF	12	17
GHI	13	17
JKL	15	16
MNO	19	13
PQR	20	13
STU	21	12
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ABC	1991	2	25
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ABC	1993	5	22
ABC	1994	6	22
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The temporal dependence among observations requires that we maintain the sequencing of the time series data.

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- For intraday day, be sure you are sorting on the entire date/time stamp, and not just the date/day/year/hr, etc..

Date Time Stamp vs Time Index

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Company	Date Time Stamp	Time Index	# Distribution Facilities	Order Fullfillment Time
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- Using the Time Index is easy to implement and interpret for simple linear time trends on equally spaced data
- But if data has temporal patterns (e.g. seasonality, unequal spacing, etc..), then modeling directly on the Date Time Stamp is preferred.

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Don't blindly drop rows

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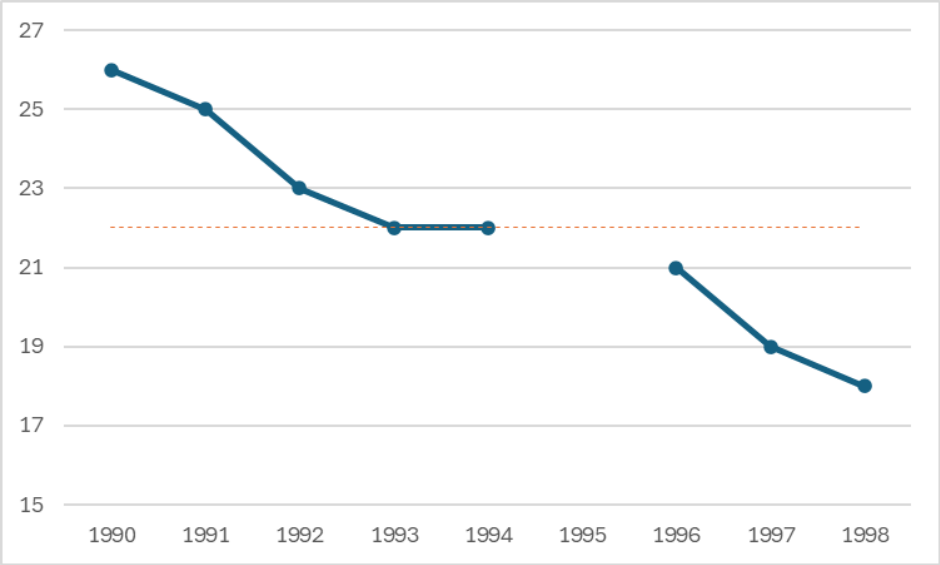
The temporal dependence of the series is disrupted when you drop rows.

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Filling with Sample Mean can be Dangerous

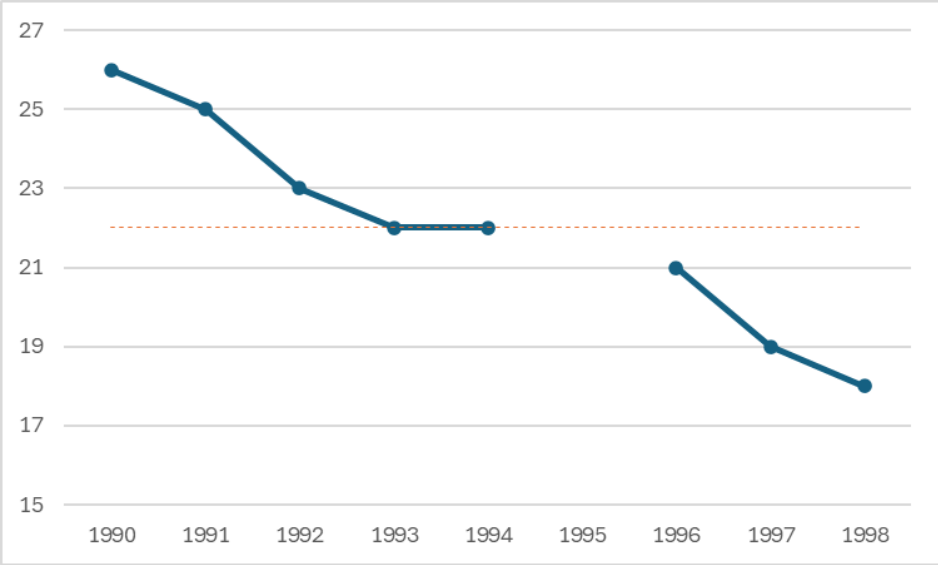
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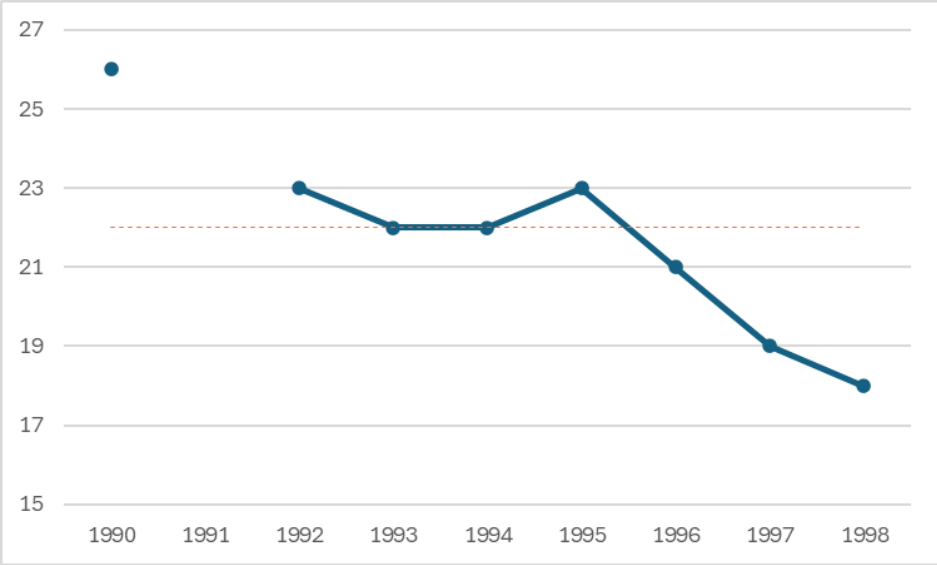
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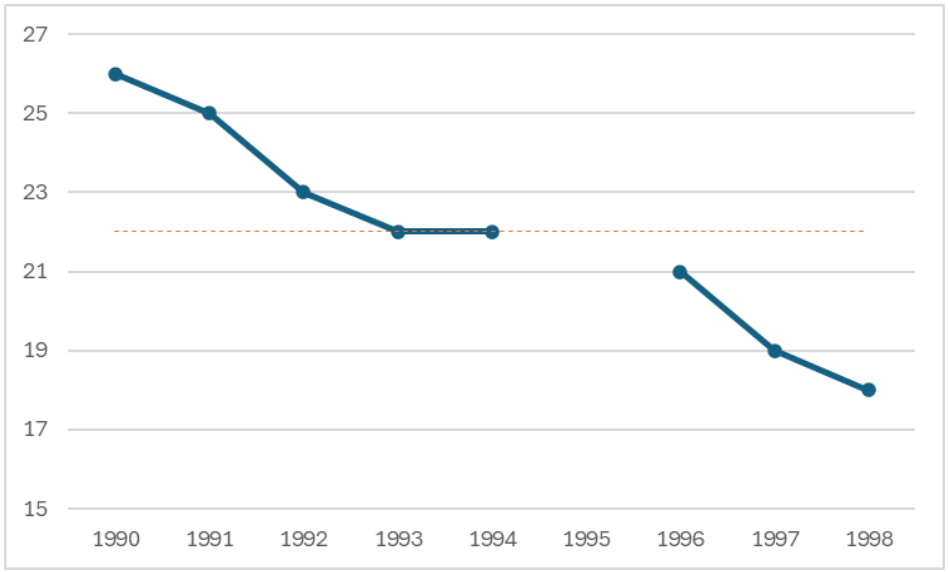
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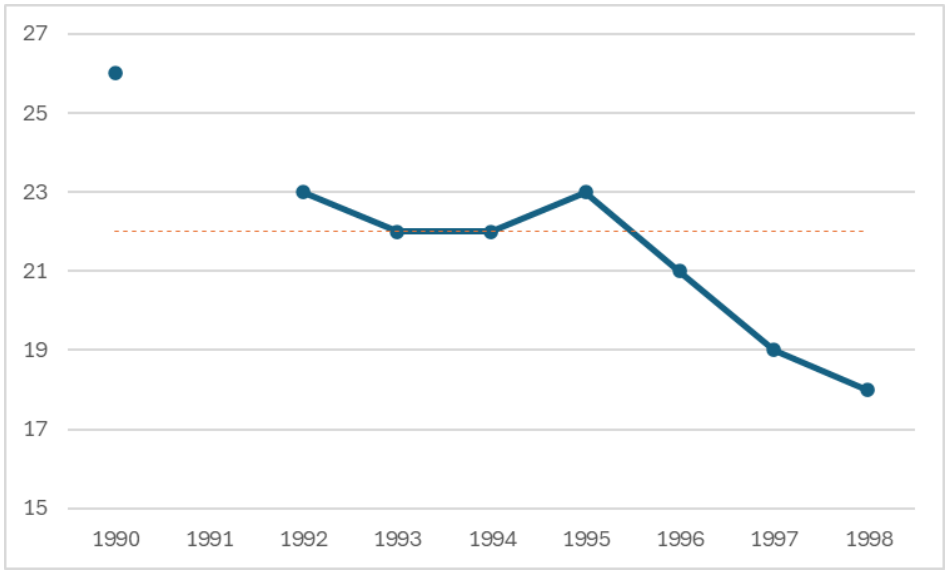




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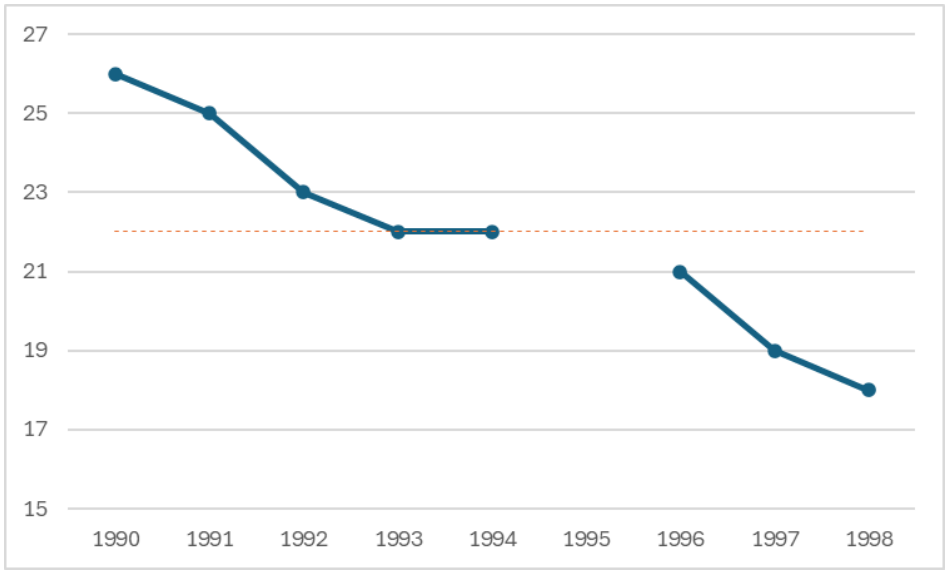


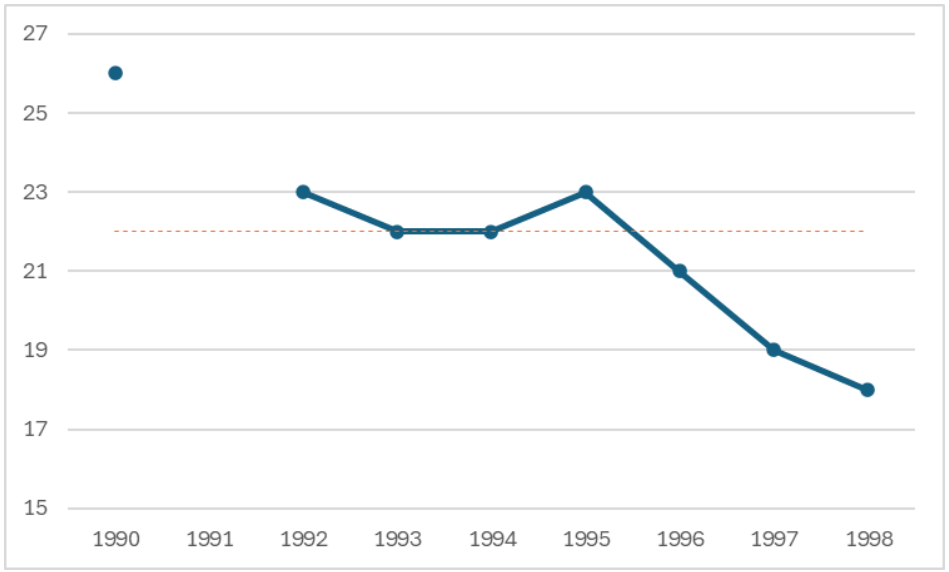


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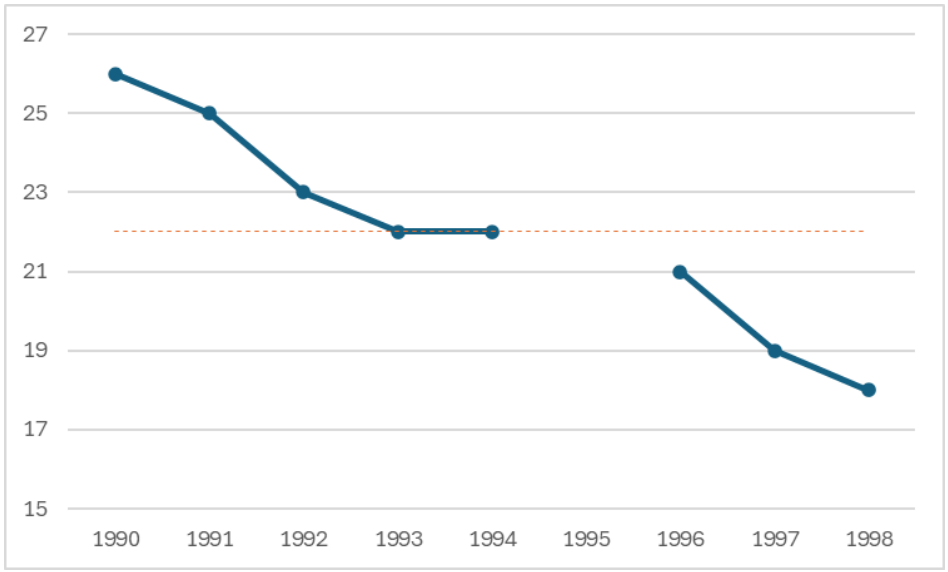


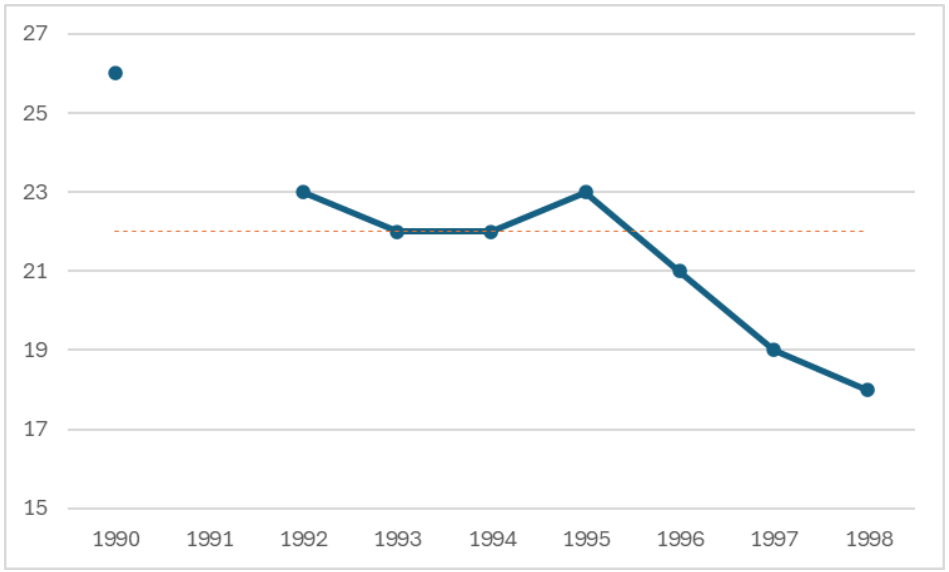
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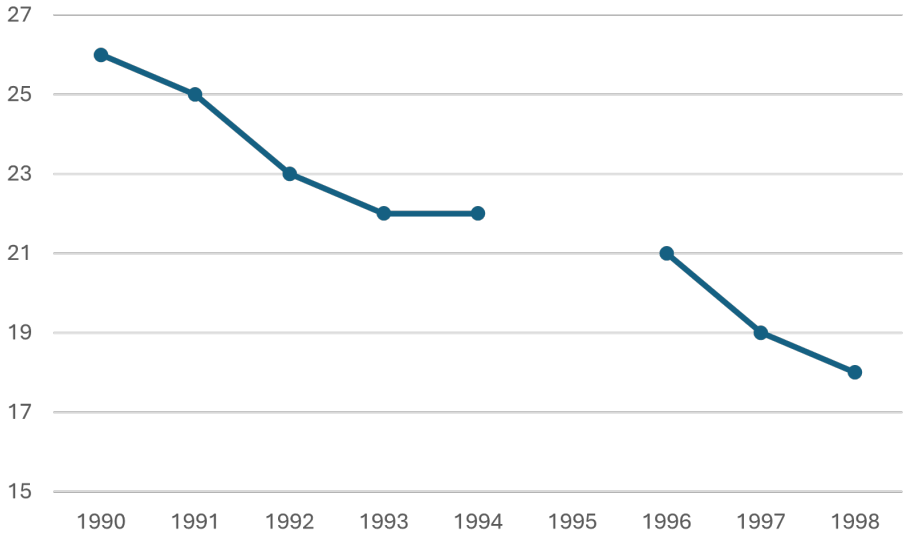
- While the mean may be a suitable proxy for the series on the left, it is not for the series on the right.
- In general, filling with the sample mean is especially dangerous on trending series

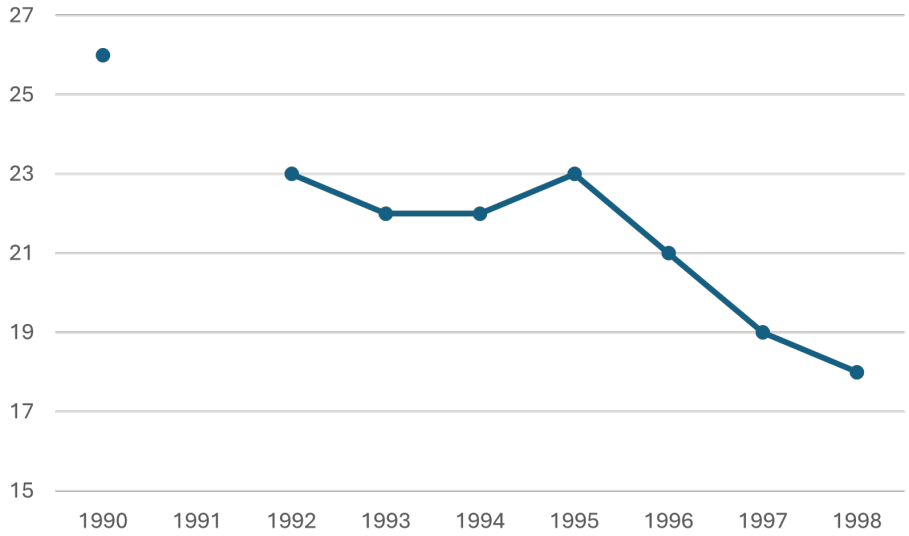
Cleaning Time Series

Interpolating is a common approach

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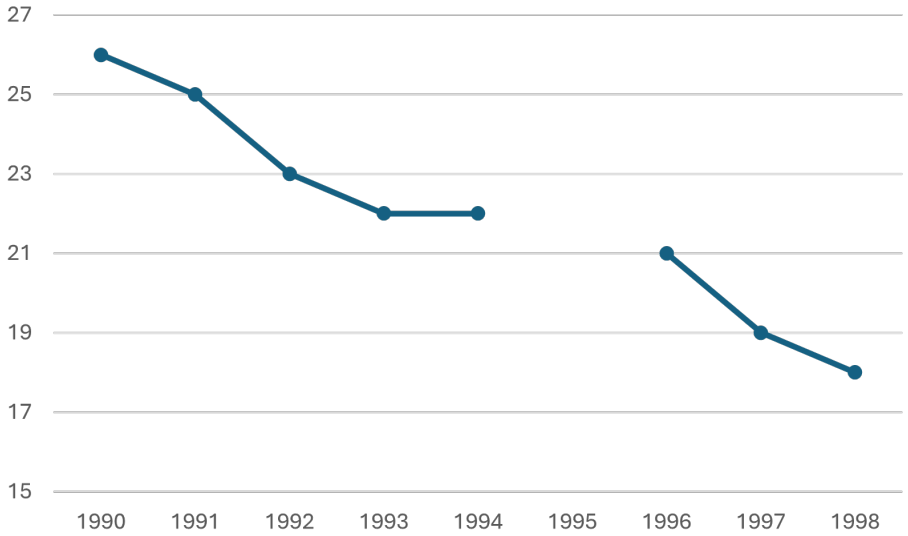
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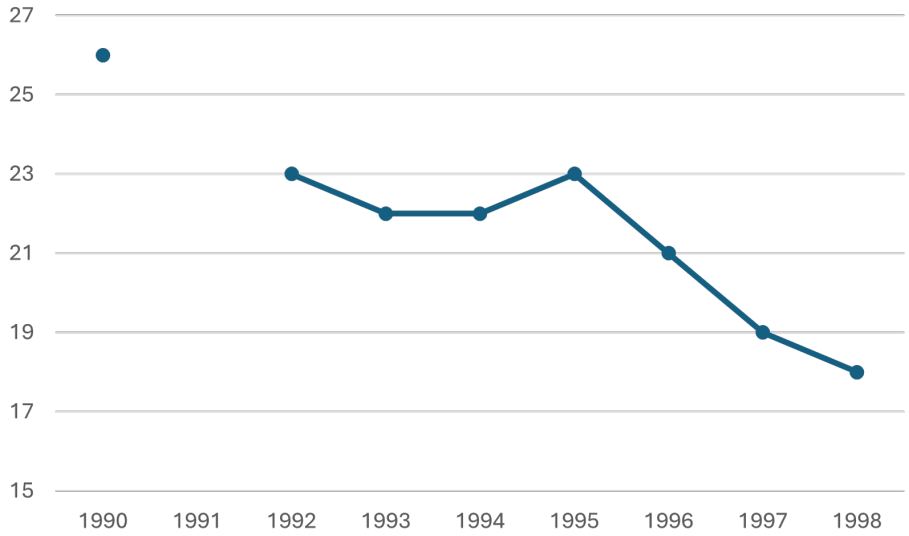




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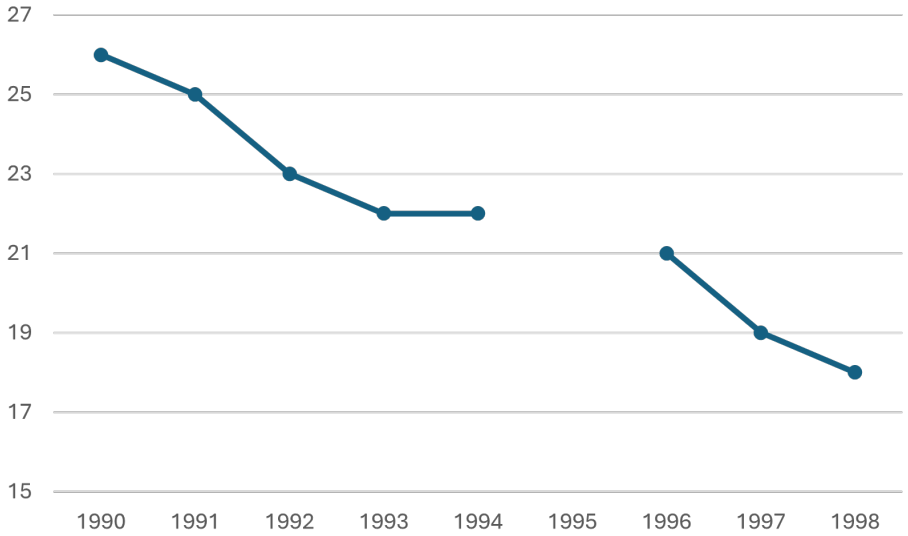


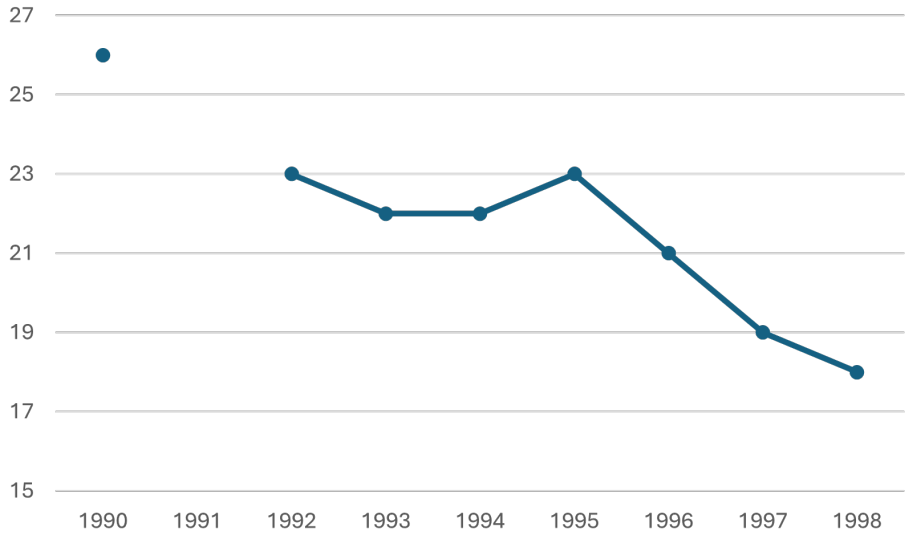


- Interpolating with a local midpoint is common

Cleaning Time Series

Interpolating is a common approach

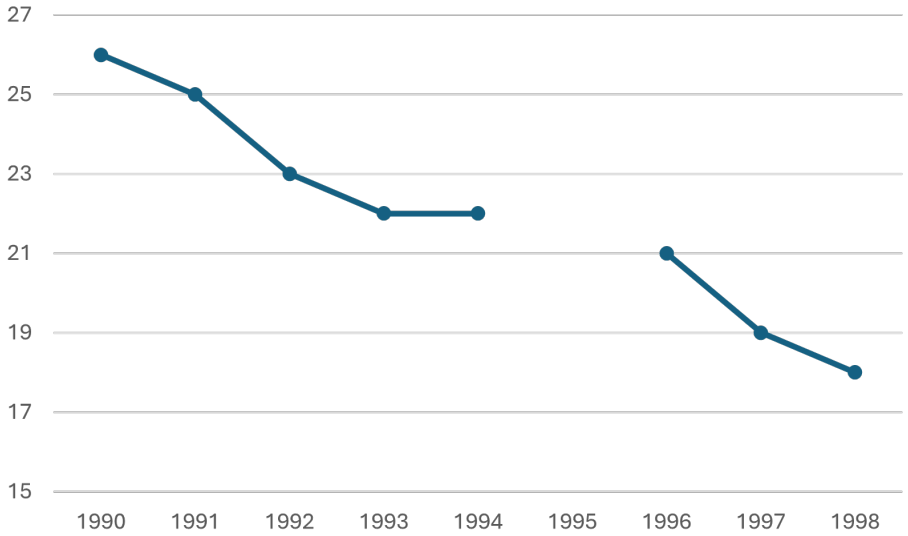


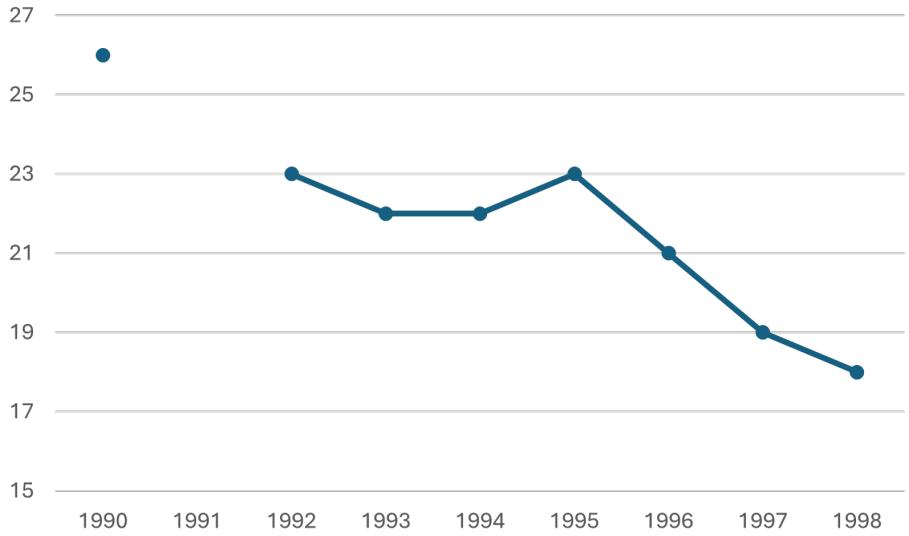


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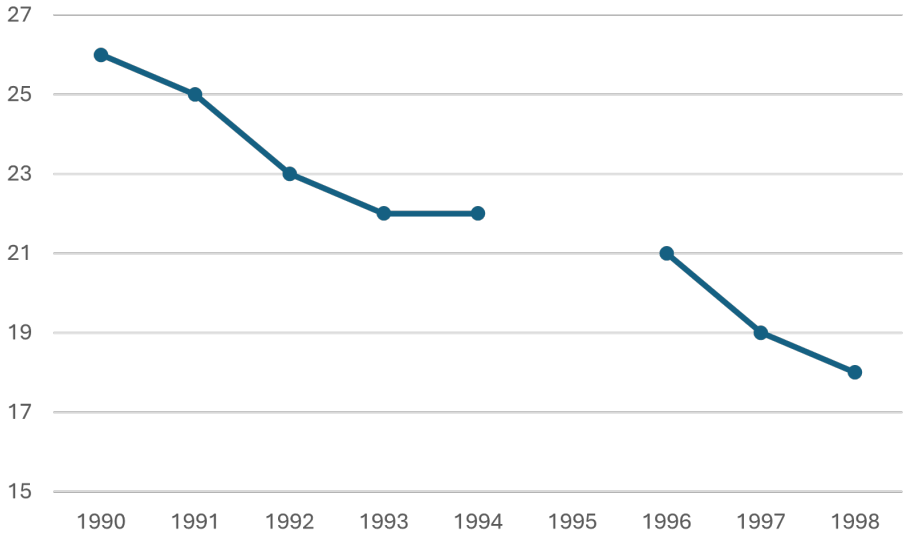


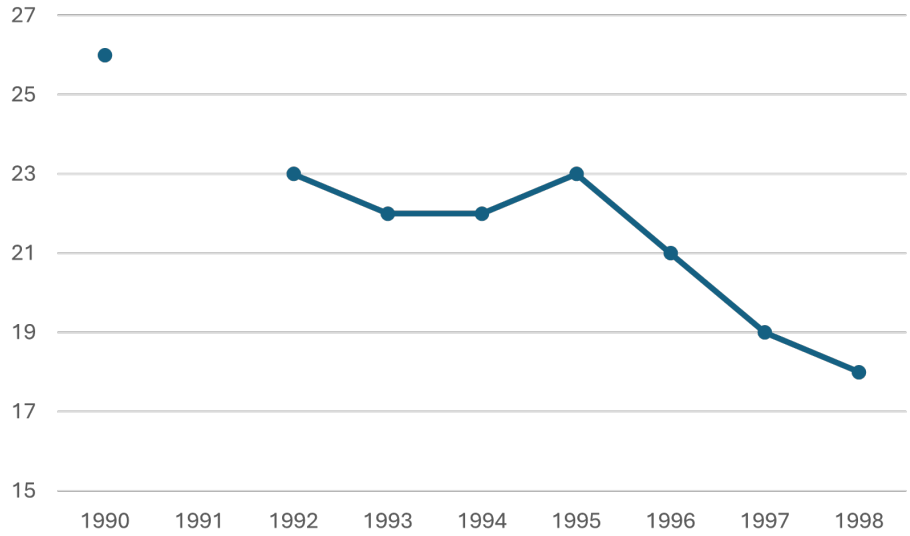


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Cleaning Time Series

Interpolating is a common approach





- Interpolating with a local midpoint is common
- Using sophisticated structures (e.g ARMA) is possible, but requires i) more parametric assumptions, and ii) more data (e.g. modeling the missing point in the right panel is more difficult)

Cleaning Time Series

Forward Filling Values

Cleaning Time Series

Forward Filling Values

Date Time Stamp	Trade Price	Last Trade Price
1/3/2025 10:01	**25.25**	**25.25**
1/3/2025 10:02	**25.35**	**25.35**
1/3/2025 10:03	**26.05**	**26.05**
1/3/2025 10:04		**26.05**
1/3/2025 10:05	**26.10**	**26.10**
1/3/2025 10:06	**26.10**	**26.10**
1/3/2025 10:07	**26.05**	**26.05**
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Cleaning Time Series

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Cleaning Time Series

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- Forward filling values can be sensible in some applications
- Here, the last trade at 10:03 is the best information available at 10:04

Mixed Frequencies

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Date	UR	GDP
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Feb-20	4.3	
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May-20	12.6	
Jun-20	10	-5
Jul-20	9	
Aug-20	9.2	
Sep-20	8	4
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  1. Average of monthly data within quarter
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- Leave as is and use mixed frequency modeling techniques (e.g. MIDAS)

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- Use Case: When earnings is released I will use my model to predict the stock price reaction in order to guide my trade.
- Simple Linear Model: $\mathit{Price} = \alpha + \beta \mathit{Earnings} + u$.
  Note: We usually don't model price directly. We also need to be more precise about forecast horizon. But let us ignore for now.

Avoid Lookahead Bias

Q: What's wrong with setting up our data this way?

Date	Quarter	EPS	Stock Price
31-Mar	Q1	6.12	225
30-Jun	Q2	6.14	236
30-Sep	Q3	6.19	212
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- Our model estimates suffer from Lookahead Bias
- Solution: align release dates; e.g. if Q1 earnings were released on 24Apr, then put that EPS in the same row as the stock price on that day

Outline

1 Introduction

2 Data Prep

3 Explore

4 Explain

5 Forecast

Outline

3 Explore
- AutoCorrelation Functions
- Clustering
- Dimension Reduction
- Stationarity
- Nonstationarity and Cointegration
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EDA for Time Series

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We encounter a few special challenges due to the nature of the data.

We also have a few new tools.

Time Series Tools for EDA

- Autocorrelation Function
- Partial Autocorrelation Function
- Cross Correlation Function

Autocorrelation Functions

Definition

The kth autocorrelation is defined as

$$\rho_y(k) = \frac{Cov(y_t, y_{t-k})}{V(y_t)}$$

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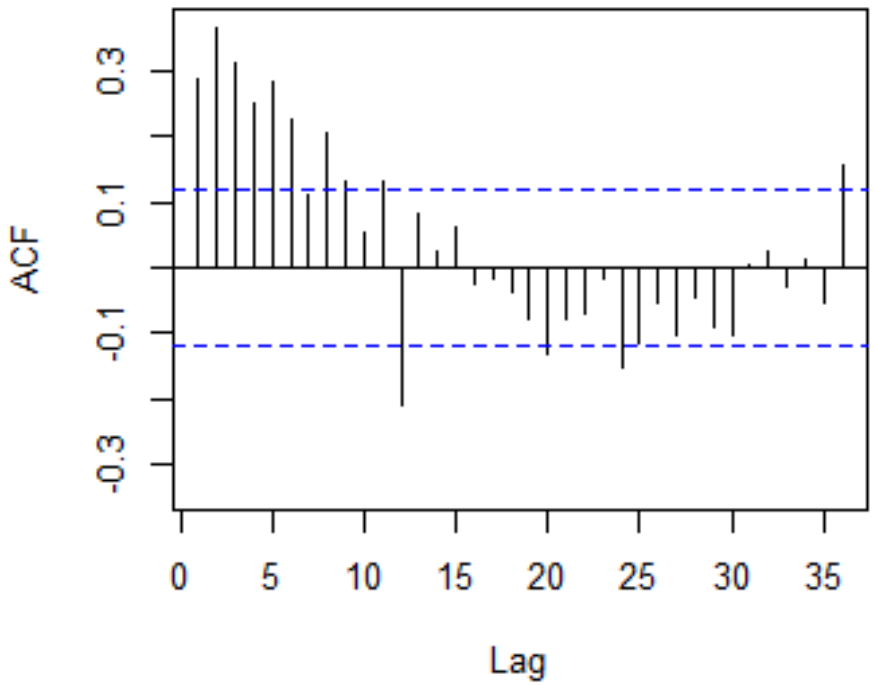
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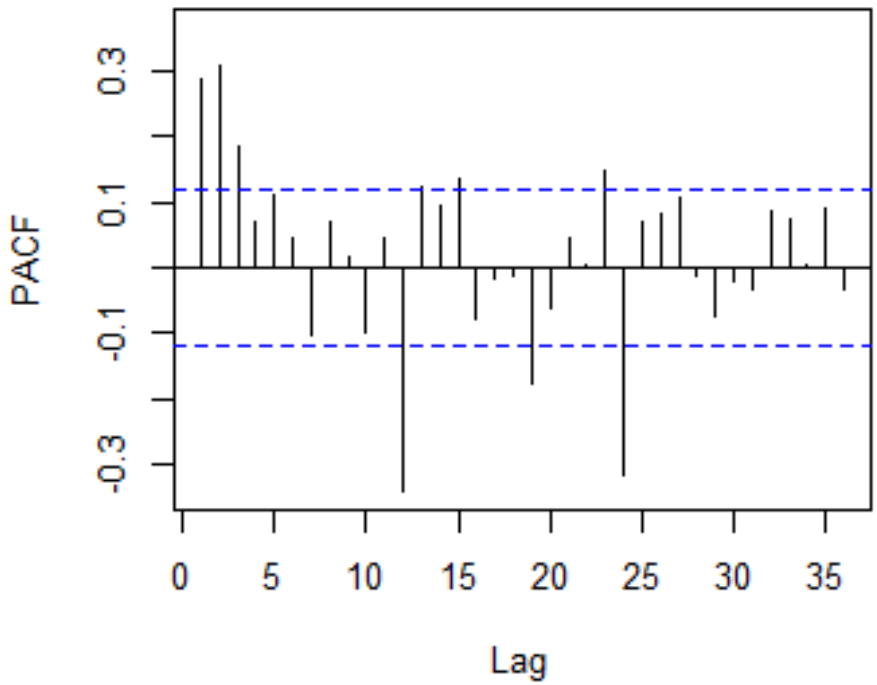
Definition

The kth partial autocorrelation is defined as the estimate of θ_k in the AR(k) autoregression. For instance

- $pac_y(1) = \hat{\theta}_1$ from $y_t = \theta_1 y_{t-1} + \varepsilon_t$
- $pac_y(2) = \hat{\theta}_2$ from $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t$

ACF and PACF


ACF
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-0.1
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0
5
10
15
20
25
30
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Lag


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Since PACF(5) $<$ ACF(5), most of the dependence is transmitted through shorter lags rather than a direct effect.

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Q: What does $\rho_{r_{IBM}, r_{SP500}}(2) = 0.1$ mean?
A: Correlation between SP500 returns on "Monday" and IBM returns on "Wednesday"is 0.1; i.e. IBM responds with a two day lag.

Outline

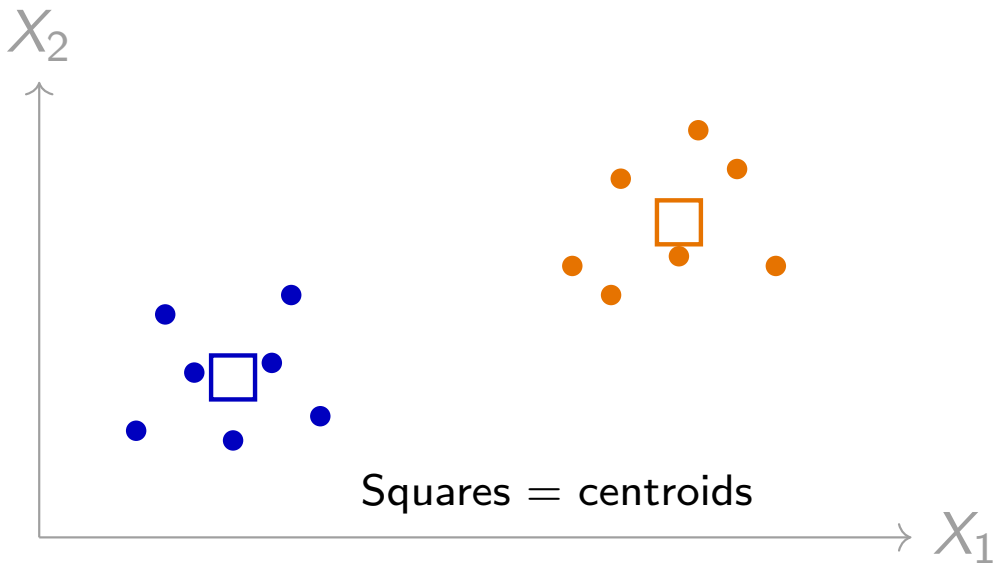
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k-Means on Cross-Sectional Data

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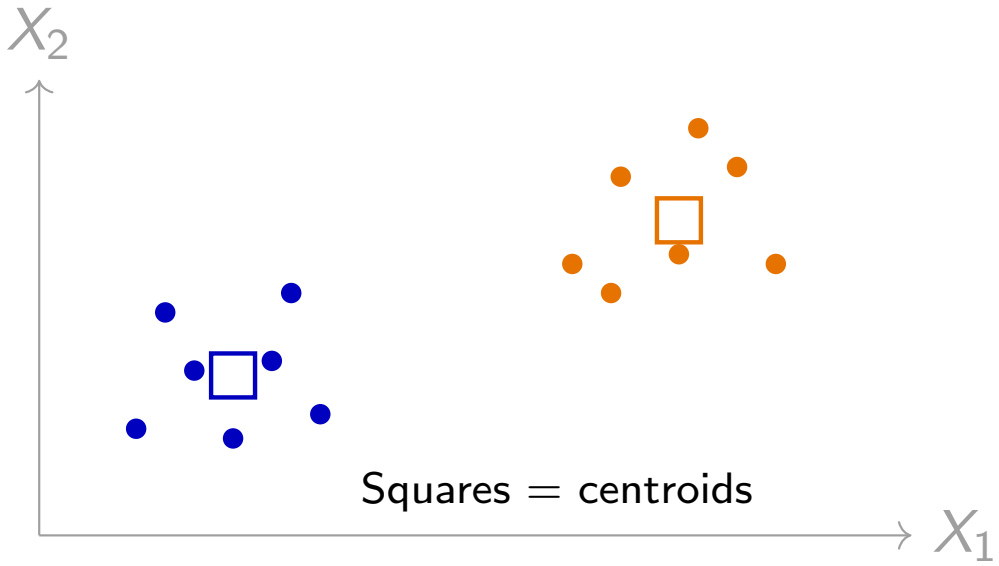
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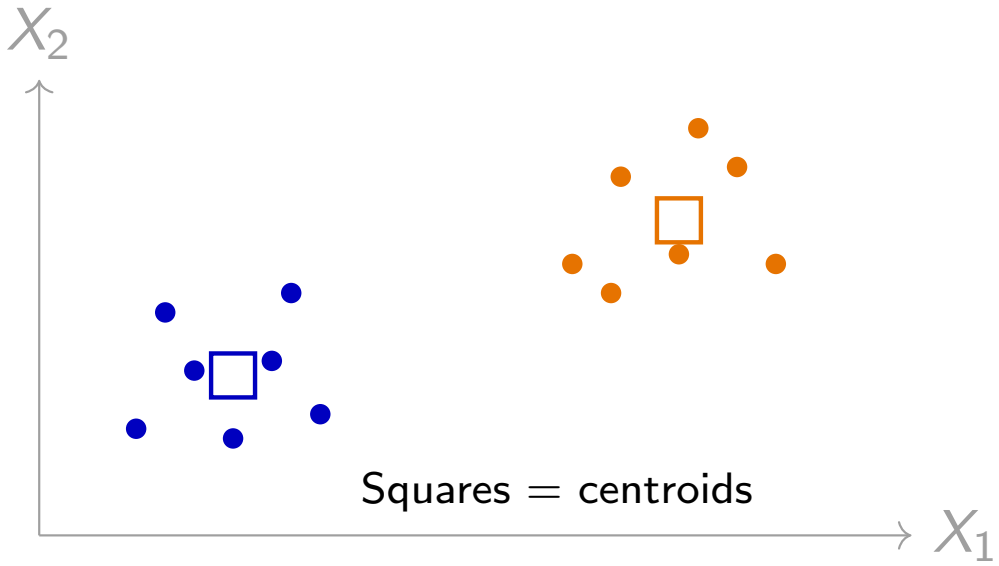
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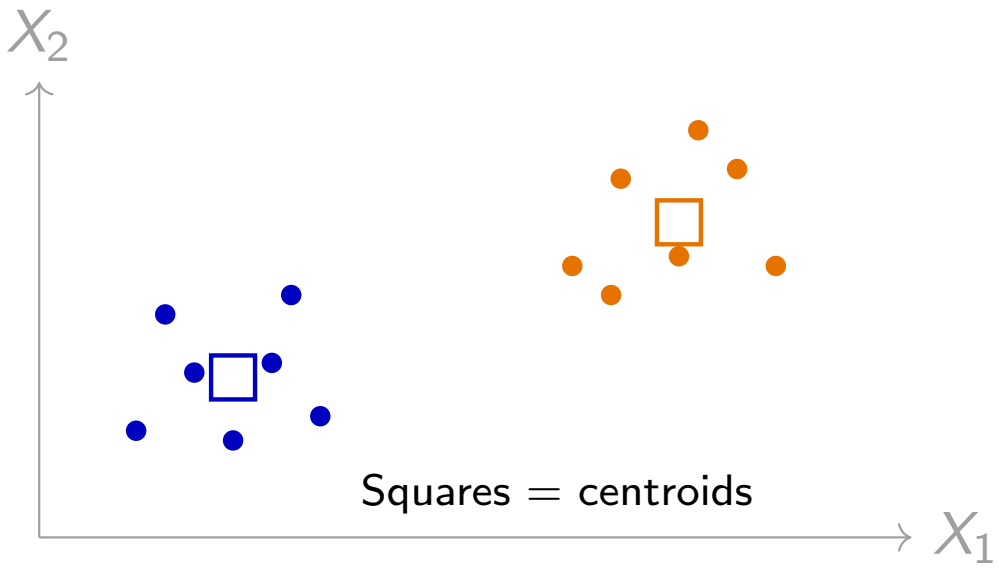
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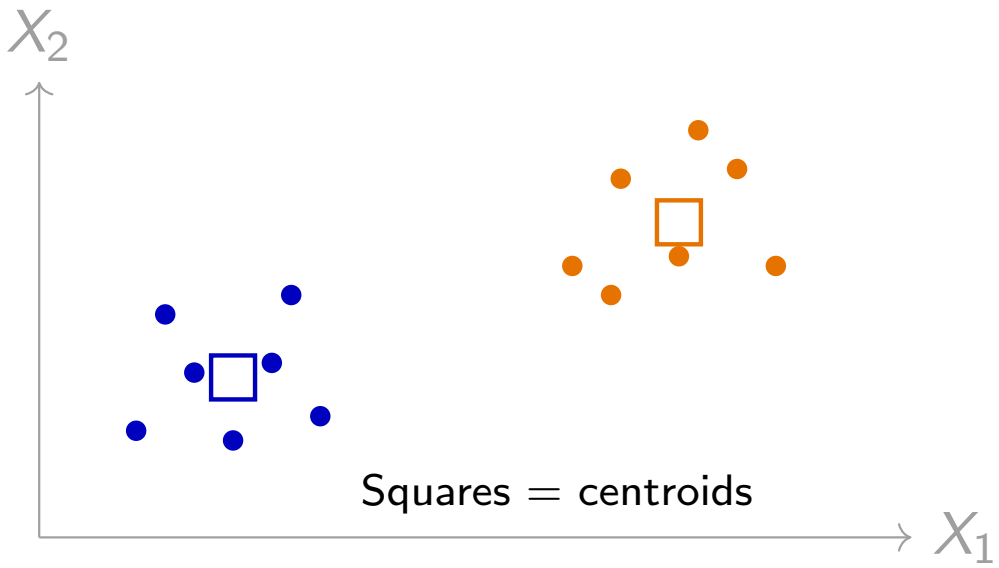
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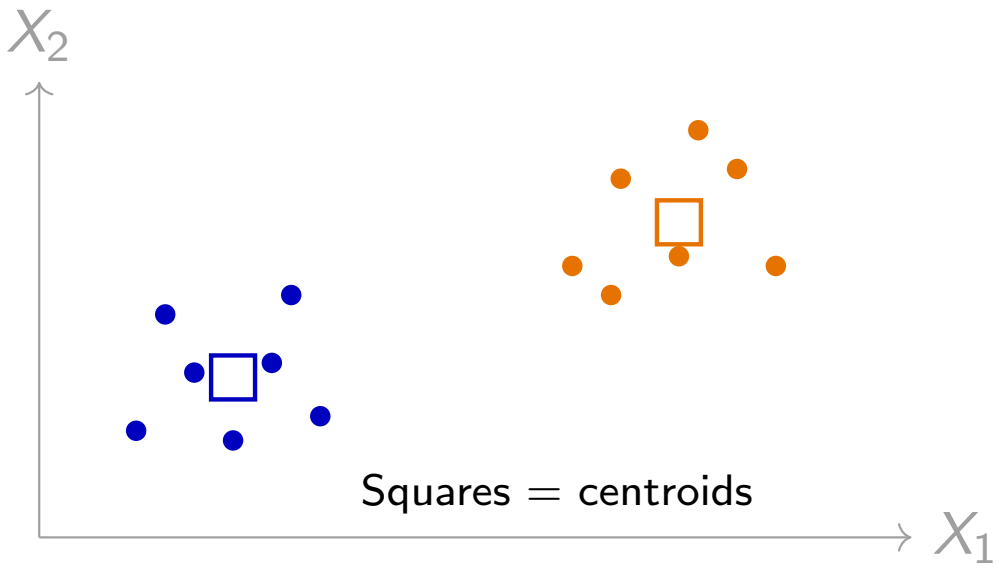
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k-Means on Cross-Sectional Data

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For time series, the key challenge is deciding what "near" means.

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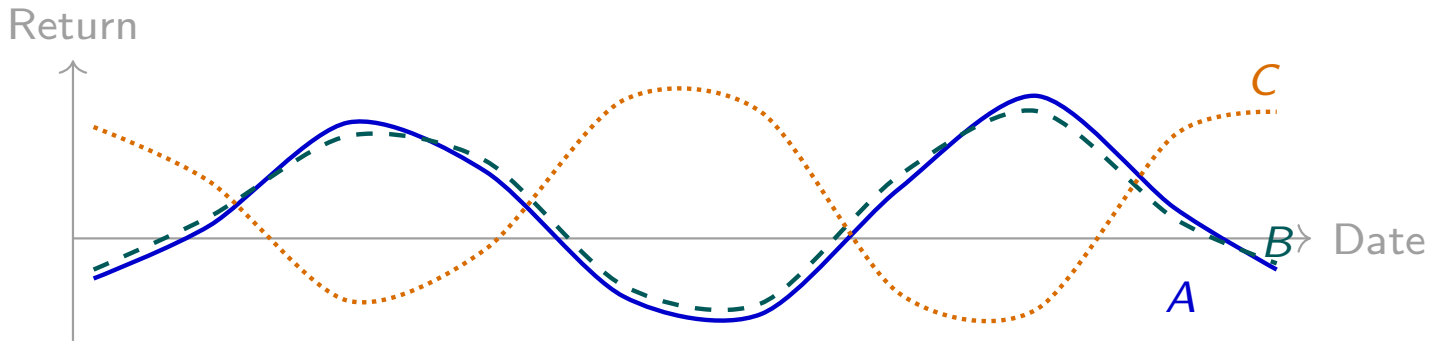
Important: These are different definitions of similarity, not successive improvements to one measure.

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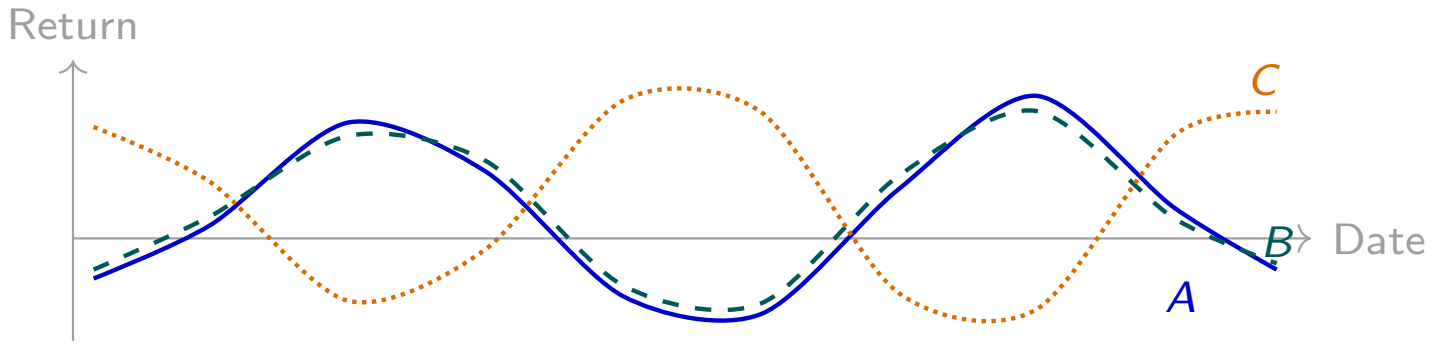
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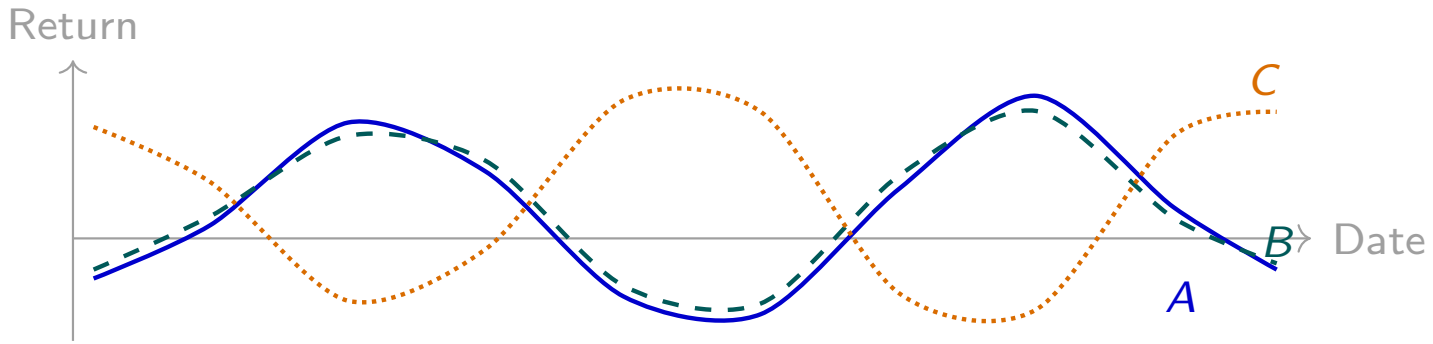
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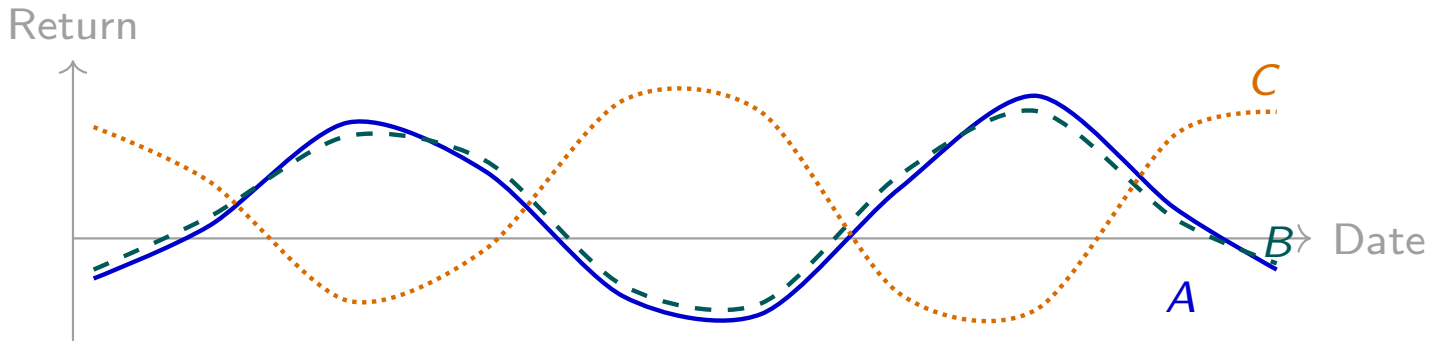
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- When dates are aligned, **contemporaneous correlation** summarizes this co-movement across entire return histories.

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Interpretation: A and B are close; C is far from both because it tends to move in the opposite direction.

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Advantage: We can change the time-series dissimilarity measure without changing the clustering algorithm.

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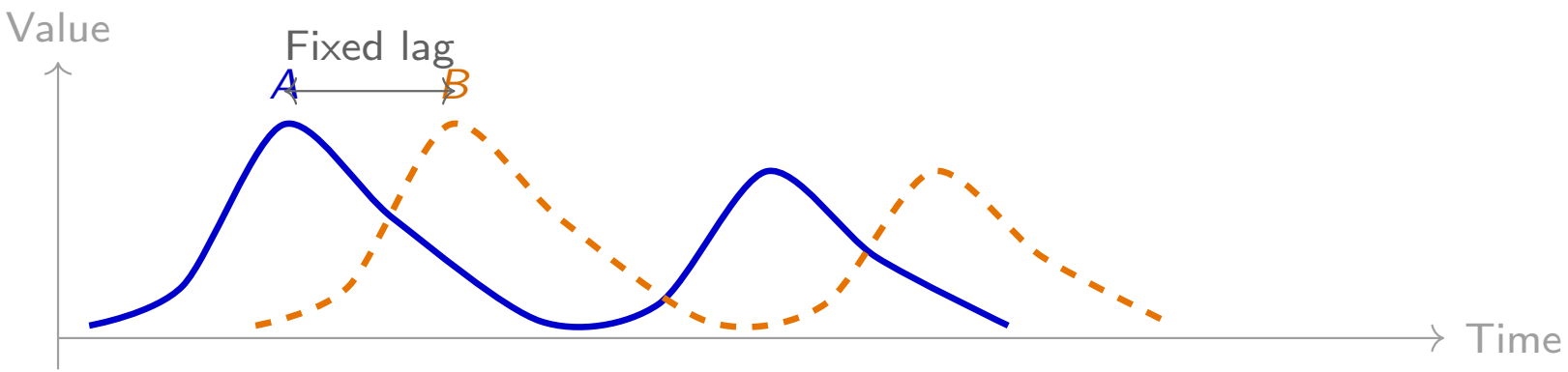
For aligned series, ordinary k-means on standardized return histories is also possible. We use k-medoids so that lagged-correlation and DTW dissimilarities can be used later without changing the clustering algorithm.

Clustering Time Series with Leads/Lags

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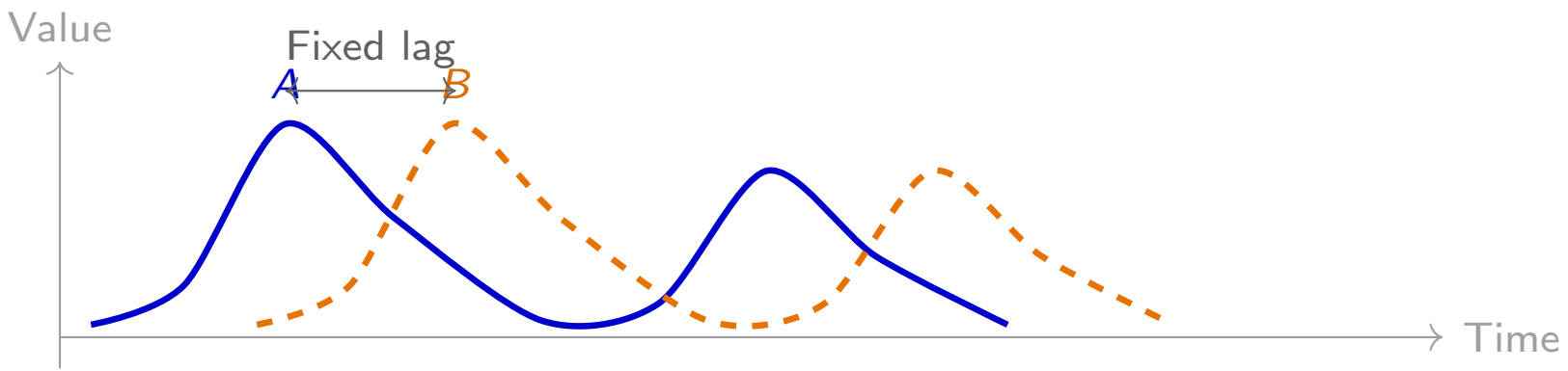
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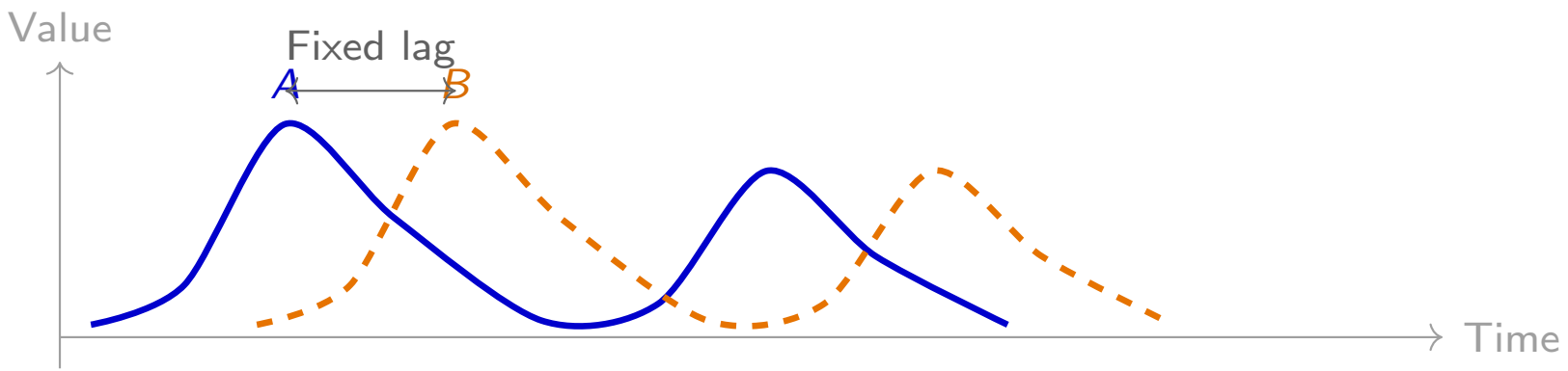


Search over plausible fixed lags $\ell \in \{-L, \ldots, L\}$:

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Positive ℓ means j lags i. The maximum is computed over permissible lags using overlapping observations.

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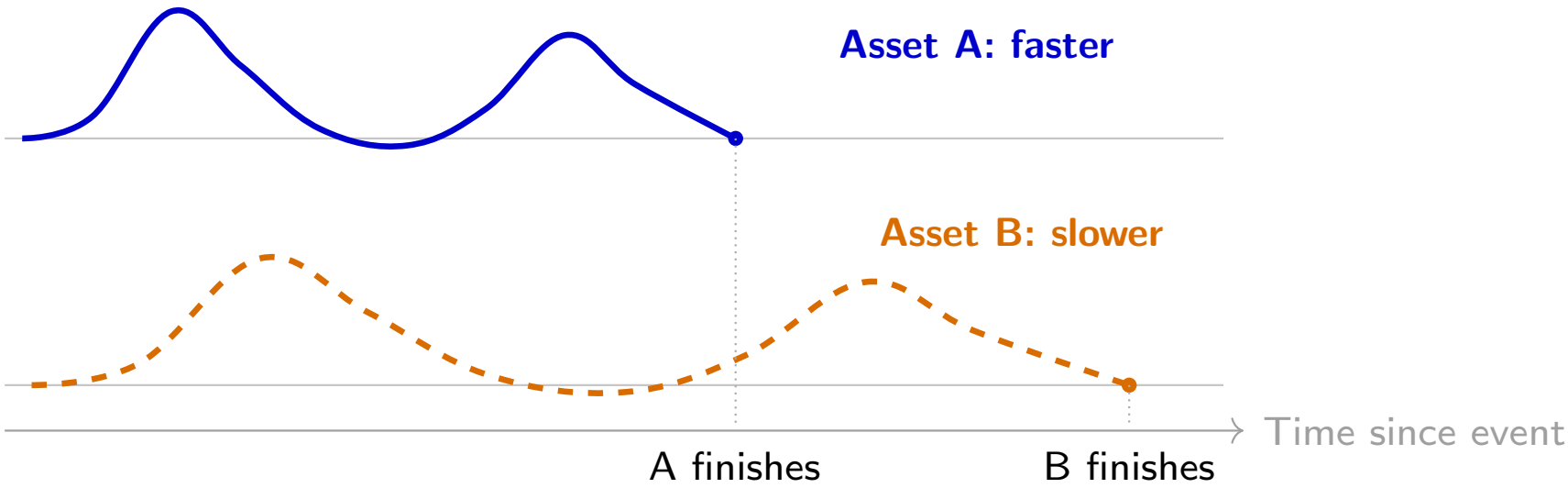
Takeaway: We changed the pairwise dissimilarities, not the k-medoids procedure.

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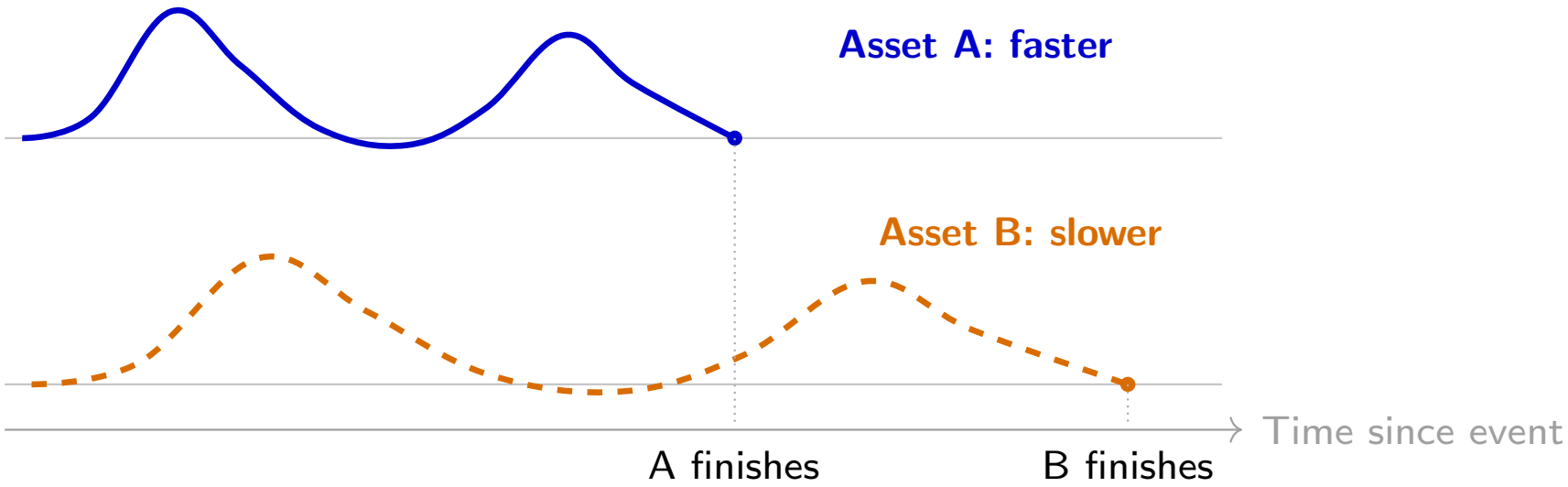
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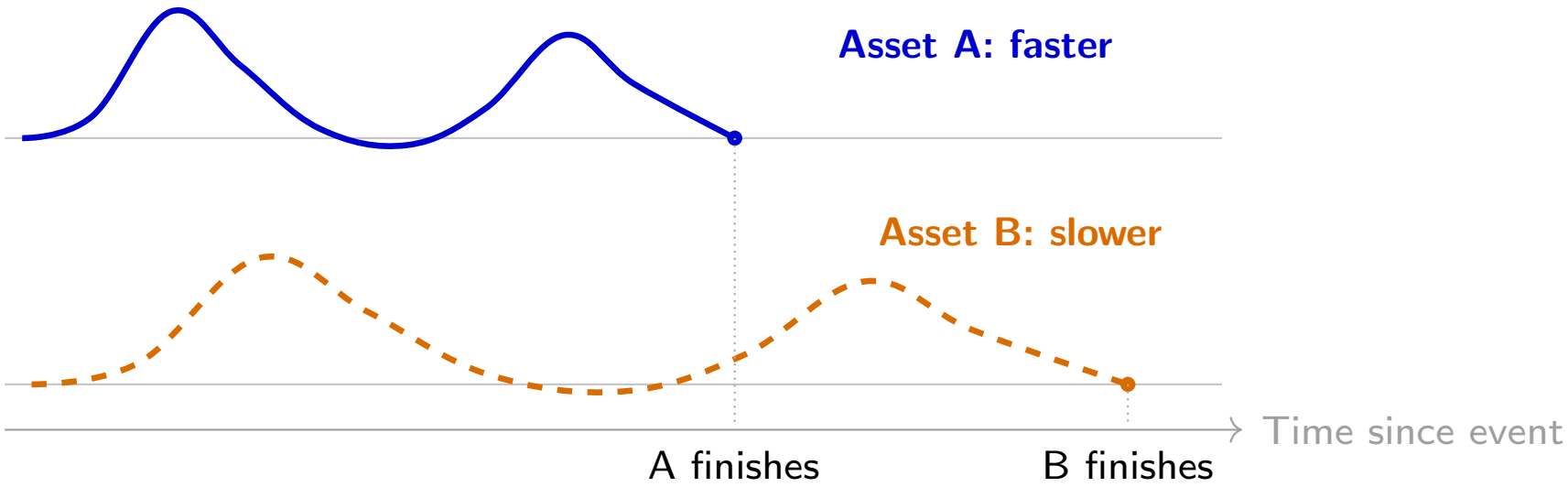
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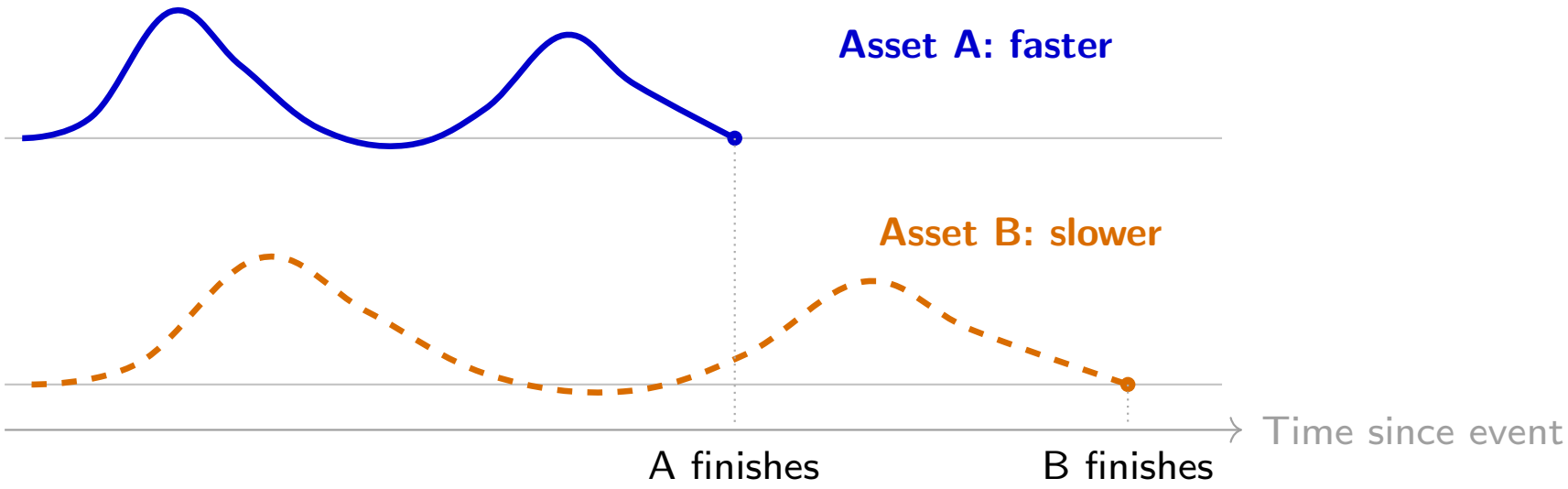
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- A fixed lag only *shifts* a series. It cannot *stretch or compress* time. DTW can.

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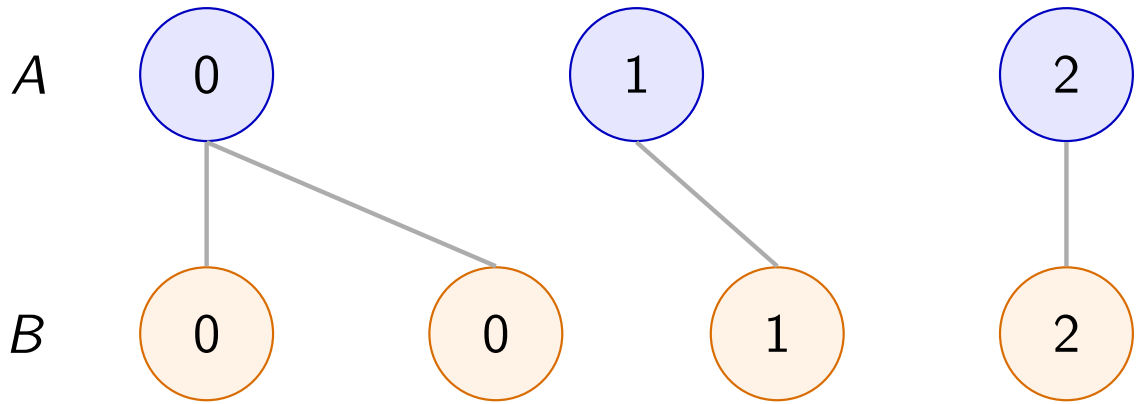
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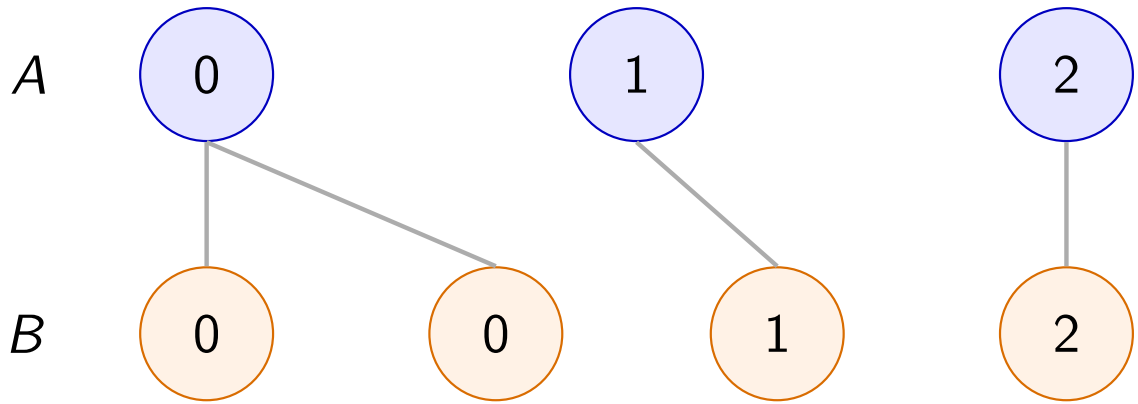


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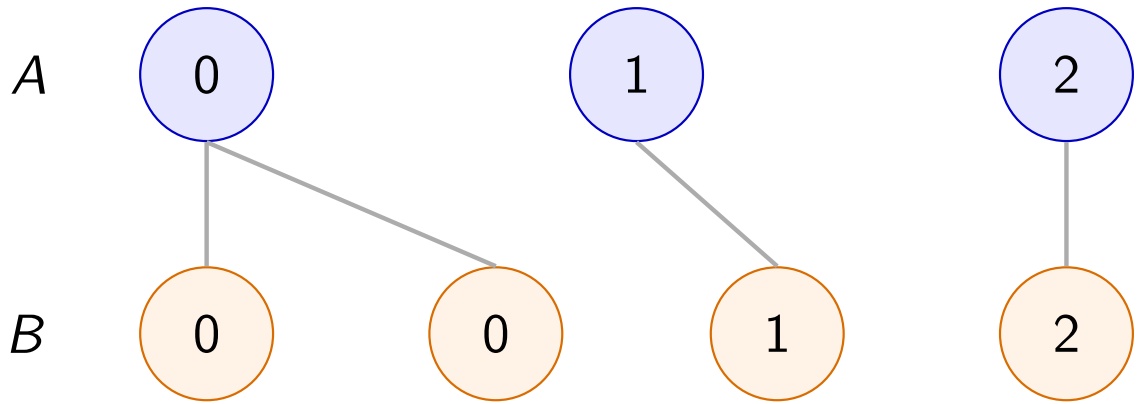
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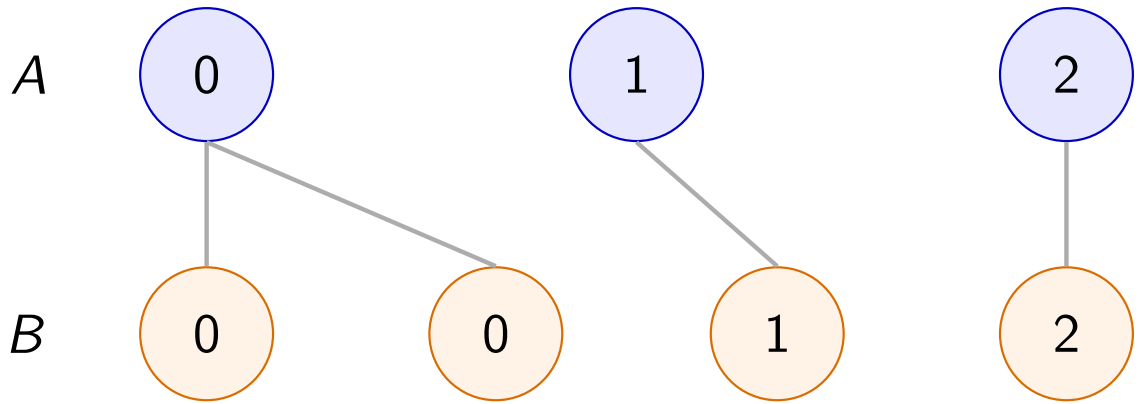
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- DTW can match one observation in A to multiple observations in B.

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- Both sequences follow the same progression; B stays at zero for an extra observation.
- DTW can match one observation in A to multiple observations in B.
- It finds a monotone alignment that minimizes total *value mismatch*.

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Important: Standard DTW penalizes *differences between matched values*, not the amount of warping itself. We may impose a maximum warping window.

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This is **dynamic programming**: at each cell, use the cheapest of the three allowable preceding paths.

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Interpretation: The extra zero in B can be matched to the first zero in A without an additional mismatch cost.

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What changed? The *distance measure*, not the basic k-medoids logic.

Alternatives: hierarchical clustering; specialized centroid-based methods using DTW barycenters (DBA). k-Shape is a related shape-based method built around normalized cross-correlation.

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- **Interpretation:** If the exact event date matters, allowing flexible timing may erase economically important differences.

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Example: Two assets can have weak return correlation but similar persistence or volatility dynamics. They may belong together under a feature- or model-based definition of similarity.

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Takeaway: First decide what makes series similar; then cluster the assets using a compatible algorithm.

Outline

3 Explore
- AutoCorrelation Functions
- Clustering
- Dimension Reduction
- Stationarity
- Nonstationarity and Cointegration
- Decomposition, Filters, and Seasonality
- From Diagnostics to Models

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Goals:

- Condense related features into a few informative factors.
- Summarize the historical behavior of a series.
- Identify common drivers of multiple evolving series.

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3. **Multiple series evolving together**
   - A panel of asset returns $\longrightarrow$ a few common factor series.
   - Method: dynamic factor model (DFM).

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Goal: Produce a low-dimensional score, such as $z_{i1,t}$, that captures important common variation across horizons.

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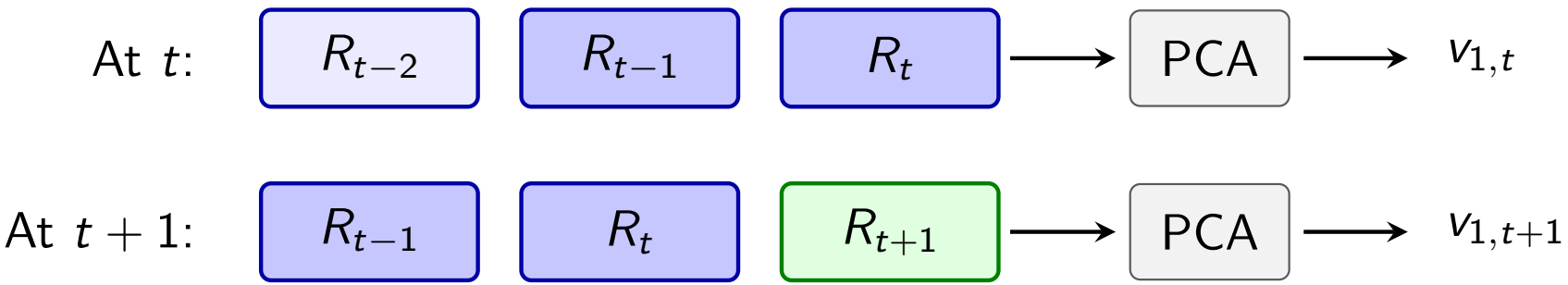


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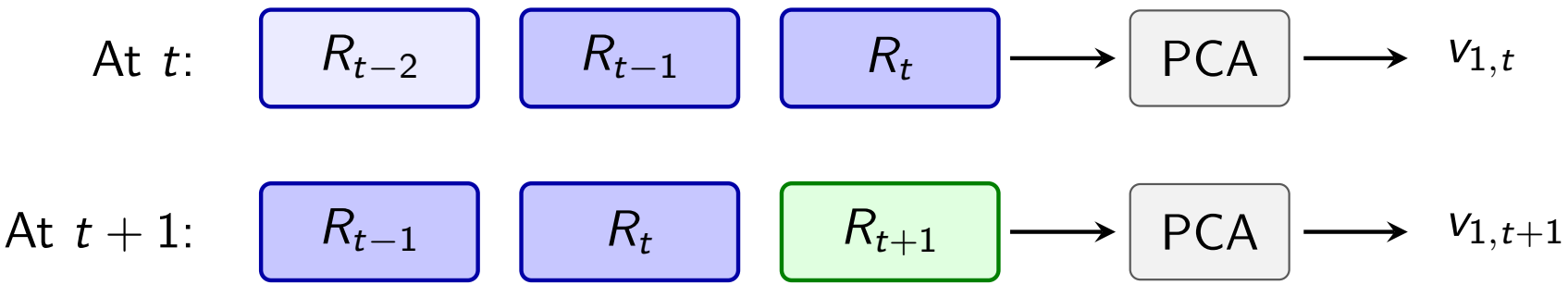


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- $W = 1$: Use only R_t: cross-sectional PCA at date t.

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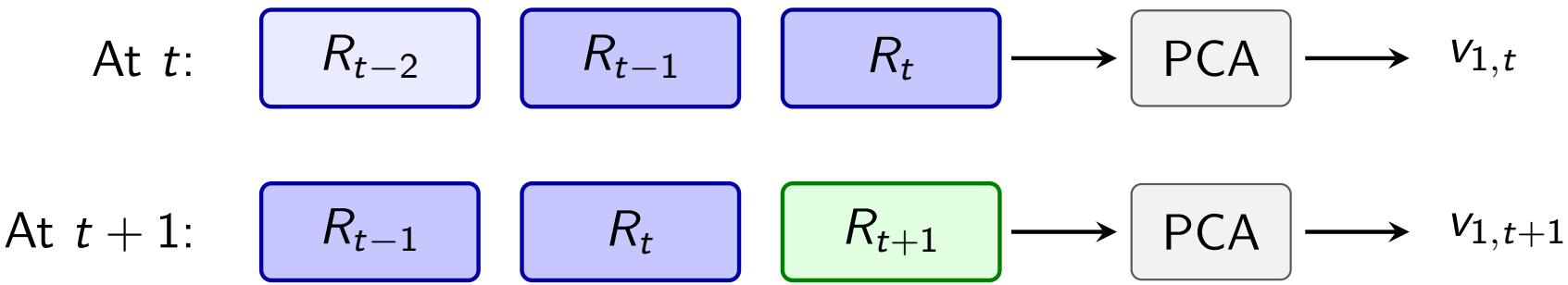


Illustration: $W = 3$ dates. Each box contains N asset observations.

- $W = 1$: Use only R_t: cross-sectional PCA at date t.
- $W > 1$: Pool W dates. At $t + 1$, drop the oldest date, add the newest, and reestimate PCA.

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Practical note: Eigenvectors may switch signs across windows without changing the component; align signs before comparing scores.

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Contrast: Ordinary feature PCA compresses related indicators; lagged PCA compresses information from a historical window.

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Recall: Under suitable conditions, a linear autoencoder with squared-error loss recovers the PCA reconstruction subspace.

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Key distinction: A standard autoencoder can compress features nonlinearly, but its architecture does not explicitly model their sequential evolution.

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Limitation for forecasting: A full-history embedding summarizes that completed sample; it does not provide a historical real-time signal at earlier dates.

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- Training windows and transformations must exclude information unavailable at the date the code is used.

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Q: What distinguishes a sequence autoencoder from a standard one?

Recall: A standard autoencoder compresses features observed at one date:

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Limitation: Encoding assets separately does not explicitly model their contemporaneous comovement. A multivariate sequence autoencoder can do both.

Dynamic Factor Models: Motivation

Q: What dimension-reduction problem are we trying to solve?

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Connection: PCA can identify common factors across assets. A DFM additionally specifies how those factors evolve over time.

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Observed returns and common factors:

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- Unlike PCA, which summarizes common variation across asset returns without explicitly modeling factor dynamics, a DFM also specifies how those common factors evolve over time.

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For live forecasts, avoid using smoothed historical factors that incorporate future observations.

Dimension Reduction: Choosing the Method

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Method	**Main structure**	**Typical output**
PCA	Linear feature covariance	Scores by asset/date
Standard AE	Nonlinear feature compression	Codes by asset/date
Lagged PCA / SSA	Relationships across lags	Window scores
Sequence AE	Ordered temporal patterns	Window codes
DFM	Common factors + time dynamics	Factor time series

These methods solve different compression problems; they are not simply more complex versions of one another.

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Caution: High explained variance or low reconstruction loss need not imply better forecasts or investment signals. Evaluate any resulting signal out of sample, without look-ahead bias.

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- AutoCorrelation Functions
- Clustering
- Dimension Reduction
- Stationarity
- Nonstationarity and Cointegration
- Decomposition, Filters, and Seasonality
- From Diagnostics to Models

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Our diagnostics guide our modeling choices:

- Autocorrelation $\rightarrow$ Autoregressive structure (AR, ARIMA)
- Trends / nonstationarity $\rightarrow$ Differencing or trend models
- Seasonality $\rightarrow$ Seasonal components (SARIMA, decomposition)
- Lagged effects $\rightarrow$ Distributed lag / ADL models
- Long-run relationships $\rightarrow$ Error Correction Models (ECM)
- Multiple series $\rightarrow$ VAR or panel models

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Diagnostics are not just descriptive—they determine the model

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4 Explain

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   - Linear Models (ADL, FDL)
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   - Linear Model Violations (Serial Correlation)
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Modeling Temporal Dependence

From our analysis, we observed:

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Key idea:

- The past helps predict the present

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One way to capture dependence:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$$

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We now generalize this idea with richer models

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Comparing Time Series Modeling Approaches

Approach	**Strengths**	**Limitations**
Linear Models	Interpretable Economic structure Works with small data	Restrictive Linear dynamics
State Space Models	Flexible Captures latent components Unifying framework	More complex Requires specification
Machine Learning	Strong predictive performance Captures nonlinearities	Less interpretable Data intensive
Global / Foundation Models	Leverages many series Pretrained / scalable	Less transparent Harder to control

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- Forecasting is NOT EDA
- Forecasting is NOT Causal Estimation & Inference
- Forecasting IS a type of prediction: predicting outcomes at future dates from currently available information.

Forecasting vs Other Prediction Tasks

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- A future conditional scenario becomes a deployable forecast only if that future market movement is known, predicted, or explicitly treated as a scenario.
- Many ML applications predict observed outcomes that are not in the future; **time-series forecasting** emphasizes future, out-of-sample dates and information availability.

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- **Additional forecast requirements:** predictive uncertainty, coherent aggregation, and updating as data arrive.

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- Note: We will occasionally switch to different applications (e.g. financial returns) to illustrate specialized tasks such as volatility forecasting.

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- $\widehat{s}_{t+h|t}^{(i)}$: forecast made at t of sales at $t + h$.
- All transformations and covariates must respect information availability; future planned variables may be used if they were already known at t.

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- Data Properties and availability (what can you get, regular time intervals, suffers from look ahead bias, trustworthy, etc...)

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  - Proxy model / behavior e.g. "I want to know if the CoT data can forecast next month's Treasury yield"

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If the answer is NO to any of these questions, you need to iterate back to Steps 1,2,3

6. Forecast

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How Evaluate?

- Summary metrics, e.g. MSFE (Mean squared forecast error)
- Forecast error plots with time and covariates
- Forecast error densities

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- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
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- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

Challenges with Forecasting

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All

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All Forecasts

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- Advice: never settle for a single point forecast; provide a range of values

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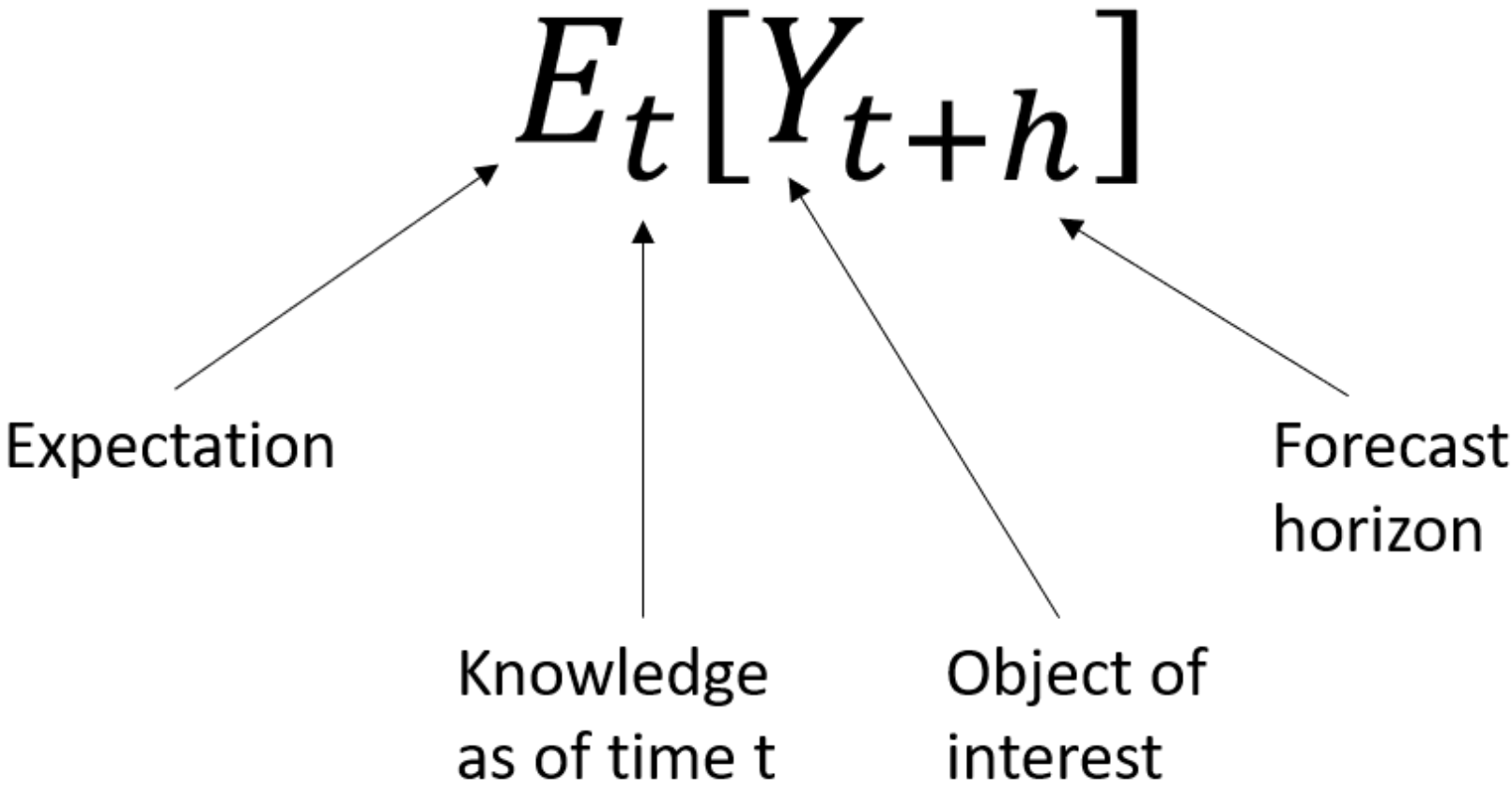
Anatomy of a Forecast

Q: Can you explain the components of this forecast?

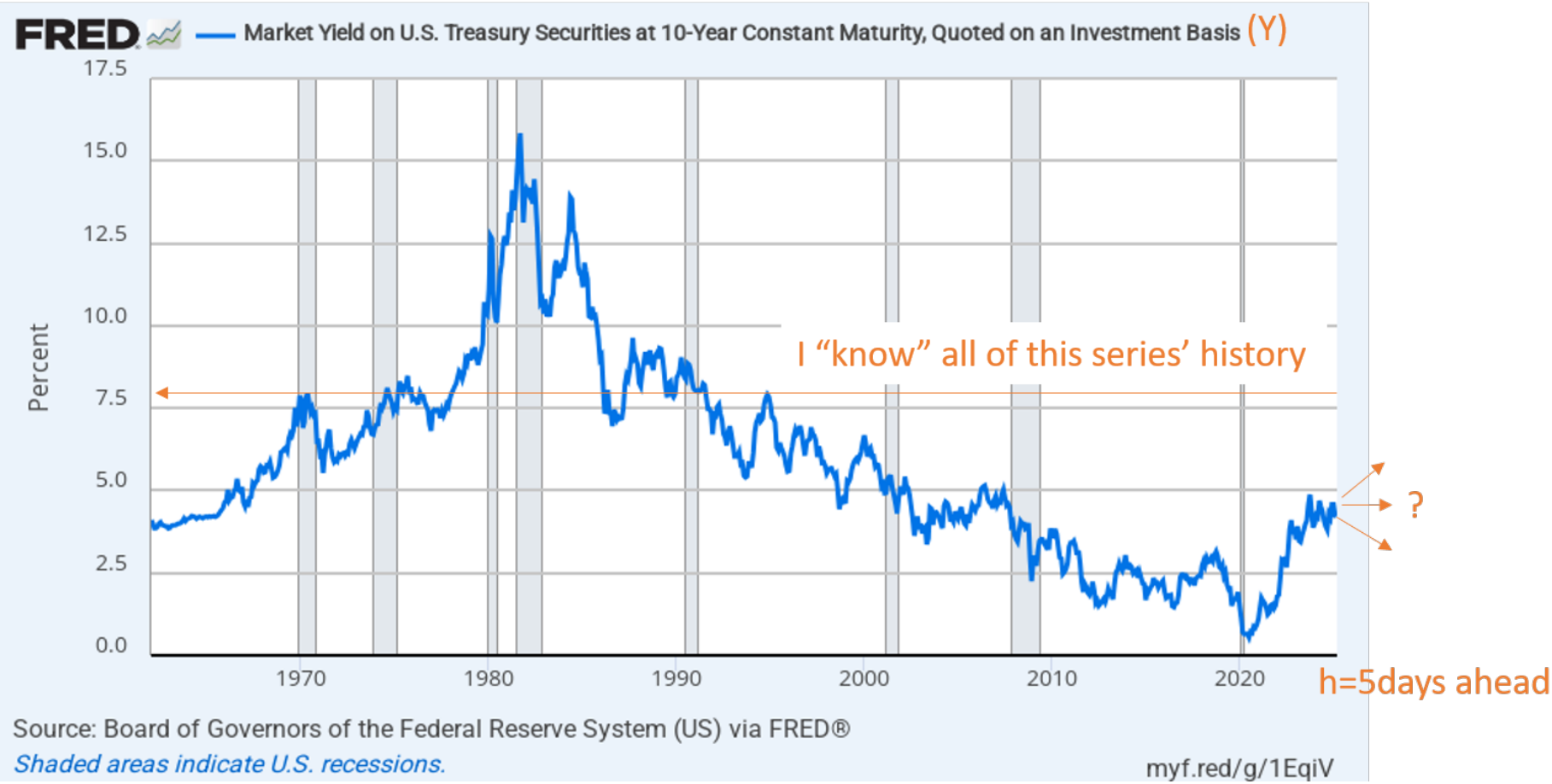
$$E_t[Y_{t+h}]$$

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5day ahead US Treasury yield forecast



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  - aka Out of Sample

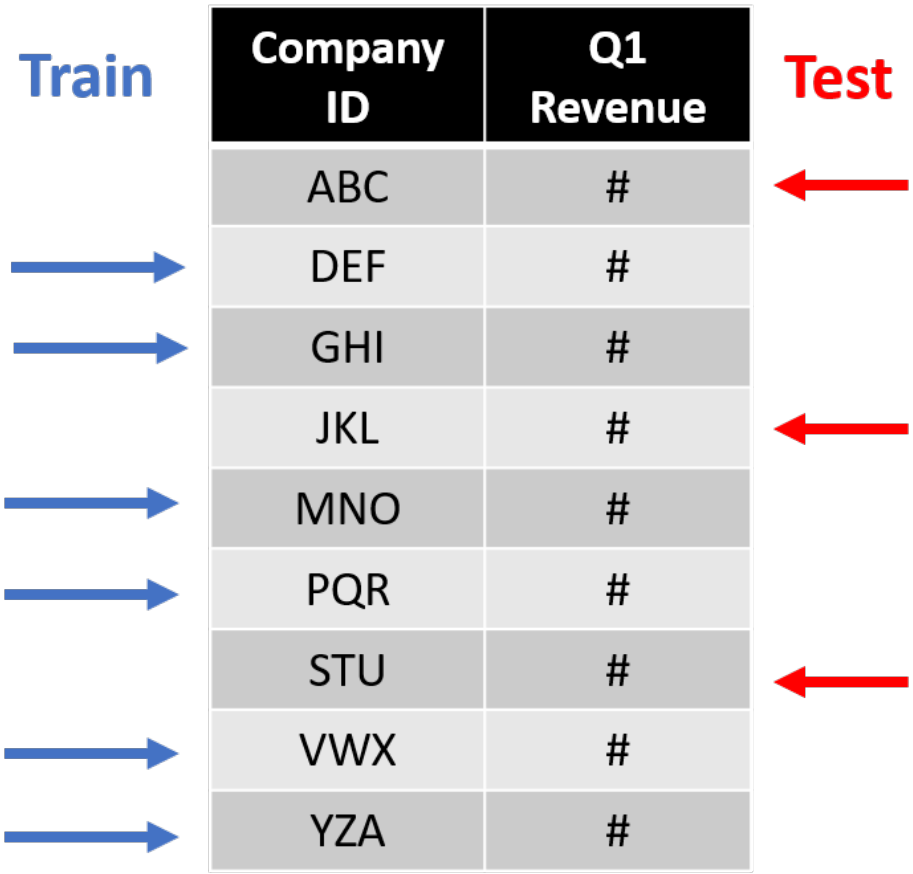
Typical Train / Test Split

Q: Consider the data below. How do we determine the train/test split?

Company ID	Q1 Revenue
ABC	#
DEF	#
GHI	#
JKL	#
MNO	#
PQR	#
STU	#
VWX	#
YZA	#

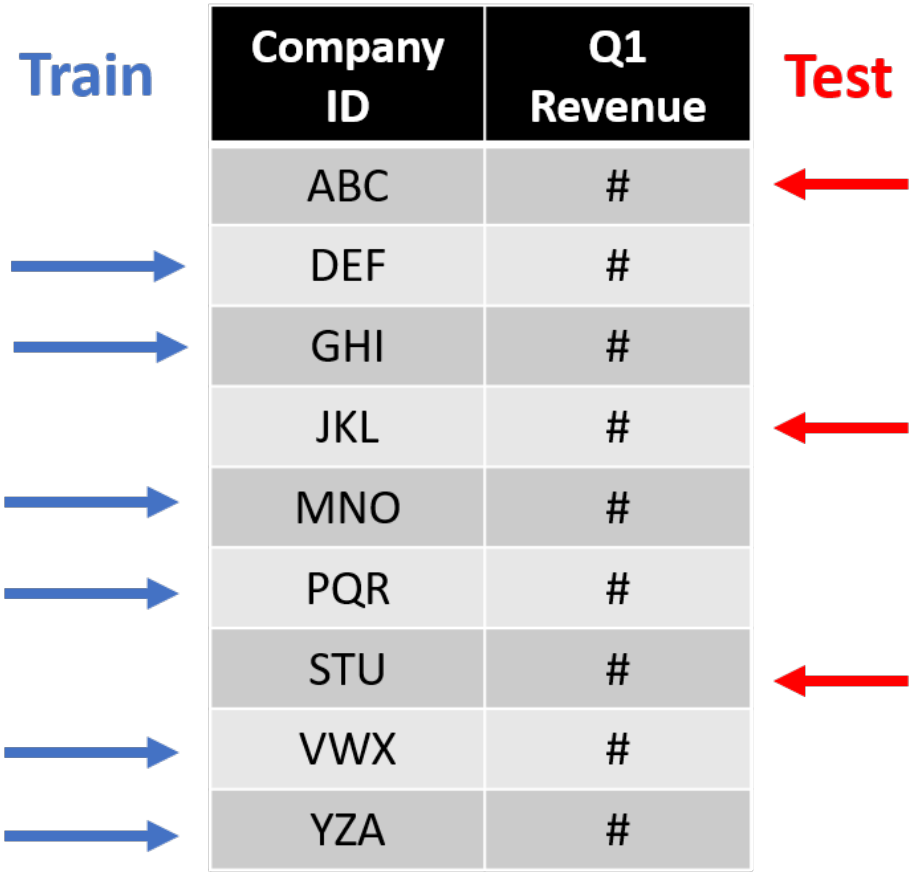
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A: Random assignment of each "row".



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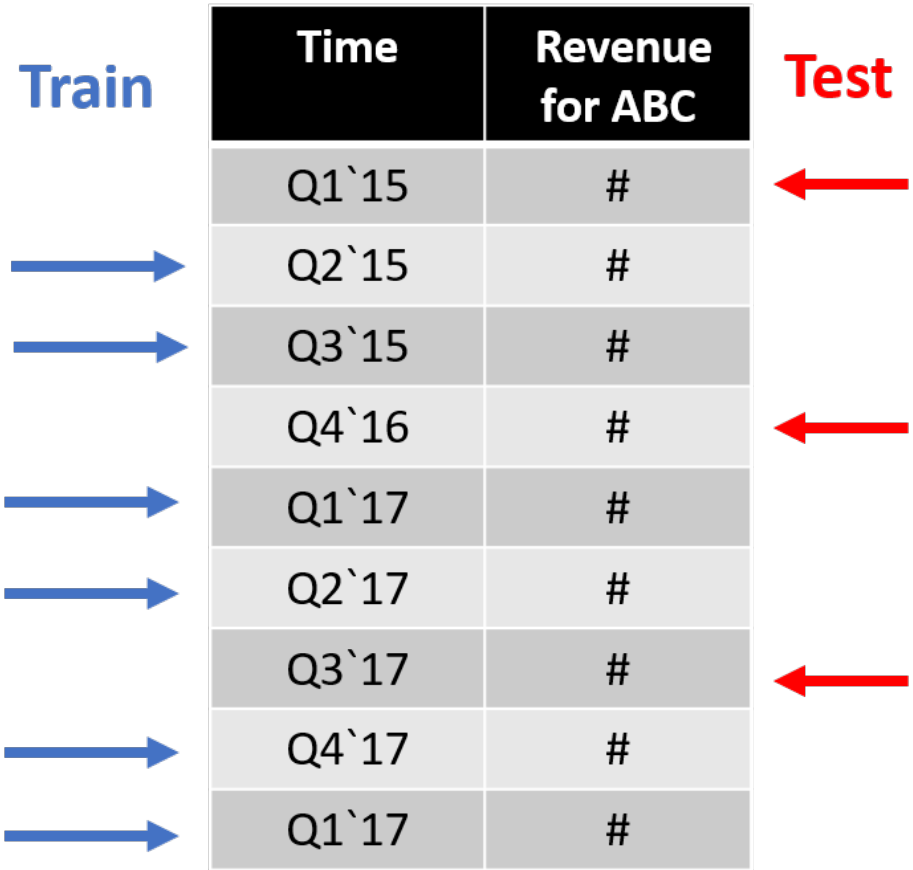
The fraction of the sample assigned to train is usually higher than to split; e.g. 80% to 20%. This is a nuisance parameter that you should stress test.

Time Series Train / Test Split

Q: Why isn't the typical approach (outlined on prior slide) appropriate for our usual forecasting applications?

Time Series Train / Test Split

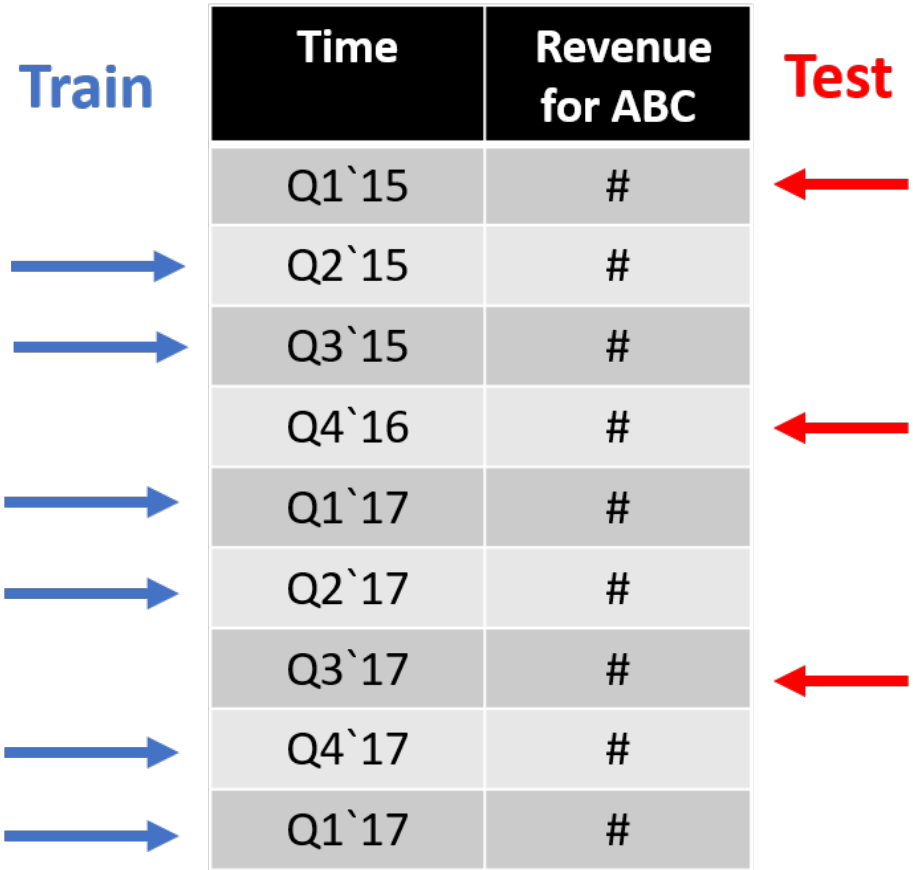
Q: Why isn't the typical approach (outlined on prior slide) appropriate for our usual forecasting applications?



Time	Revenue for ABC
Q1`15	#
Q2`15	#
Q3`15	#
Q4`16	#
Q1`17	#
Q2`17	#
Q3`17	#
Q4`17	#
Q1`17	#

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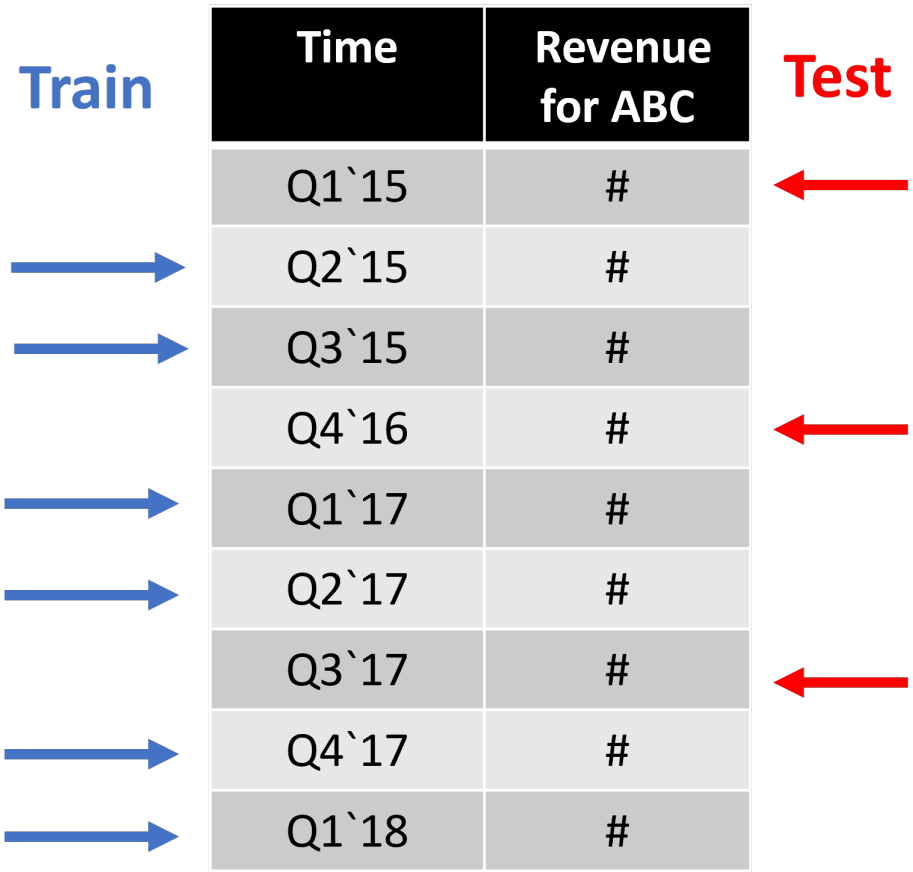
A: Randomization disrupts the temporal dependence.

Proper Splitting with Time Series data

Q: What's the proper way to split train/test with time series data?

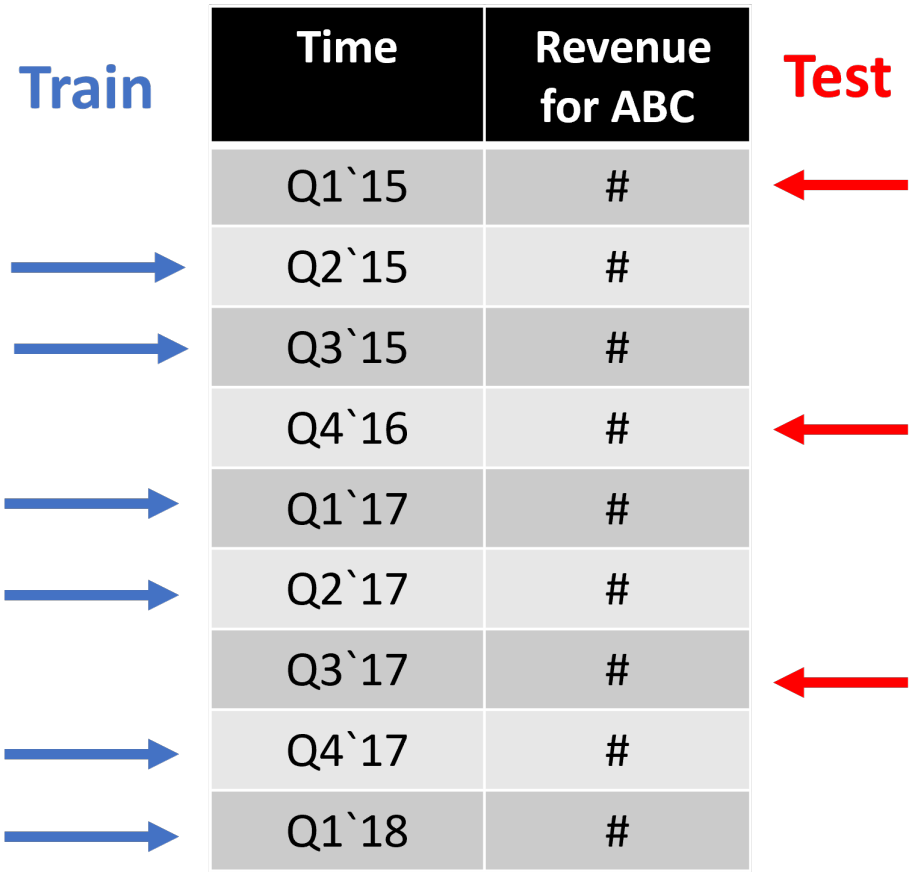
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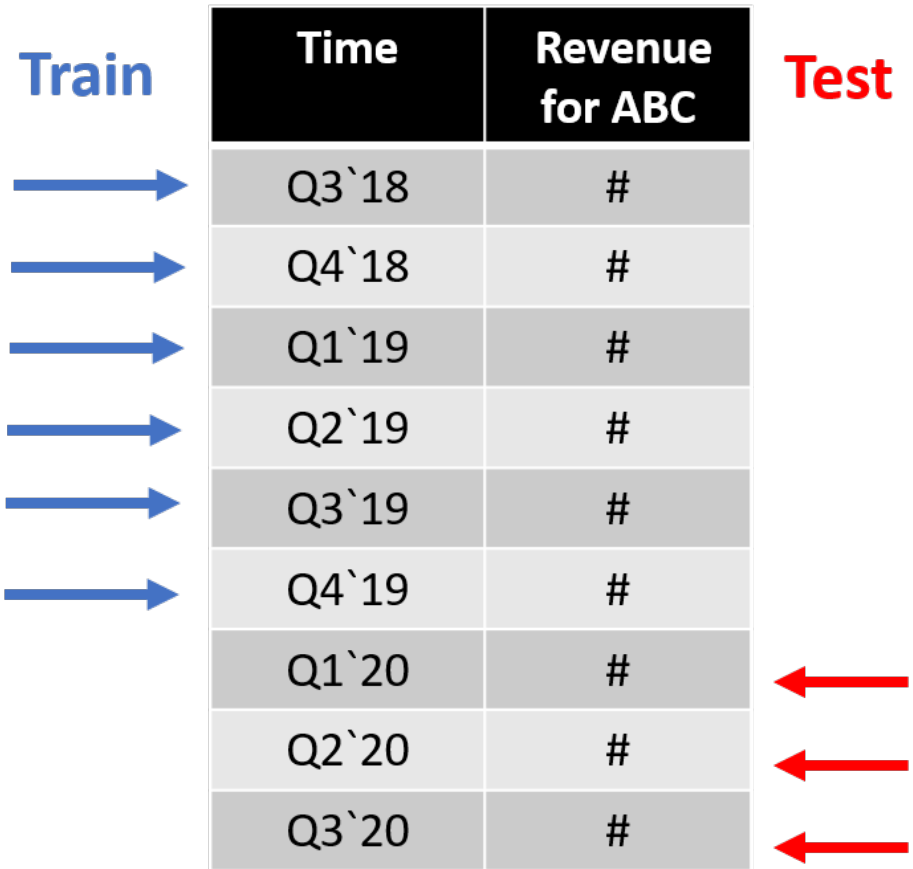
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Again, the size of the split needs to be stress tested.

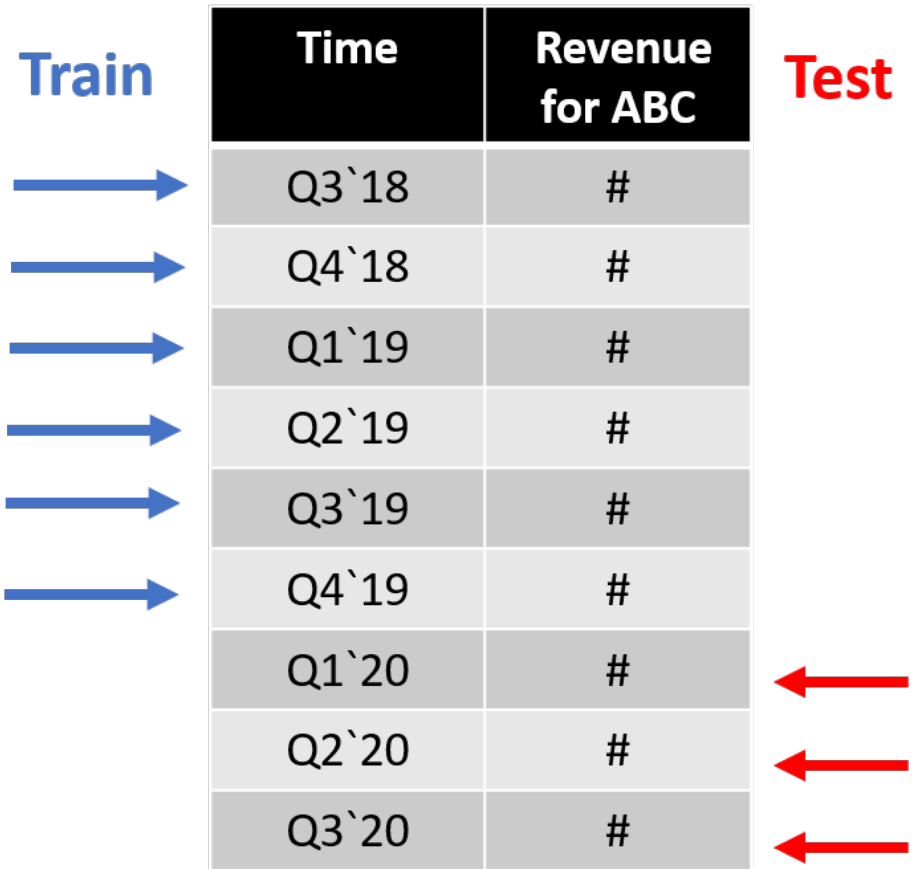
Dangers with Test/Train Splitting

Q: What's the danger in this split? (Notice the different dates)



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Time	Revenue for ABC
Q3`18	#
Q4`18	#
Q1`19	#
Q2`19	#
Q3`19	#
Q4`19	#
Q1`20	#
Q2`20	#
Q3`20	#

A: COVID (2020) likely disrupted the historical patterns. The training sample is not representative of the testing sample.

Cross Validation in Time Series

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A: Step 1: Separate the data into k-folds

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	Q1`19	#
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Note: we should stress test fold size.

Cross Validation in Time Series

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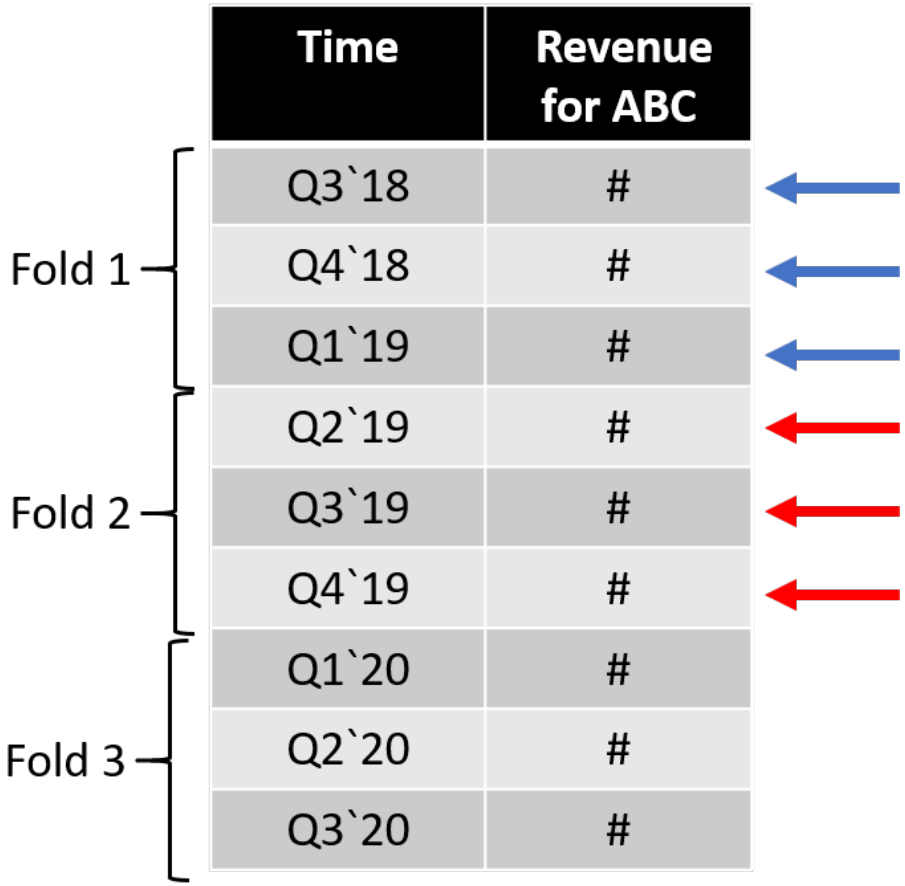
Step 2:

Cross Validation in Time Series

Step 2: Train on Fold 1 and Test on Fold 2

Cross Validation in Time Series

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Note: there are many variants, which alter size and shape of testing folds, etc...

Cross Validation in Time Series

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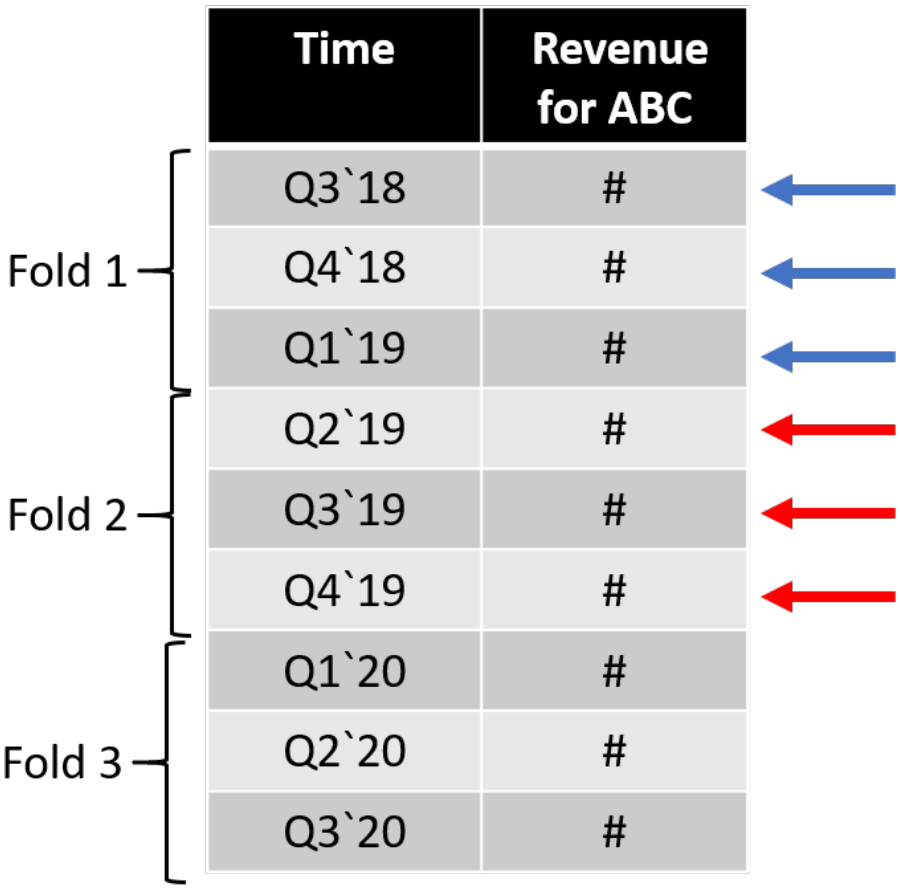
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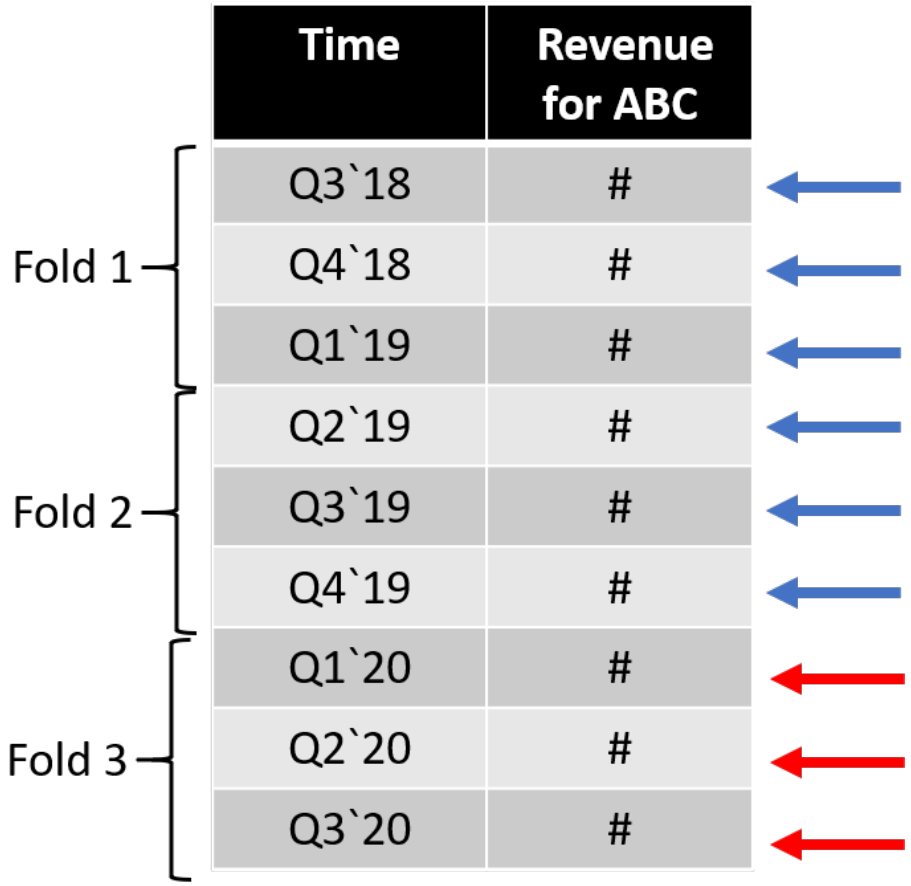
Cross Validation in Time Series

Step 3: Train on Folds 1 and 2 combined, Test on Fold 3

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Repeat for all of your k folds.

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- Other sample metrics are computed easily.

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\begin{aligned}
Y_{t+h|t} &= \beta_1 + \beta_2 X_t + u_{t+h} \\
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- $ME = \frac{1}{T} \sum_{t=1}^{T} \hat{u}_{t+h|t}$
- $MSE = \frac{1}{T} \sum_{t=1}^{T} \hat{u}_{t+h|t}^2$
- $RMSE = \sqrt{MSE}$
- $MAE = \frac{1}{T} \sum_{t=1}^{T} |\hat{u}_{t+h|t}|$
- $MAPE = \frac{1}{T} \sum_{t=1}^{T} |\frac{\hat{u}_{t+h|t}}{Y_{t+h}}|$

Forecast Evaluation Test

Diebold-Mariano

Consider the errors from forecasting strategies: $\hat{u}^{(A)}_{t+h|t}$ and $\hat{u}^{(B)}_{t+h|t}$.

Impose a loss function: $L(\hat{u}^{(i)}_{t+h|t}) = (\hat{u}^{(i)}_{t+h|t})^2$

Define $\bar{d} = \frac{1}{T}\sum_{t=1}^{T}\left[L_t^{(A)} - L_t^{(B)}\right]$

$$H_0 : E[d] = 0$$
$$H_a : E[d] \neq 0$$

$$DM = \frac{\bar{d}}{V(\bar{d})^{1/2}} \sim N(0,1)$$

Evaluating Errors

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- Diagnostic plots of the errors are very useful. If you spot any type of pattern, that implies your model is missing something.
  - Plot Errors vs Time

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How Forecast?

Q: You've found the model with the best error properties. How forecast out of sample?

How Forecast?

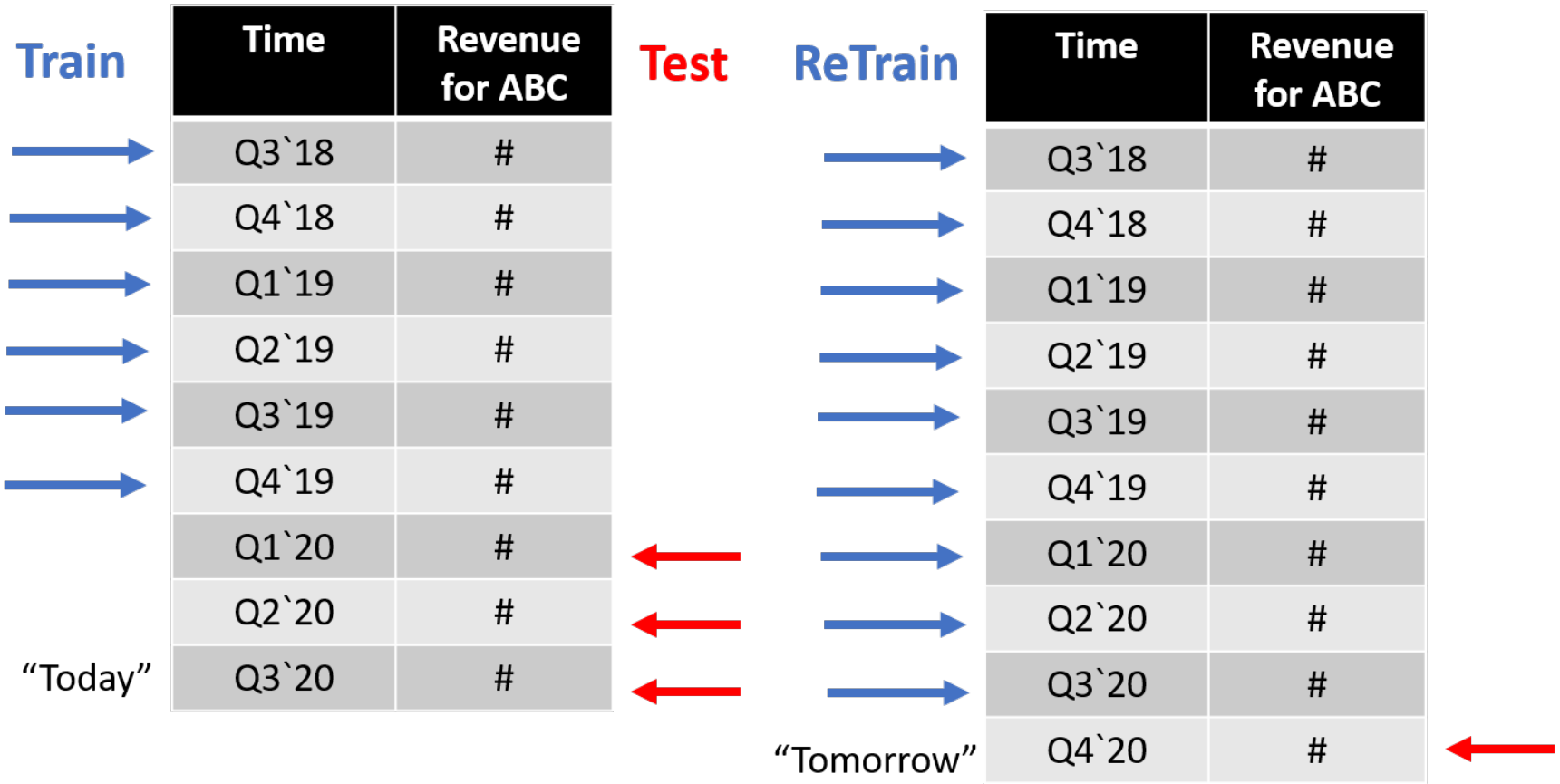
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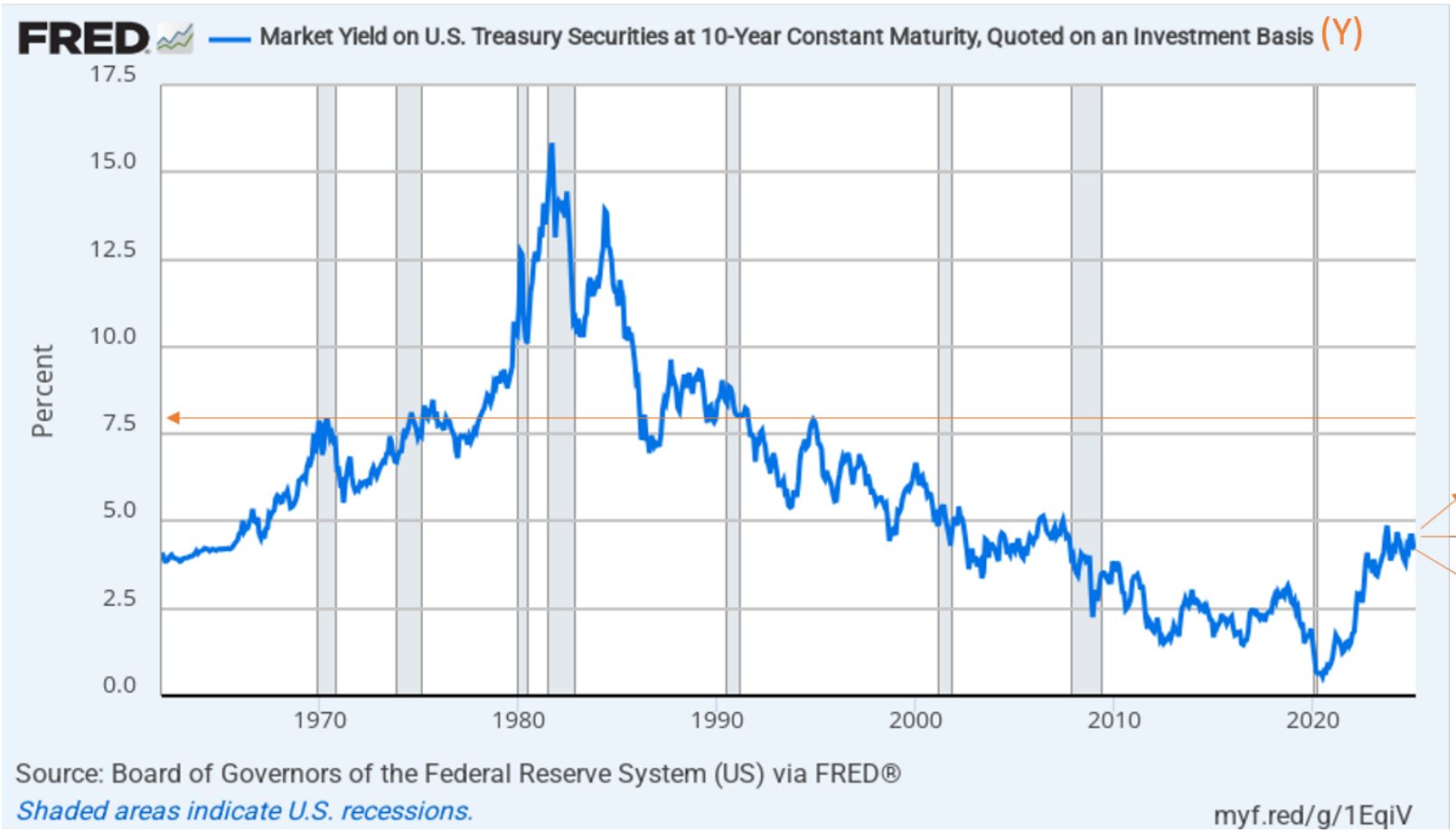
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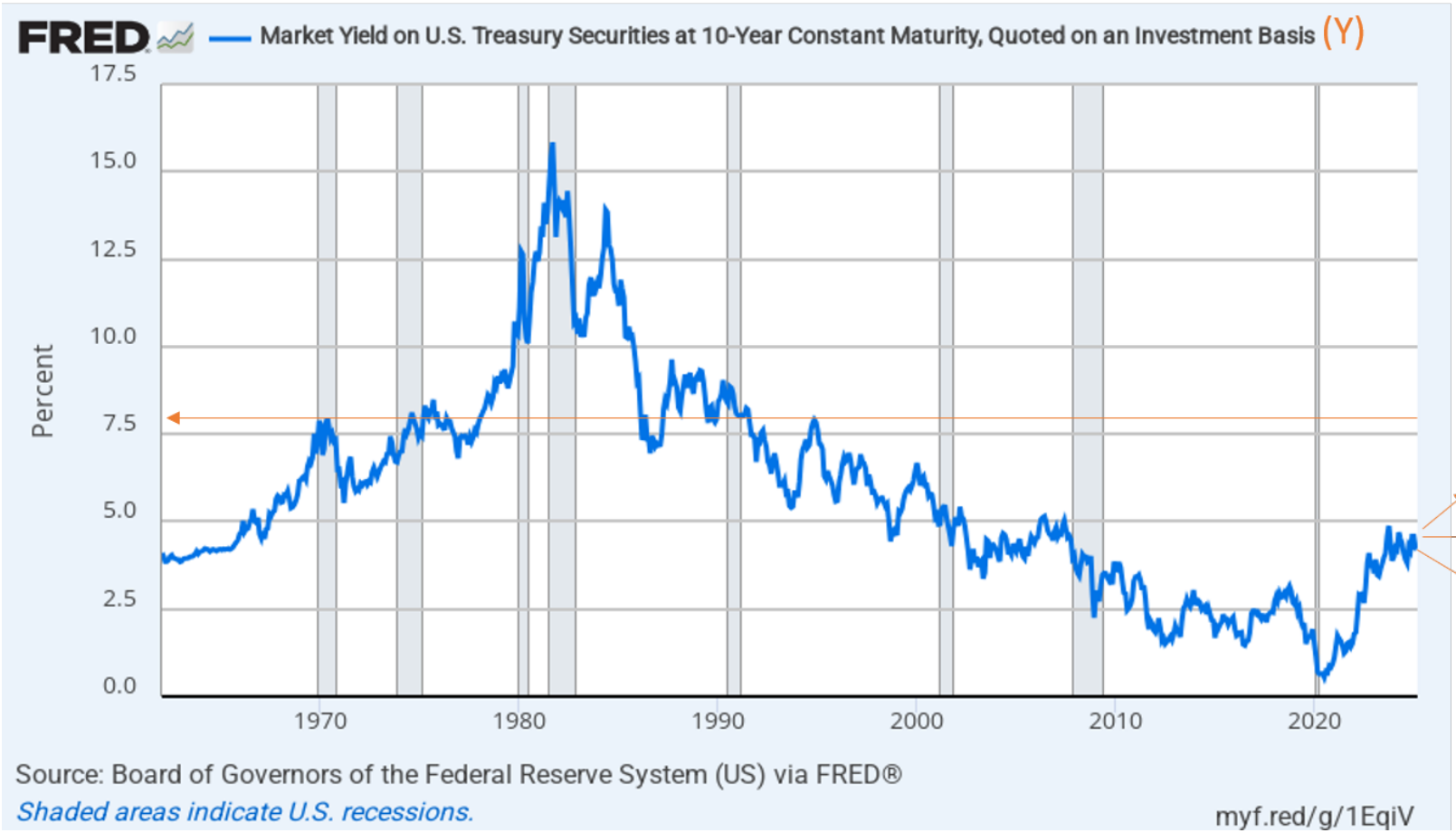
Prediction Intervals

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A: There is a 95% chance that the interval (4-5.5) will cover the future observation.

Forecast vs Confidence Intervals

Q: Which is wider?

Forecast vs Confidence Intervals

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A: The confidence interval is:

$$\hat{Y} \pm t_{\alpha/2,n-2} \cdot se_{errors} \cdot \sqrt{\left(\frac{(x-\bar{X})^2}{SS_X} + \frac{1}{n}\right)}$$

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- Forecast Intervals are wider (notice the extra "1"). Confidence intervals reflects uncertainty about the mean prediction. Forecast intervals reflects uncertainty about the mean and the individual observation for which you are forecasting.

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One Step Ahead Forecasting

One Step Ahead could literally mean tomorrow...

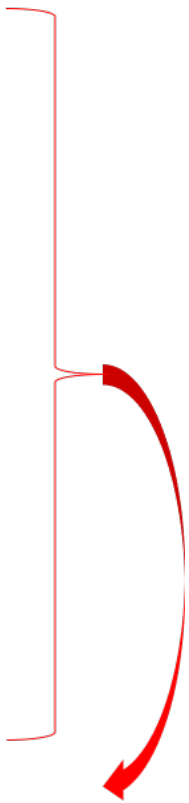
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Time	Revenue for ABC
1Jan	#
2Jan	#
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9Jan	#
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Mar	#
Apr	#
May	#
Jun	#
Jul	#
Aug	#
Sep	#
Oct	#

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Q4`18	#
Q1`19	#
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Q3`19	#
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Q2`20	#
Q3`20	#
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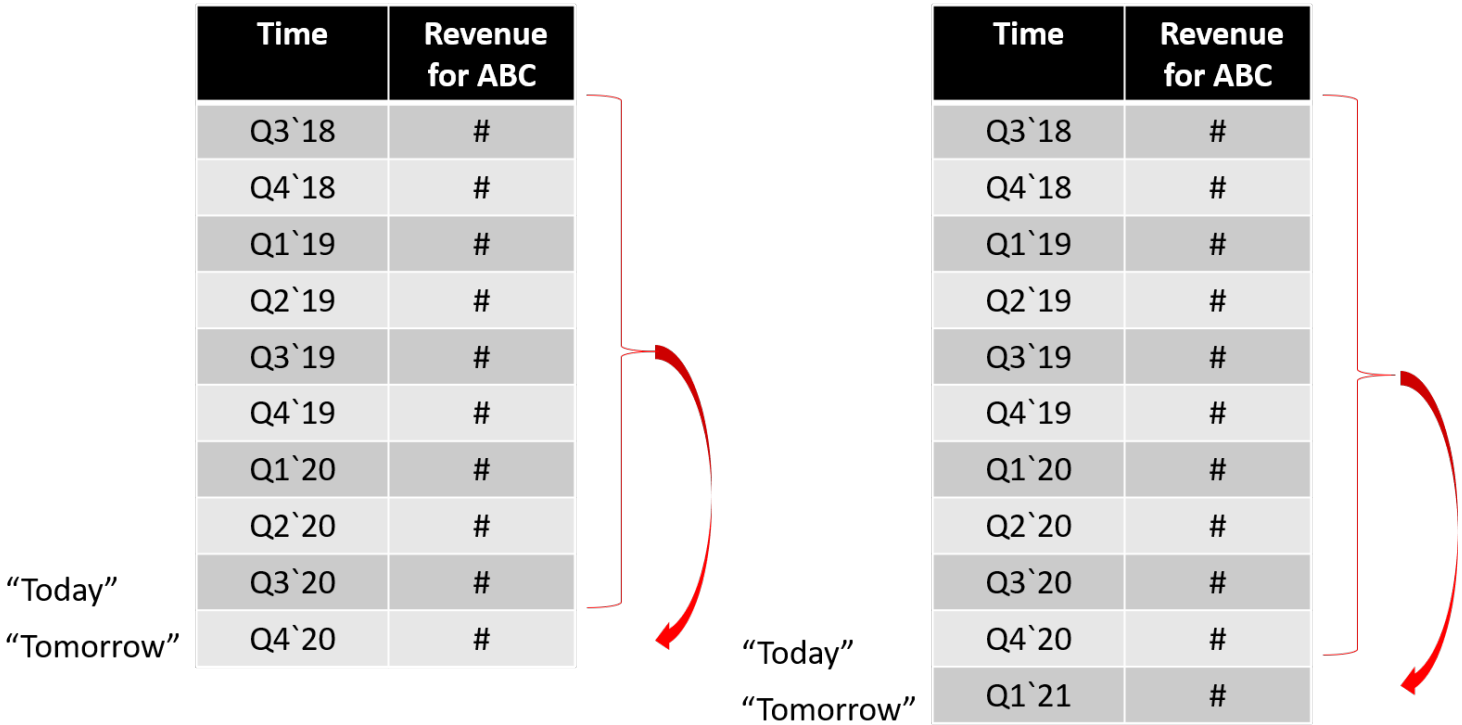
Rolling Forecasts

Q: What is a one step ahead rolling forecast?

Rolling Forecasts

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A: Each period we re-estimate the model to produce a one step ahead forecast.



Note: The training windows can i) retain a fixed length and slide forward or ii) grow (as shown here).

Rolling via Re-Forecasting or Re-Estimating

Suppose we had a model to forecast revenues $Rev_{Q4'20} = \hat{\beta}^{Q3'18-Q3'20} NGDP_{Q3'20}$

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If the model is stable, these should yield similar results.

Multi Step Ahead Forecasts - Static

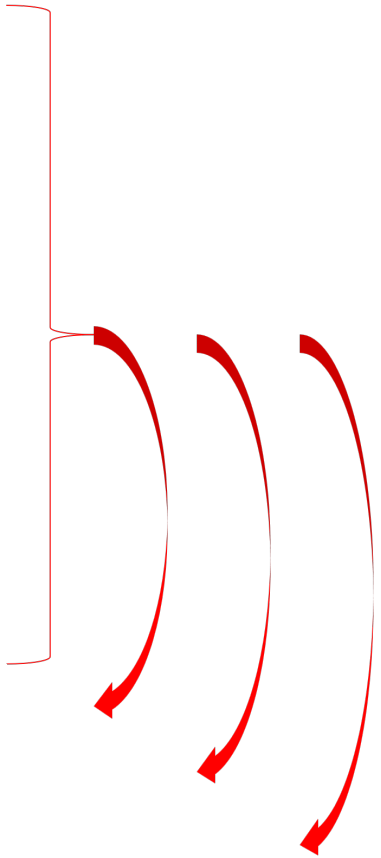
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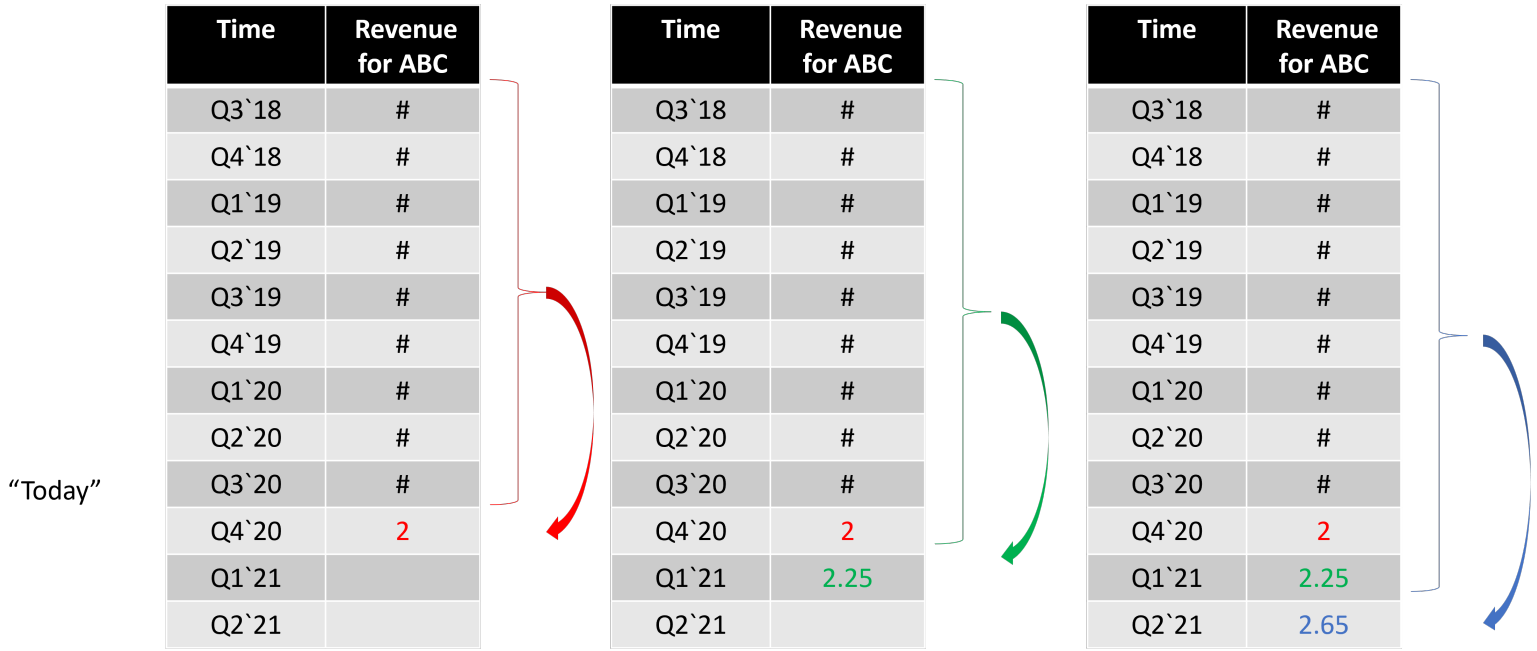


Multi Step Ahead Forecasts - Dynamic

Dynamic forecasts use the one step ahead estimates to forecast two steps ahead, etc..

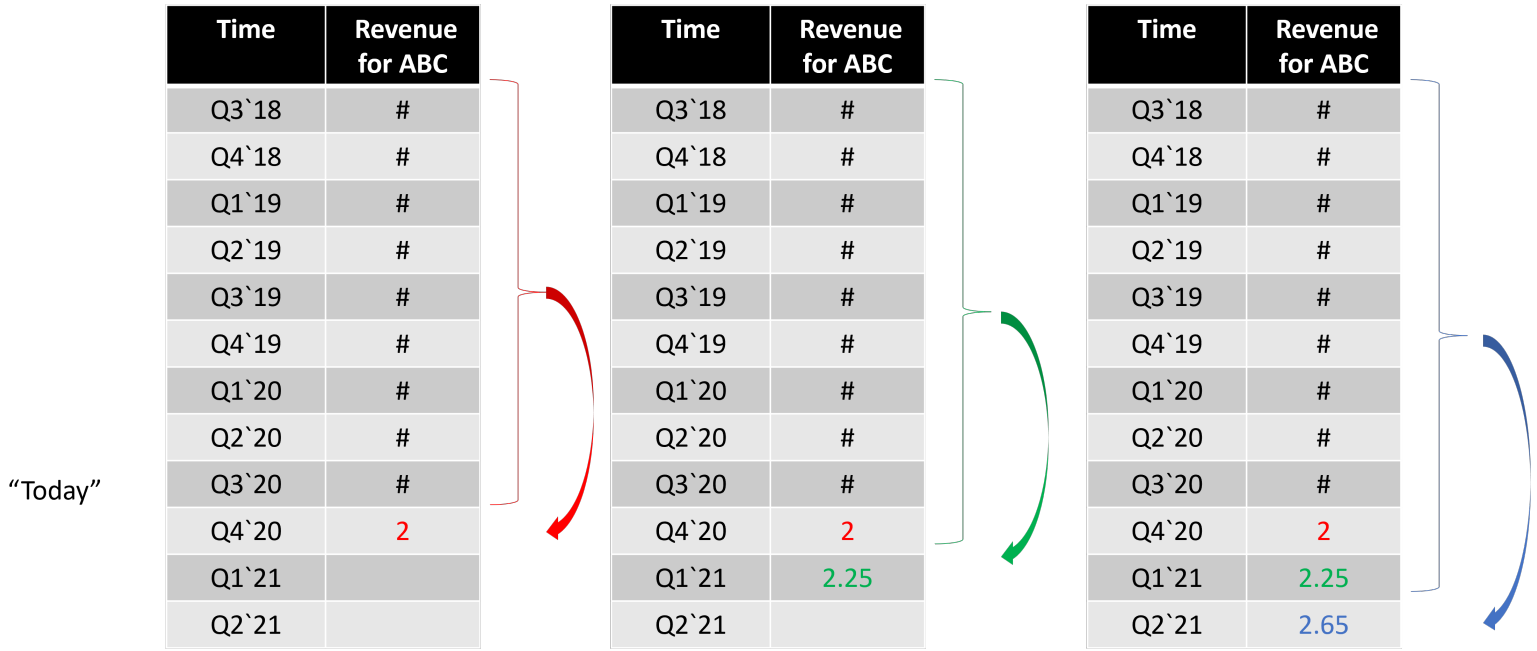
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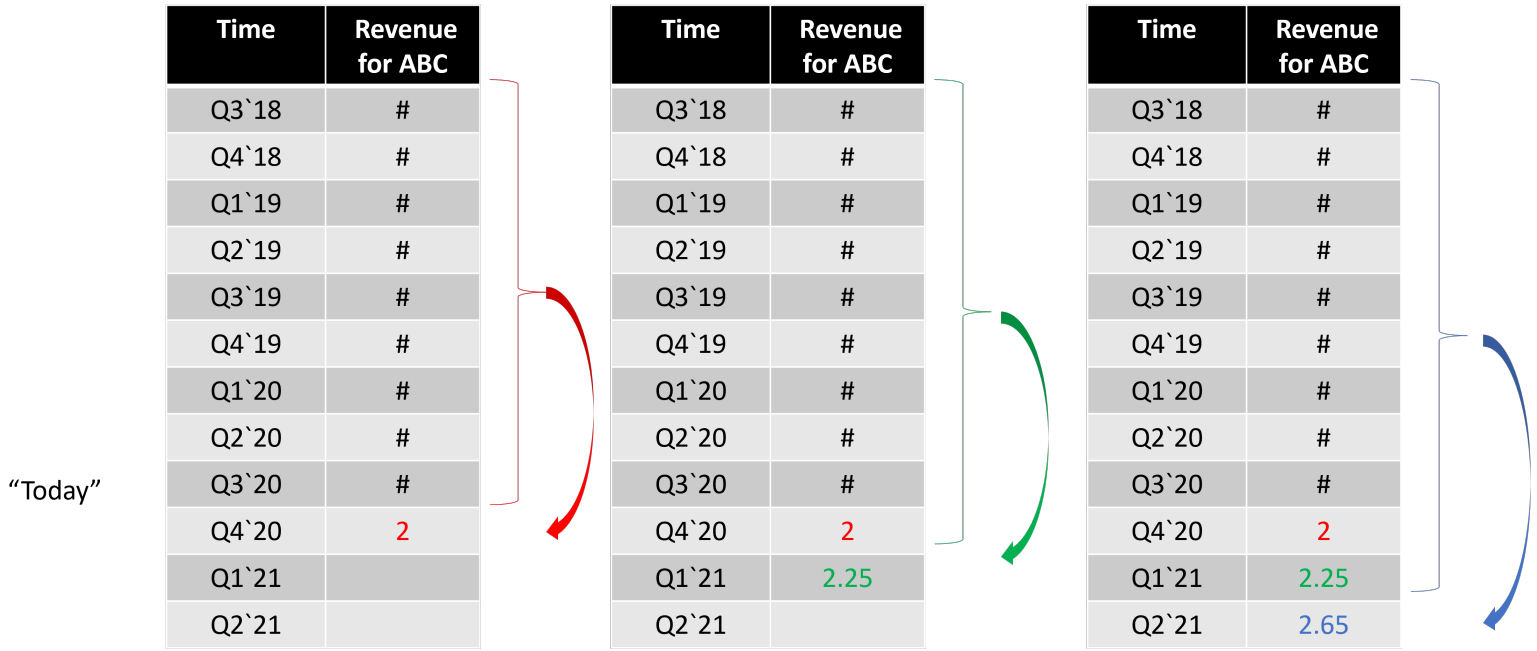
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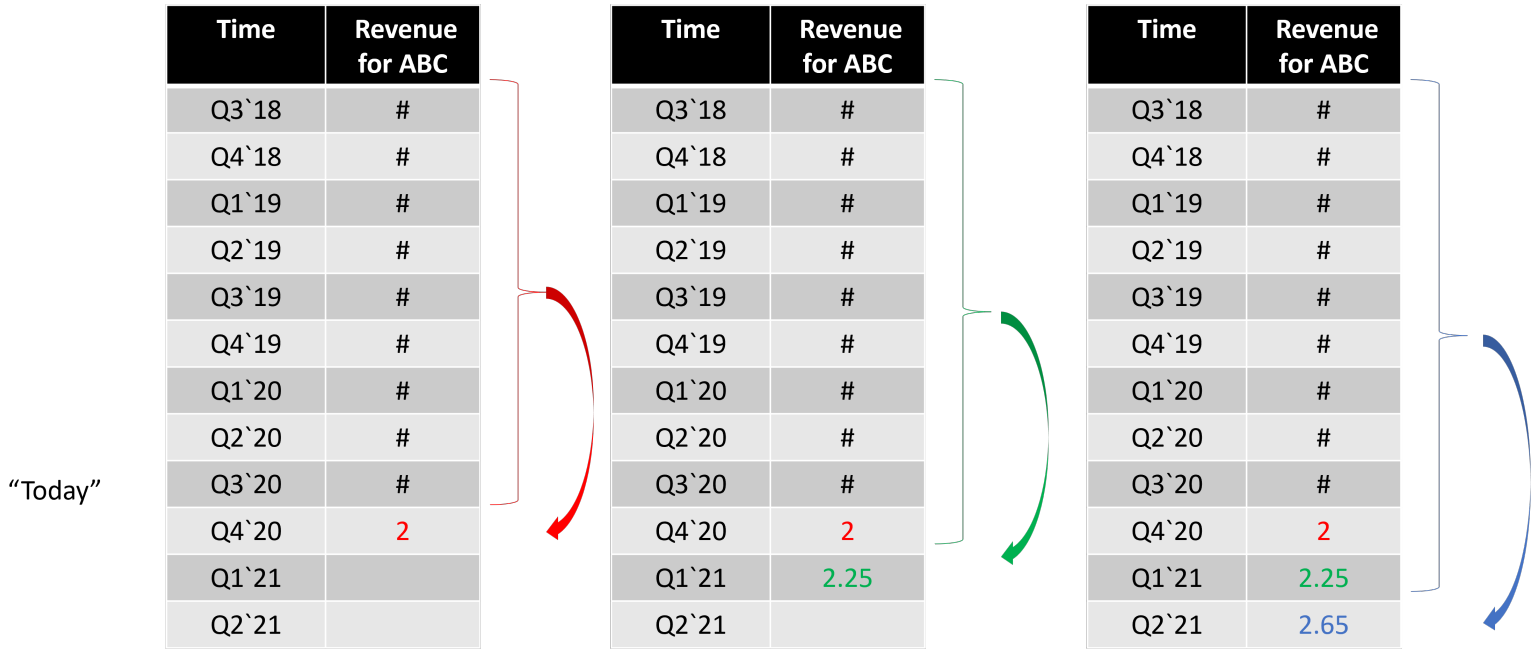
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Q: What are (dis)advantages of Dynamic relative to Static?

A:

- Advantages: Can potentially extend far into the future; adapt quickly to new settings
- Disadvantage: Errors compound on each other

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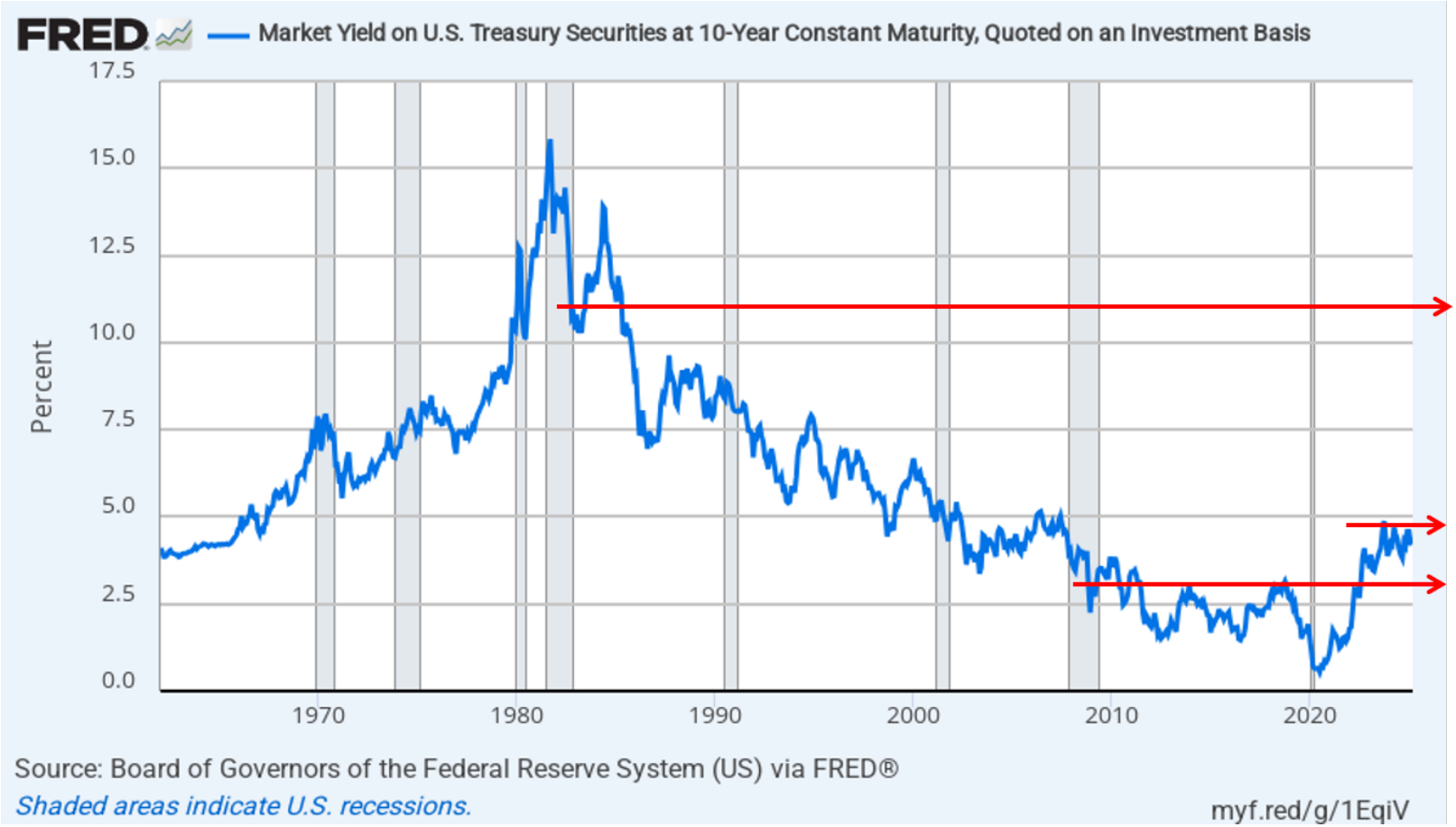
The two "simplest" models can serve as naive benchmarks:

$$\text{Mean Model: } E_t\left[Y_{t+h}\right] = \frac{1}{T}\sum_{t=1}^{T} Y_t$$

$$\text{Linear Time Trend Model: } E_t\left[Y_{t+h}\right] = \hat{\alpha} + \hat{\beta} \times Time_{t+h}$$

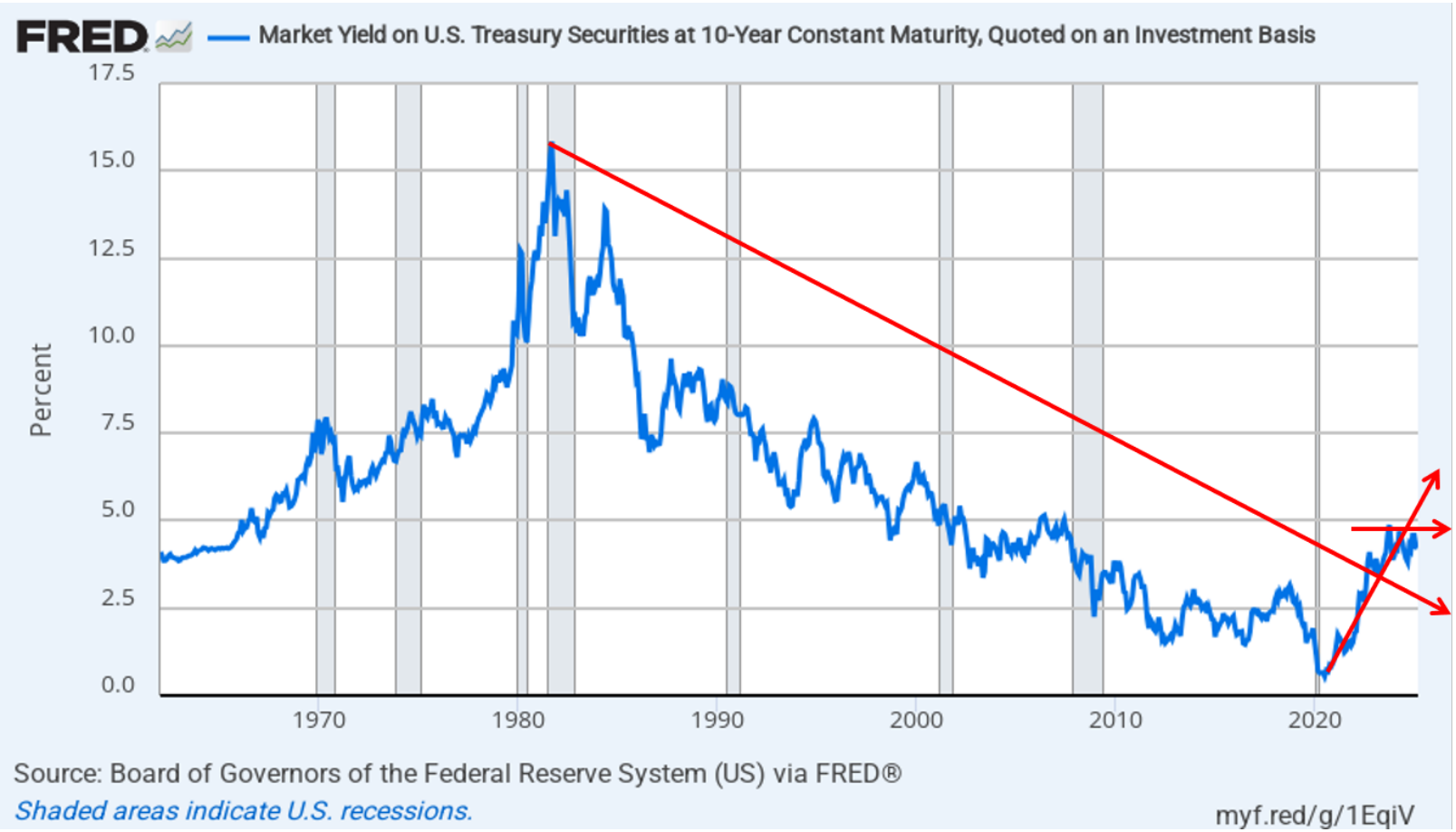
Mean Model

Though easy to compute, the size of training sample must be tuned



Linear Time Trend Model

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- Try several sample splits and examine the range of forecasts for stability

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- Imagery: every series has its own unique temporal "fingerprint" defined by the mean, variance, and autocorrelation of the series.
- If we can identify the fingerprint, we can model the series.

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Example (MA(1))

- Find the $E[Y_t]$.
- Find the $Var(Y_t)$.
- Find the autocovariance of Y over 1 lag i.e. $Cov(Y_t, Y_{t-1})$.
- Find the autocovariance of Y over 2 lags i.e. $Cov(Y_t, Y_{t-2})$.

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Moving Average Process

Definition

A moving average process of order q [MA(q)] can be represented by

$$Y_t = \mu + \varepsilon_t + \sum_{j=1}^{q} \alpha_j \varepsilon_{t-j}$$

where $\varepsilon \sim N(0, \sigma^2)$ iid

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Autoregressive Process

Consider the $MA(\infty)$

$$Y_t = \mu + \sum_{j=0}^{\infty} \alpha_j \varepsilon_{t-j}$$

where $\alpha_0 \equiv 1$, and $\varepsilon \sim N(0, \sigma^2)$ iid. Assume that the α weights converge to zero as j gets large.

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where $\alpha_0 \equiv 1$, and $\varepsilon \sim N(0, \sigma^2)$ iid. Assume that the α weights converge to zero as j gets large. Rewrite $\alpha_j = \theta^j$ for some θ, $|\theta| < 1$.

$$Y_t = \mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j}$$

Autoregressive Process

Notice from the last line above that

$$Y_{t-1} = \mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j-1}$$

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and so we can write

$$Y_t - \theta Y_{t-1} = \mu - \theta\mu + \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j} - \theta \sum_{j=0}^{\infty} \theta^j \varepsilon_{t-j-1}$$

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where $\delta = \mu - \theta\mu$.

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where $\delta = \mu - \theta\mu$. Defining $y_t = Y_t - \mu$ yields an AR(1) $y_t = \theta y_{t-1} + \varepsilon_t$

Example (AR(1))

Consider the AR(1) $y_t = \theta y_{t-1} + \varepsilon_t$, where $\varepsilon \sim N(0, \sigma^2)$ iid.

- Find the $E[y_t]$.
- Find the $Var(y_t)$.
- Find the autocovariance for Y_t and Y_{t-1}.
- Find the autocovariance for Y_t and Y_{t-2}.

Example (AR(1))

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$$E[y_t] = E[\theta y_{t-1} + \varepsilon_t]$$

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Autoregressive Process

Definition

An autoregressive process of order p [AR(p)] can be represented by

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ARMA

Definition

An ARMA(p,q) process can be represented by

$$y_t = \sum_{j=1}^{p} \theta_j y_{t-j} + \varepsilon_t + \sum_{j=1}^{q} \alpha_j \varepsilon_{t-j}$$

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Notice this is non linear. Maximum Likelihood is the common alternative.

Model Choice via Box Jenkins

Rules of Thumb

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An AR(p) process is described by

1. An ACF with slow decay
2. A PACF that is (close to) zero for lags larger than p

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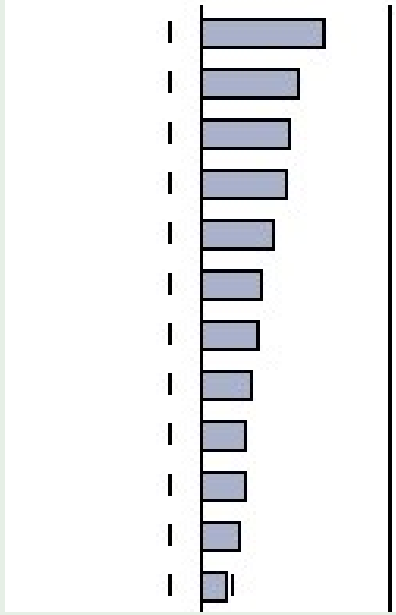
An ARMA process is described by

- Slow decay in both ACF and PACF

Example (Reading the Correlogram)

Q: What model specification seems appropriate given the correlogram below?

Date: 07/11/07 Time: 11:27
Sample: 1996M04 2007M06
Included observations: 135

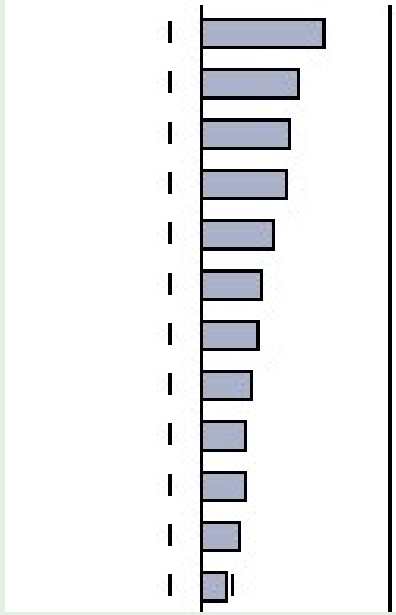


Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.680	0.680	63.865	0.000
		2	0.549	0.160	105.71	0.000
		3	0.495	0.139	140.06	0.000
		4	0.466	0.108	170.72	0.000
		5	0.402	0.001	193.72	0.000
		6	0.331	-0.031	209.45	0.000
		7	0.305	0.034	222.91	0.000
		8	0.271	-0.003	233.64	0.000
		9	0.248	0.021	242.68	0.000
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A: The slowly decaying ACF and the spike at 1 lag for the PACF suggest an AR(1) might be appropriate.

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   - H_0 : Residuals are W.N. vs H_a :Residuals are not W.N.
   - Test Statistic: $Q_k = T(T+2)\sum_{k=1}^{K} \frac{1}{T-k} r_k^2$, where r_k are the estimated autocorrelation coefficients of the residuals $\hat{\varepsilon}_t$.
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   - Critical Value: $Q_k \sim \chi^2(k-p-q)$
4. Search for the smallest AIC and/or BIC values
   - $AIC = \log(\hat{\sigma}^2) + 2(p+q)/T$
   - $BIC = \log(\hat{\sigma}^2) + log(T)(p+q)/T$

ARIMA

ARIMA is an extension of the ARMA structure,

AR: Autoregressive

I: Integrated

MA: Moving Average

The order of integration indicates the number of times the series must by differenced in order to achieve stationarity. A process that is $I(0)$ is stationary. A process that is $I(1)$ is said to be "integrated of order 1", implying that it must be differenced once to achieve stationarity.

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Extension: Seasonal ARIMA

- SARIMA$(p, d, q) \times (P, D, Q)_s$
- Captures both short-run dynamics and seasonal patterns
- Seasonal differencing: $\Delta_s Y_t = Y_t - Y_{t-s}$
- Connection: Seasonal ARIMA underlies many decomposition tools (e.g. X-13 ARIMA-SEATS)

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ARIMAX Motivation: Diagnosing Missing Structure

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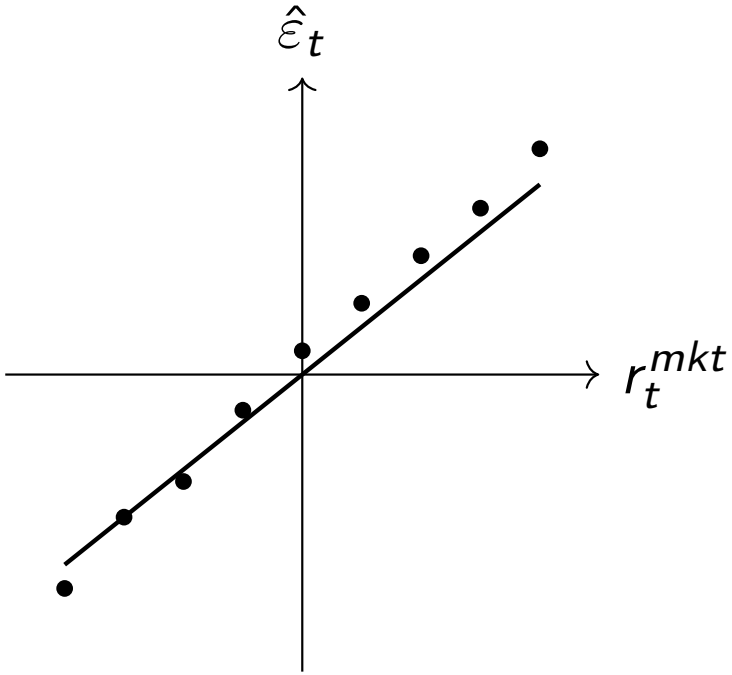
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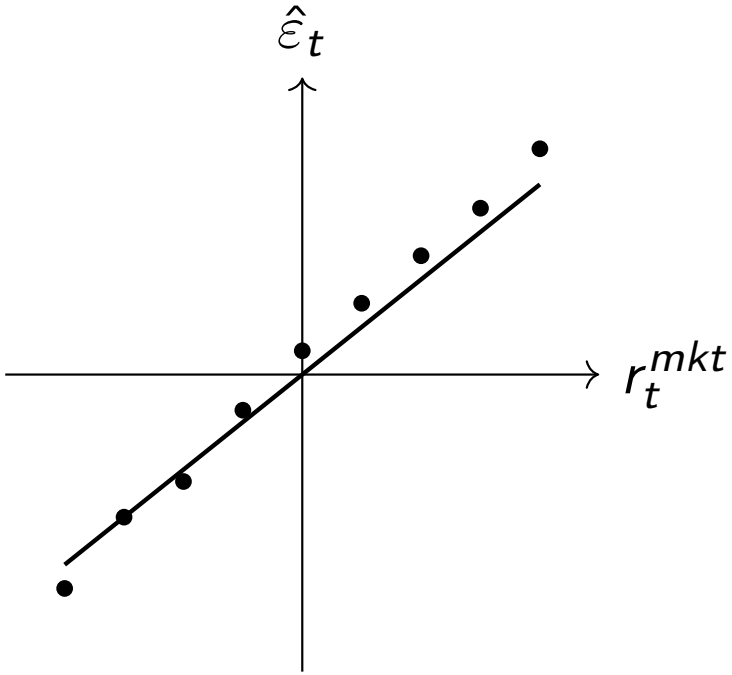


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- If residuals contain structure $\Rightarrow$ model is misspecified
- Add X_t to capture this variation
- $\Rightarrow$ ARIMAX

ARIMAX Model

- Extend ARIMA to include exogenous variables

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- Examples of X_t:
  - Market returns
  - Interest rates
  - Macro indicators
- Interpretation:
  - AR term: internal dynamics
  - X_t: external information

ARIMAX Model

- Extend ARIMA to include exogenous variables

$$r_t = \alpha + \phi r_{t-1} + \beta X_t + \varepsilon_t$$

- Examples of X_t:
  - Market returns
  - Interest rates
  - Macro indicators
- Interpretation:
  - AR term: internal dynamics
  - X_t: external information
- Important:
  - Must forecast X_{t+h} or use only lagged X
  - If forecasting X_{t+h}, then forecast errors may compound

Outline

5 Forecast
- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

Case Study Background

Consider the Phillips Curve, which relates inflation (π) to unemployment (UR). Suppose each follows an AR(1), and by theory is related to the other.

$$\begin{aligned}\pi_t &= \delta_1 + \theta_{11}\pi_{t-1} + \theta_{12}UR_{t-1} + \epsilon_{1t} \\ UR_t &= \delta_2 + \theta_{21}\pi_{t-1} + \theta_{22}UR_{t-1} + \epsilon_{2t}\end{aligned}$$

where ϵ_{1t} and ϵ_{2t} are potentially correlated white noise processes, each independent of the history of UR and π.

The Case for VAR

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- Possibly lead to more accurate forecasts.
- Greene (2000) suggests that simultaneous estimates might be more efficient in some situations.

A VAR(1) System

The VAR(1) specification can be written as

$$\begin{pmatrix} Y_t \\ X_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ X_{t-1} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

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or written compactly as

$$\overrightarrow{Y}_t = \delta + \Phi_1 \overrightarrow{Y}_{t-1} + \overrightarrow{\epsilon}_t$$

where $\overrightarrow{Y}_t = (Y_t, X_t)'$ and $\overrightarrow{\epsilon}_t = (\epsilon_{1t}, \epsilon_{2t})'$.

A VAR(p) System

Define $\overrightarrow{Y}_t$ as a $t \times k$ matrix. Then the VAR(p) model can be written as

$$\overrightarrow{Y}_t = \delta + \Phi_1 \overrightarrow{Y}_{t-1} + \cdots + \Phi_p \overrightarrow{Y}_{t-p} + \overrightarrow{\epsilon}_t$$

where each parameter matrix Φ is a $k \times k$ and the white noise errors $\overrightarrow{\epsilon}$ is a $k-$dimensional vector with covariance matrix Σ.

Expected Value

The expected value can be written as

$$E\{\overrightarrow{Y}_t\} = \delta + \Phi_1 E\{\overrightarrow{Y}_{t-1}\} + \cdots + \Phi_p E\{\overrightarrow{Y}_{t-p}\}$$

If we define the lag operator L and I_k as the $k-$dimensional Identity matrix, then we can rewrite as

$$\mu = E\{\overrightarrow{Y}_t\} = (I_k - \Phi_1 - \cdots - \Phi_p)^{-1}\delta = \Phi(1)^{-1}\delta$$

Note that stationarity requires that $\Phi(1)$ is invertible.

Covariance Estimation

Denote $e_{1t}, \cdots, e_{kt}$ as the residuals from each of the $k-$equations in the system. Then we can estimate the (i,j)-element of Σ as

$$\hat{\sigma}_{ij} = \frac{1}{T-p} \Sigma_{t=p+1}^{T} e_{it} e_{jt}$$

This permits us to form

$$\hat{\Sigma} = \frac{1}{T-p} \Sigma_{t=p+1}^{T} \overrightarrow{e}_t \overrightarrow{e}'_t$$

where $\overrightarrow{e}_t = (e_{1t}, \cdots, e_{kt})$.

Forecasting 1-step ahead

If we demean via $\overrightarrow{y}_t = \overrightarrow{Y}_t - \mu$, we can form

$$\overrightarrow{y}_t = \Phi_1 \overrightarrow{y}_{t-1} + \cdots + \Phi_p \overrightarrow{y}_{t-p} + \overrightarrow{\epsilon}_t$$

then we can form optimal one-period ahead forecasts via

$$\overrightarrow{y}_{T+1|T} = E\{\overrightarrow{y}_{T+1} | \overrightarrow{y}_T, \overrightarrow{y}_{T-1}, ...\} = \Phi_1 \overrightarrow{y}_T + \cdots + \Phi_p \overrightarrow{y}_{T-p+1}$$

with the accompanying forecast error variance

$$V\{\overrightarrow{y}_{T+1} | \overrightarrow{y}_T, \overrightarrow{y}_{T-1}, \cdots\} = \Sigma$$

Forecasting multiple steps ahead

Forecasting is accomplished via recursive substitution. For instance, for a

$$\begin{aligned}\overrightarrow{y}_{T+2|T} &= \Phi_1 \overrightarrow{y}_{T+1|T} + \cdots + \Phi_p \overrightarrow{y}_{T-p-2} \\ &= \Phi_1(\Phi_1 \overrightarrow{y}_T + \cdots + \Phi_p \overrightarrow{y}_{T-p+1}) + \cdots + \Phi_p \overrightarrow{y}_{T-p+2}\end{aligned}$$

Determining the Lag Length (p)

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A reasonable strategy (see Verbeek 2000) is to

1. Estimate the VAR over a range of p values
2. Pick the specification with the smallest AIC/BIC.

Impulse Response Function (IRF)

If $\Phi(1)$ is invertible, we can rewrite the VAR as a Vector Moving Average (VMA)

$$\overrightarrow{Y}_t = \mu + \Phi(L)^{-1}\overrightarrow{\epsilon}_t$$

Writing $\Phi(L)^{-1} = I_k + A_1 L + A_2 L^2 + \cdots$, we can see

$$\overrightarrow{Y}_t = \mu + \overrightarrow{\epsilon}_t + A_1\overrightarrow{\epsilon}_{t-1} + A_2\overrightarrow{\epsilon}_{t-2} + \cdots$$

Each element of any A matrix can be thought of as a marginal effect of a shock to the residual upon data in Y.

Impulse Response Function (IRF)

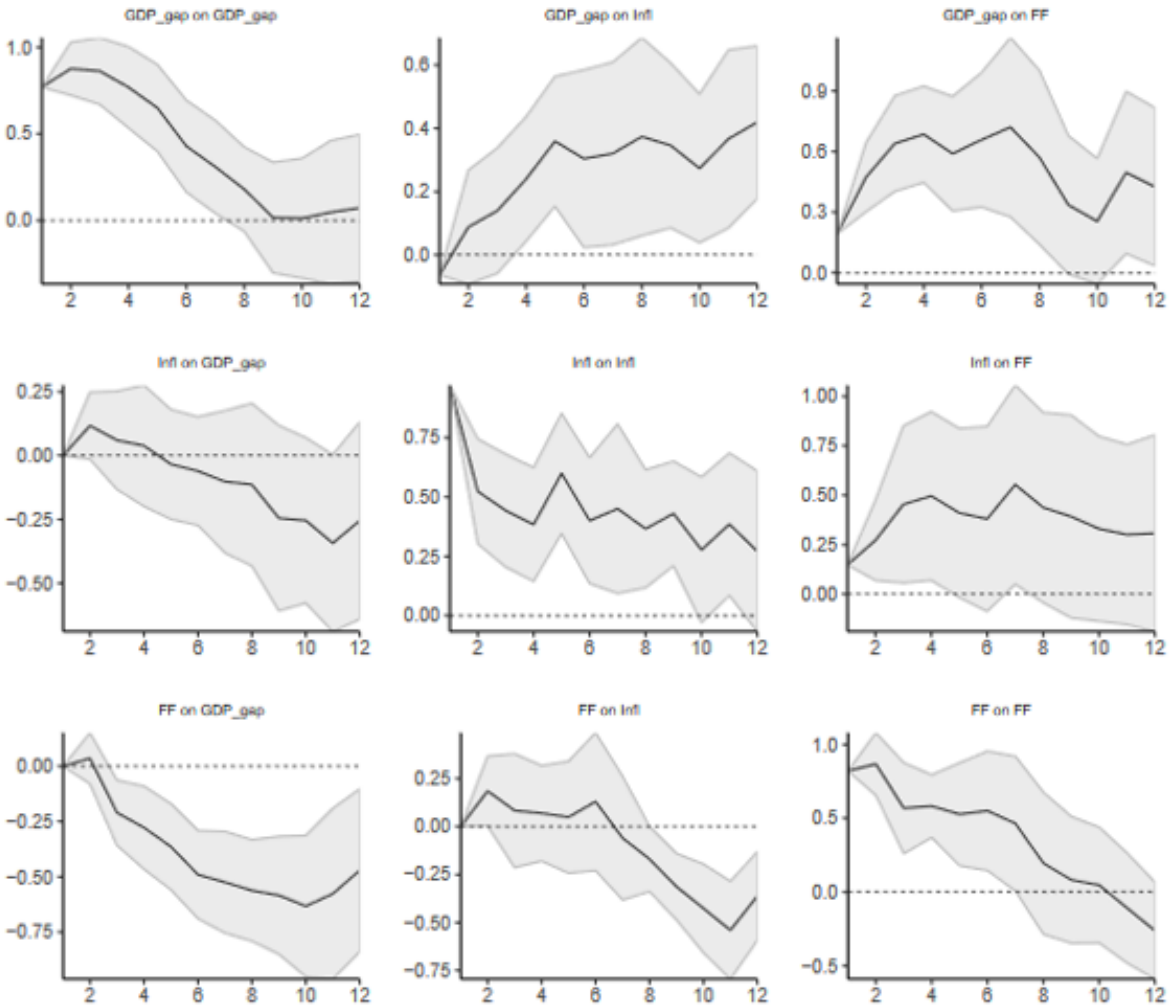
Consider again the Phillips Curve example

$$\begin{pmatrix} UR_t \\ \pi_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} UR_{t-1} \\ \pi_{t-1} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

Rewriting as a VMA we have $\overrightarrow{Y}_t = \mu + \overrightarrow{\epsilon}_t + A_1 \overrightarrow{\epsilon}_{t-1} + A_2 \overrightarrow{\epsilon}_{t-2} + \cdots$.
To examine the impact of a unit-one shock of unemployment upon inflation we consider the dynamics of the marginal effects

- $\frac{\partial Y_{2,t+1}}{\partial \epsilon_{1,t}} = A_{1,(1,2)}$
- $\frac{\partial Y_{2,t+2}}{\partial \epsilon_{1,t}} = A_{2,(1,2)}$
- $\cdots$

Impulse Response Function (IRF)



Multivariate Cointegration

Consider a binary VAR(1) model with variables integrated of order one ($\overrightarrow{Y}_t \sim I(1)$), we can write it as:

$$\overrightarrow{Y}_t = \delta + \Phi_1 \overrightarrow{Y}_{t-1} + \overrightarrow{\epsilon}_t$$

where matrix Φ_1 is a 2×2 and the white noise errors $\overrightarrow{\epsilon}$ is a $2-$dimensional vector with covariance matrix Σ.
By subtracting the lagged vectors $\overrightarrow{Y}_{t-1}$ from both side, we get:

$$\overrightarrow{Y}_t - \overrightarrow{Y}_{t-1} = \delta + \Phi_1 \overrightarrow{Y}_{t-1} - \overrightarrow{Y}_{t-1} + \epsilon_t$$
$$\Delta \overrightarrow{Y}_t = \delta + (\Phi_1 - I_2) \overrightarrow{Y}_{t-1} + \epsilon_t$$

Multivariate Cointegration

- Since we know Y_t is integrated of order one, so we have an $I(0)$ on the left hand side.
- The right hand side is the combination of two $I(0)$ (one constant vector δ and white noise vestor $\overrightarrow{\epsilon}$) and $(\Phi_1 - I_2)\overrightarrow{Y}_{t-1}$.
- If all variables are not cointegrated, then we must have:

$$(\Phi_1 - I_2) = \overrightarrow{0} \Rightarrow \Phi_1 = I_2$$

- If the variables are cointegrated, then there exists $r(0 < r \leq 2)$ cointegrated relations, $(\Phi_1 - I_2)\overrightarrow{Y}_{t-1}$ can be written as a $I(0)$ variable:

$$(\Phi_1 - I_2)\overrightarrow{Y}_{t-1} = \lambda\gamma'\overrightarrow{Y}_{t-1}$$

where γ' is the $(r \times 2)$ matrix of cointegration coefficients and λ is a $(2 \times r)$ matrix. Specifically, if $r = 2$, then all variables in $\overrightarrow{Y}_t$ must be stationary.

Multivariate Cointegration

The above structure can be extended to VAR(p) model with k variables with integrated of order d ($\overrightarrow{Y}_t \sim I(d)$):

$$\overrightarrow{Y}_t = \delta + \Phi_1 \overrightarrow{Y}_{t-1} + \cdots + \Phi_p \overrightarrow{Y}_{t-p} + \overrightarrow{\epsilon}_t$$

where each parameter matrix Φ is a $k \times k$ and the white noise errors $\overrightarrow{\epsilon}$ is a $k-$dimensional vector with covariance matrix Σ.
Apply difference operator $\Delta = 1 - L$ to both side, we get:

$$\begin{aligned}\Delta \overrightarrow{Y}_t =& \delta + (\sum_{j=1}^{p} \Phi_i - I_k) \overrightarrow{Y}_{t-1} + \Gamma_1 \Delta \overrightarrow{Y}_{t-1} + \cdots + \Delta \Gamma_p \overrightarrow{Y}_{t-p-1} + \overrightarrow{\epsilon}_t \\ =& \delta + \Pi \overrightarrow{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \overrightarrow{Y}_{t-j} + \overrightarrow{\epsilon}_t\end{aligned}$$

This equation also called vector error correction model (VECM).

Multivariate Cointegration

$$\Delta \overrightarrow{Y}_t = \delta + \Pi \overrightarrow{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \overrightarrow{Y}_{t-j} + \overrightarrow{\epsilon}_t$$

The rank of matrix Π determines number of cointegration vectors:

- If rank$(\Pi) = 0$, all variables in vector $\overrightarrow{Y}_t$ are non-stationary. There is no linear combination of variables that leads to stationary
- If Π has full rank k, then all variables in $\overrightarrow{Y}_t$ must be stationary
- If Π has reduce rank r $(0 < r < k)$, then there are cointegration relations among $\overrightarrow{Y}_t$

Vector Error Correction Model

Consider VECM equation:

$$\Delta \overrightarrow{Y}_t = \delta + \Pi \overrightarrow{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \overrightarrow{Y}_{t-j} + \overrightarrow{\epsilon}_t$$

If Π has rank $r > 0$, then Π can be written as production of A and B':

$$\Delta \overrightarrow{Y}_t = \delta + AB' \overrightarrow{Y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \overrightarrow{Y}_{t-j} + \overrightarrow{\epsilon}_t$$

where A is a $(k \times r)$ matrix contains the cointegration vectors and B is a $(k \times r)$ matrix which contains the speed of adjustment parameters.

We call this new version VECM as restricted regression in the sense that it imposes the hypothesis that rank(Π)$>$ 0, so the original VECM is unrestricted regression.

Johansen Test

As mentioned before, the Engle and Granger two-step test cannot be used to test cointegration involves more than two variables, so here we use Johansen test for multivariate cointegration testing.

Johansen test involves trace test and maximum eigenvalue test:

- Trace test:

$$\lambda_{trace} = -T \sum_{i=r+1}^{k} log(1 - \lambda_i)$$

$H_0 : r < k \Longleftrightarrow H_1 : r = k$

- Maximum eigenvalue test:

$$\lambda_{max} = -Tlog(1 - \lambda_{r+1})$$

$H_0 : r = q, q = 1, 2, \ldots, k \Longleftrightarrow H_1 : r \leq q + 1$

where λ_i is the eigenvalue value of matrix Π.

Granger Causality

In most cases, it is hard to tell the causality among variables from regression. For example:

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

the significance of coefficient β_1 indicates that there exists relationship between variable y and X, but says nothing about X cause y.

Granger Causality

For causality, we have the following concepts:

- If X causes y, the changes of X happened before and were followed by the changes of y
- If X causes y, including X in regression helps improve forecasting of y, while reverse is wrong

Granger(1969) and Sims(1972) use VAR to test for the "Causality" in above sense, thus this causal relationship is called *Granger*(or *Granger-Sims*) *causality*.

Granger Causality

Consider a binary (y and X) system to keep things simple. Based on the "Causality" concepts:

- X is said to Granger-cause y if the current or lagged term of X help improve forecast X
- X fails to Granger-cause y if predict accuracy of $y_{t+s}(s > 0)$ based on information $(y_t, y_{t-1}, \dots)$ is the same as predict based on information $(y_t, y_{t-1}, \dots, X_t, X_{t-1}, \dots)$
- If we restrict the relationship between X and y are linear, we may use MSE to measure forecasting performance and the above condition can be rewrite as: X fails to Granger-cause y if

$$MSE[\hat{E}(y_{t+s}|y_t, y_{t-1}, \dots)] = MSE[\hat{E}(y_{t+s}|y_t, y_{t-1}, \dots X_t, X_{t-1}, \dots)]$$

Granger Causality

Example (VAR Formulation)

The binary VAR(p) specification can be written as

$$\begin{pmatrix} Y_t \\ X_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \Phi_{11}^1 & \Phi_{12}^1 \\ \Phi_{21}^1 & \Phi_{22}^1 \end{pmatrix} \begin{pmatrix} Y_{t-1} \\ X_{t-1} \end{pmatrix} + \cdots + \begin{pmatrix} \Phi_{11}^p & \Phi_{12}^p \\ \Phi_{21}^p & \Phi_{22}^p \end{pmatrix} \begin{pmatrix} Y_{t-p} \\ X_{t-p} \end{pmatrix} + \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{pmatrix}$$

If X fails to Granger-cause y, then we have:

$$\Phi_{12}^1 = \Phi_{12}^2 = \cdots = \Phi_{12}^p = 0$$

that is this regression has a lower triangle coefficient matrix.

Granger Causality

Consider a linear regression of X_t on the past, current and future value of y's:

$$X_t = c + \sum_{j=0}^{\infty} b_j y_{t-j} + \sum_{j=0}^{\infty} d_j y_{t+j} + \epsilon_t$$

where the exogenous assumption is satisfied ($E[\epsilon_t y_s] = 0$ for all t and s), then X fails to Granger-cause y if and only if $d_j = 0$ for $j = 1, 2, \ldots$

Similarity, we can use this regression to test whether y Granger-cause X:

$$y_t = c + \sum_{j=0}^{\infty} b_j X_{t-j} + \sum_{j=0}^{\infty} d_j X_{t+j} + v_t$$

where the exogenous assumption is satisfied ($E[v_t X_s] = 0$ for all t and s), then y fails to Granger-cause X if and only if $d_j = 0$ for $j = 1, 2, \ldots$

Granger Causality

Causality model:

$$y_t = c_0 + \sum_{i=0}^{p_y} \alpha_i y_{t-i} + \sum_{j=0}^{p_x} \beta_j x_{t-j} + u_t$$

where p_x and p_y are the order of AR model of each series which are determined by F-test and/or AIC, BIC.

Granger-Causality test:

- Test: H_0 (X fail to Granger-cause y): $\beta_1 = \cdots = \beta_{p_x} = 0$
- Estimate the unrestricted regression, keep RSS_u
- Estimate the restricted regression ($\beta_1 = \cdots = \beta_{p_x} = 0$), keep RSS_R
- Construct F-test: $F = [(T-k)/q] \times [(RSS_R - RSS_U)/RSS_U]$, where $k = (1 + p_x + p_y)$ and $q = (1 + p_y)$

(Similar test can be conducted to test whether y Granger-cause X)

Granger Causality

There are four possible outcome for Granger-Causality test:

- X Granger-cause y, but y fails to Granger-cause X
- y Granger-cause X, but X fails to Granger-cause y
- X Granger-cause y and y Granger-cause X
- X fails to Granger-cause y and y fails to Granger-cause X

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- This model imposes a **linear, additive conditional mean** with constant coefficients, unless we enrich the specification.

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- Our simple ADL assumes that one additional dollar of promotion changes conditional expected sales by the same β, holding other inputs fixed.
- **Important:** We could extend regression with interactions, dummies, polynomial terms, or splines. ML becomes attractive when many potential nonlinearities are difficult to specify in advance.

From a Linear ADL to Flexible Machine Learning

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$$s_{t+1}^{(i)} = f\left(s_t^{(i)}, p_t^{(i)}, \text{month}_{t+1}, \ldots\right) + u_{t+1}^{(i)}.$$

Here f need not be linear or additive.

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- **But:** Flexible ML does not automatically know what happened in previous months. That depends on the information we give it.

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- Consider two hypothetical sales histories. Current sales and all other model inputs are identical, but the preceding sales trajectories differ.

	Not given to tree		**Given to tree**			
Scenario	s_{t-2}	s_{t-1}	s_t	Promotion	Price	Month
A (rising)	70	80	90	20	10	June
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- Because the tree sees identical inputs in both scenarios, it must produce the **same forecast**, despite their different histories.

Adapting ML to Time Series

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- Construct the model's input vector from information available at the forecast origin t:

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- We can also use growth rates, trailing volatility, seasonal lags, observed prices, or *already announced* future promotions.
- This is **feature engineering**: we encode aspects of the history so that standard ML algorithms can learn relationships involving time.
- The tree, forest, or boosting algorithm itself usually needs no new time-series-specific optimization procedure.

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- Unlike a linear ADL, f_h can capture nonlinearities and interactions without us specifying their exact functional forms.

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- Consecutive examples have overlapping feature windows; hence training rows need not be independent observations.
- Decide the target horizon and the feature-availability rule *before* building the supervised dataset.

Sliding Forecast Origins: An Illustration

Q: How do histories, features, and future targets line up?

Date	Y	X	YL1	YL2	XL1	XL2	XL3
Jan-00	10	~~100~~					
Feb-00	11	~~112~~	11		100		
Mar-00	12	~~115~~	12	10	112	100	
Apr-00	13	~~118~~	13	11	115	112	100
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Jun-00	15	~~130~~	15	13	120	118	115

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- Adjacent windows overlap. More rows do *not* necessarily mean an equivalent increase in independent information.

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- The tree can now split on lagged sales, recent growth, seasonality, and interactions among these features.

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- **Key insight:** The tree algorithm is unchanged. By supplying historical features, we allow it to learn nonlinear relationships involving temporal persistence.

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- **If we randomly resample these training rows, aren't we destroying their temporal ordering?**

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- Suppose we build a forest using monthly sales observations from 2018–2023.
- Consider predicting July 2021 sales using features available at the end of June 2021.
- One tree's bootstrap sample might include:

Forecast origin	**Selected for tree?**	**After June 2021?**
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- Its OOB prediction for July 2021 sales therefore does **not** replicate a forecast we could have made in June 2021.

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- **Key distinction:**
  - **Bootstrap resampling:** Randomly selects rows within the eligible training sample to fit trees.
  - **Forecast evaluation:** Tests the fitted model on subsequent observations, preserving chronological order.

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- Tune learning rate, tree complexity, and iteration count with chronological validation and early stopping (e.g., XGBoost).

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- **Example:** Boosting might learn that a promotion has a greater effect when sales have been declining for several consecutive months.

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$$\mathbf{x}_t = (s_t, s_{t-1}, s_{t-2}, \ldots).$$

- **Example:** Boosting might learn that a promotion has a greater effect when sales have been declining for several consecutive months.
- **Key insight:** Boosting changes how we *estimate* the prediction function, not how temporal information enters the model.

Outline

5 Forecast
- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

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- **For forecasting:** We will investigate different architectures for representing and processing historical observations. Depth alone does not determine how a network handles time.

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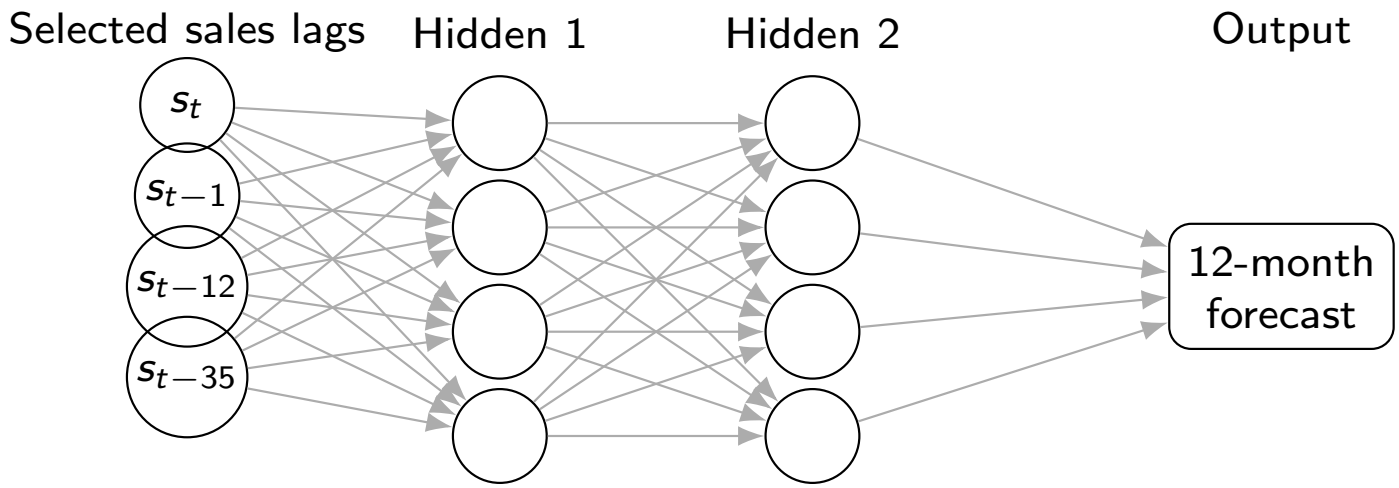
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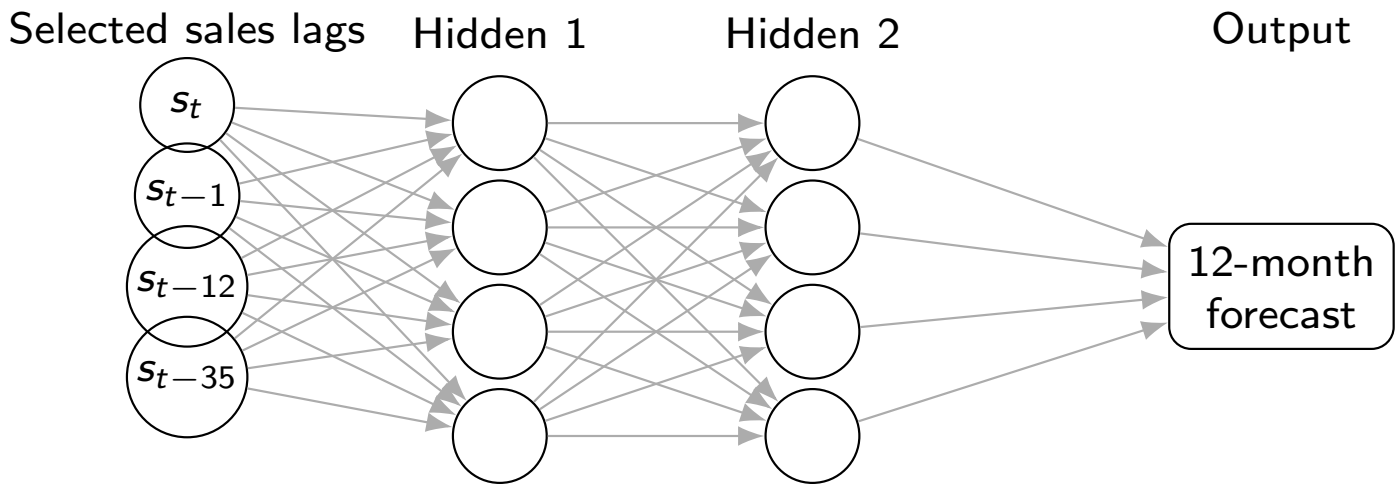
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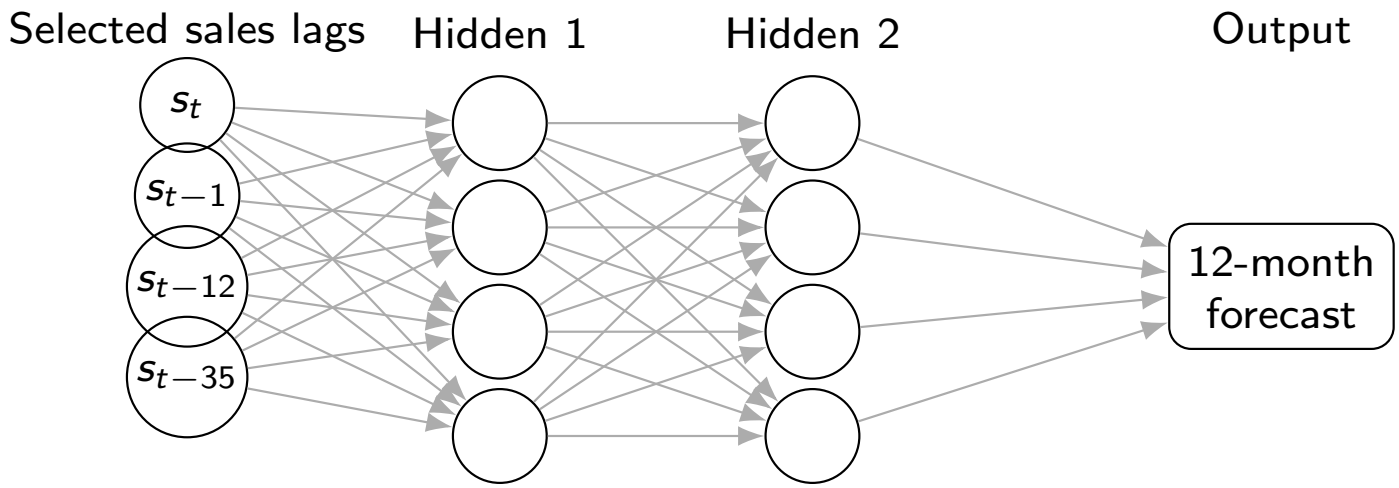
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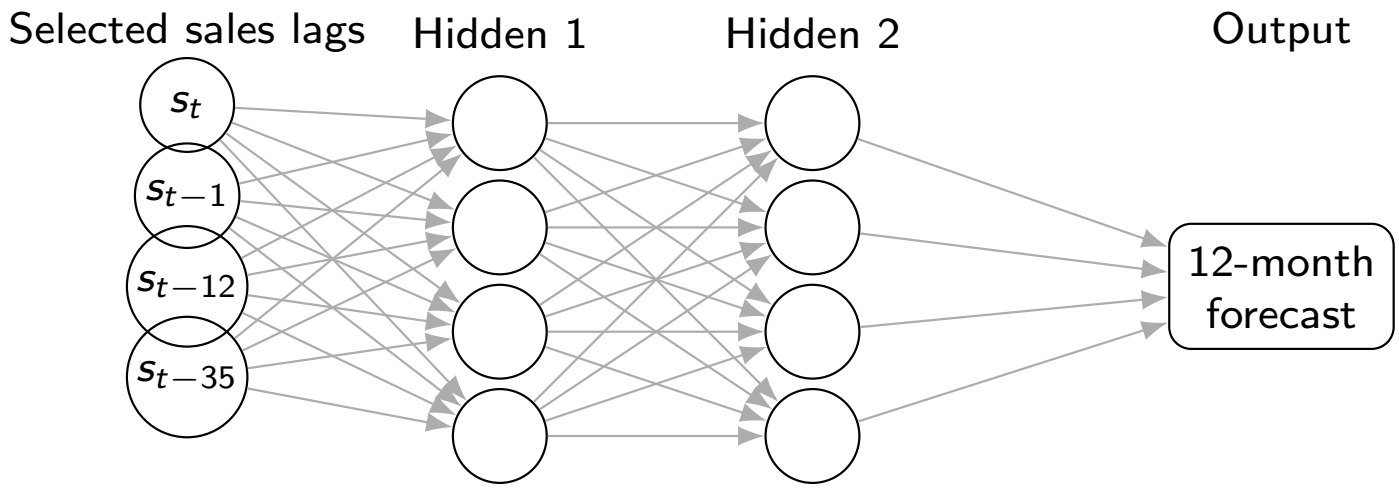
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  - **N-HiTS:** Processes historical information at multiple resolutions to represent patterns operating at different timescales.
- Both remain feedforward architectures: neither requires recurrent memory or transformer attention.

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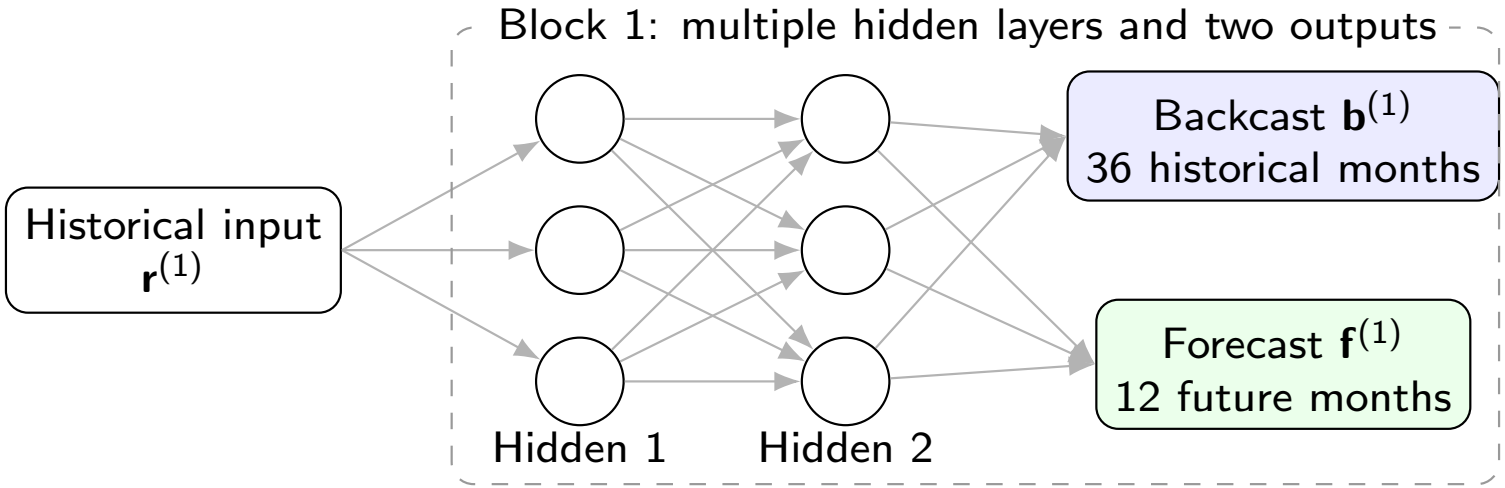
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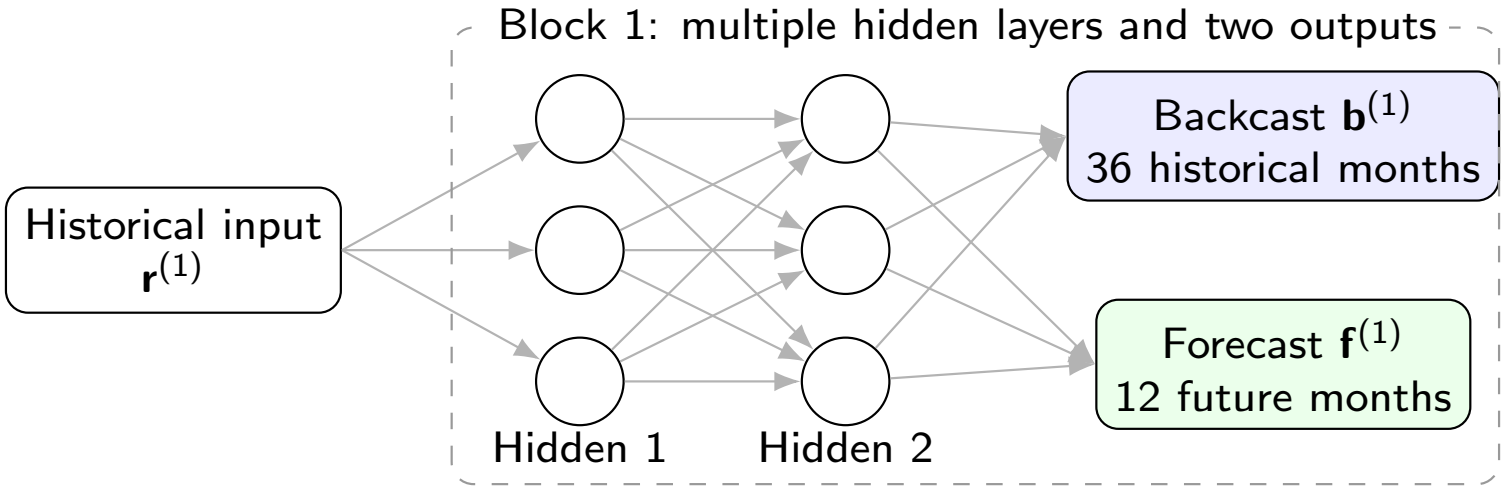
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- **Backcast:** the block's approximation of its *historical input*.
- **Forecast:** the block's contribution to *future sales*.

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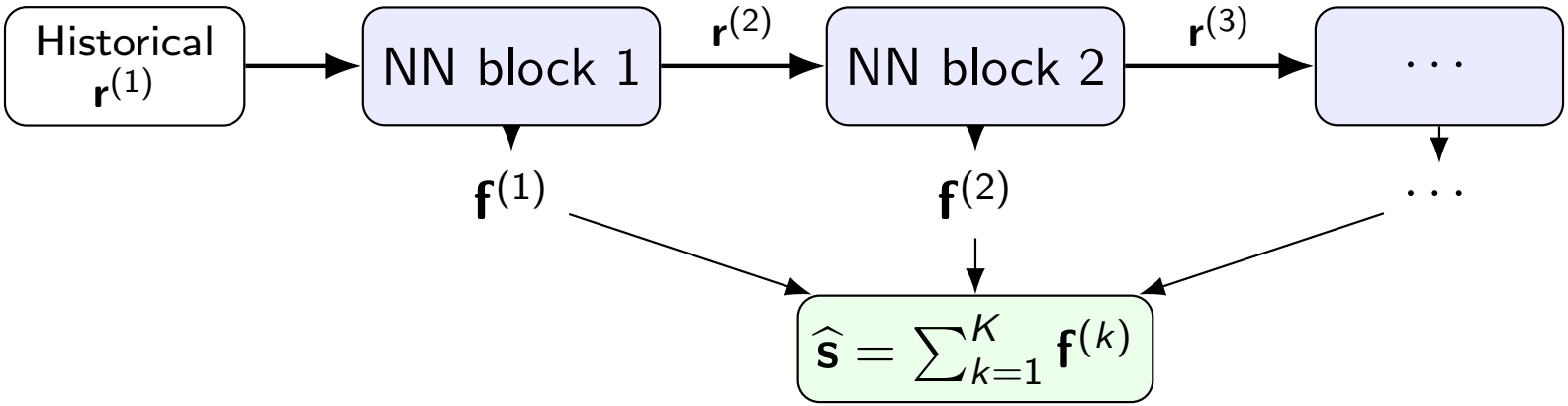
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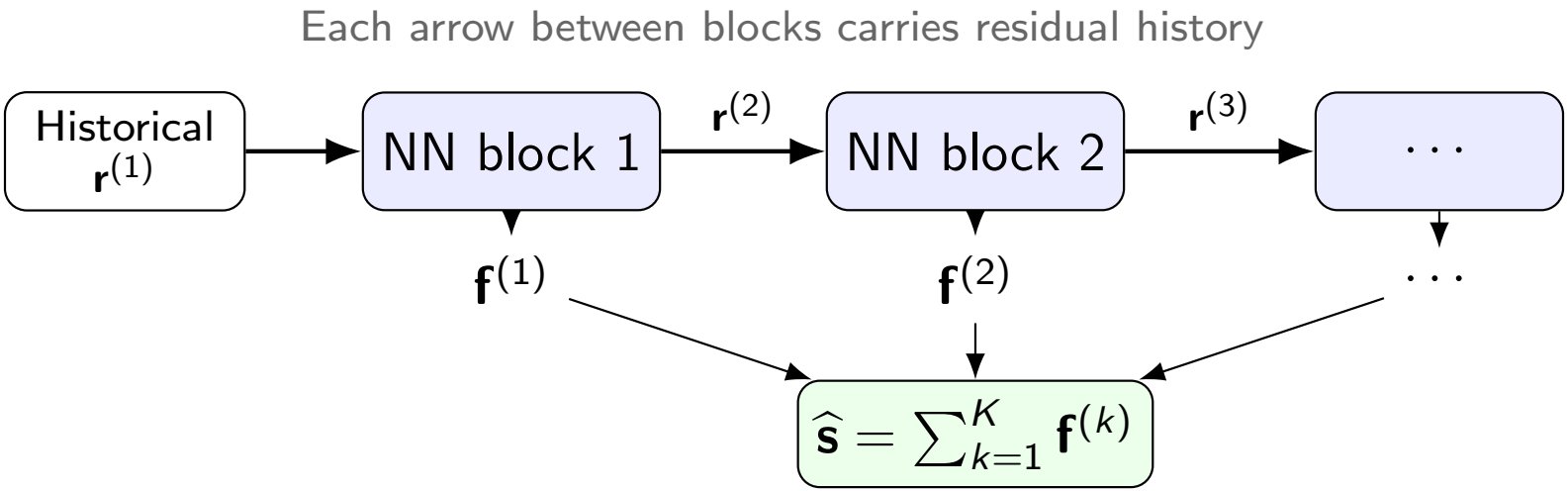
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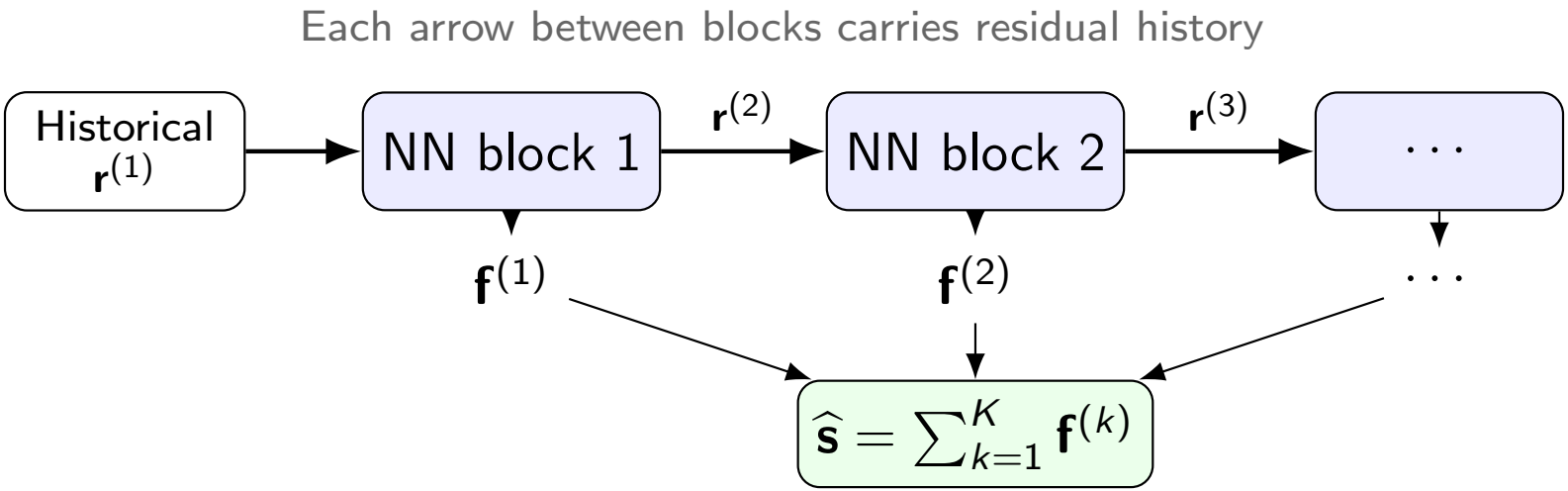
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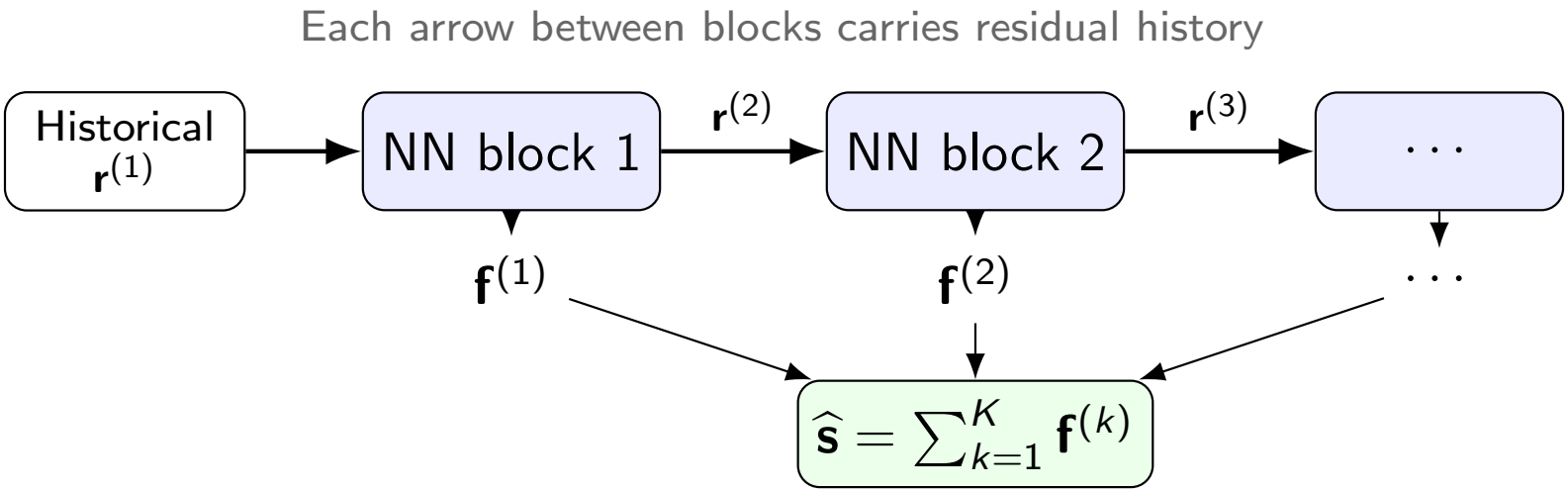
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- **Distinction:** N-BEATS emphasizes blockwise residual decomposition; N-HiTS also organizes information across resolutions. Both are feedforward architectures.

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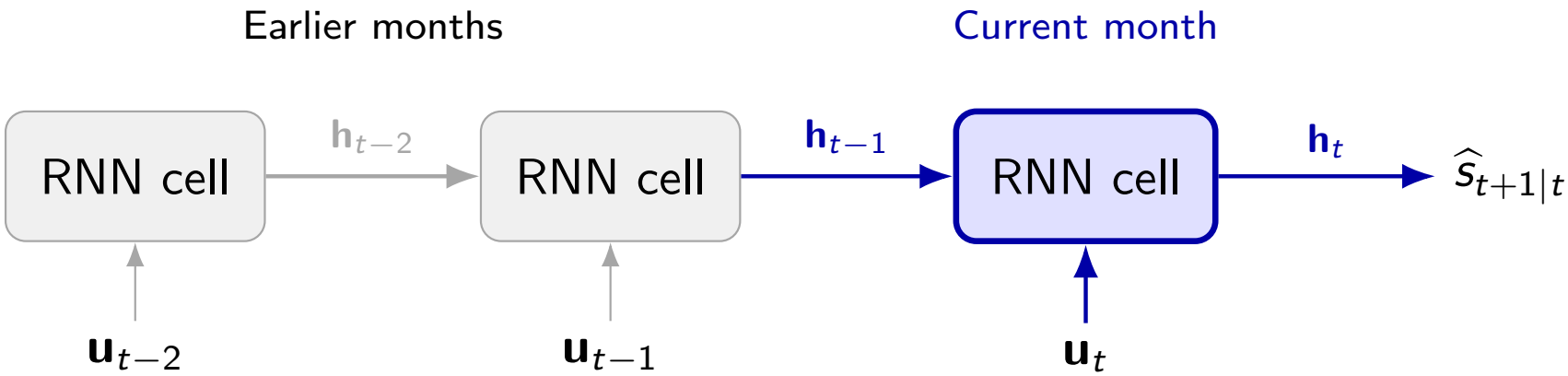
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- The same state-update rule is reused each month. The network can learn what historical information to carry forward, although its effective memory is limited.

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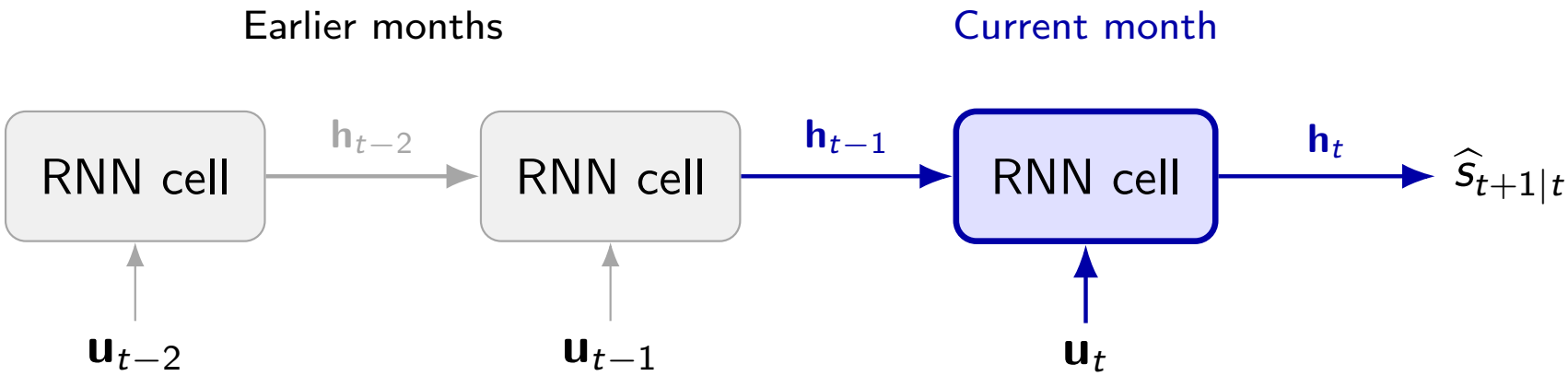
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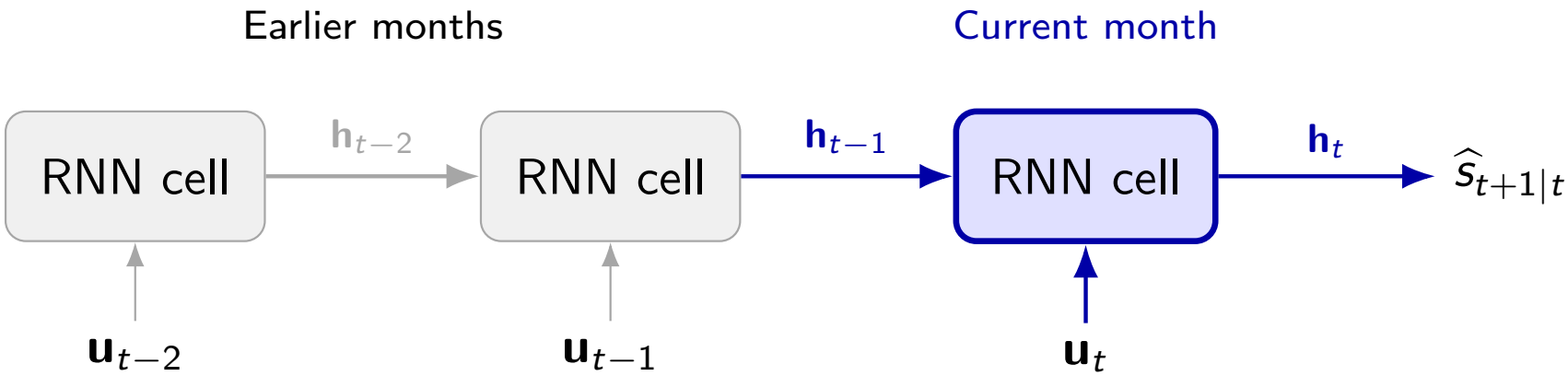
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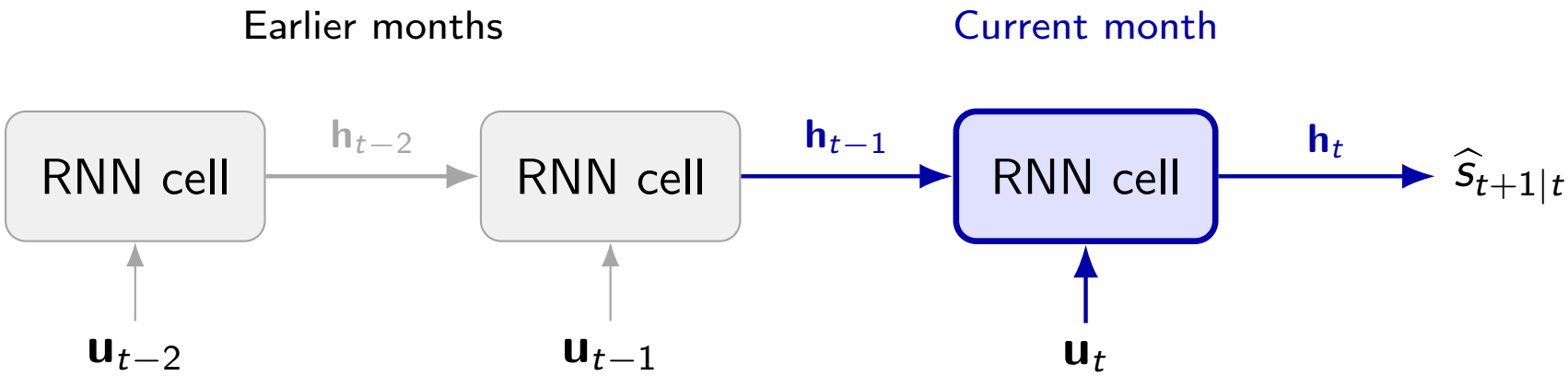
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- **Contrast with feedforward NN:** A standard feedforward network processes a supplied historical window without passing a hidden state from one month to the next.

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- **Motivation for LSTM:** Provide a more controlled memory pathway so useful information can be retained, revised, or forgotten.

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- An LSTM maintains two distinct states:
  - $\mathbf{c}_t$: **cell state**, a pathway for retaining information over time.
  - $\mathbf{h}_t$: **hidden state**, passed to the next time step and used to generate forecasts.

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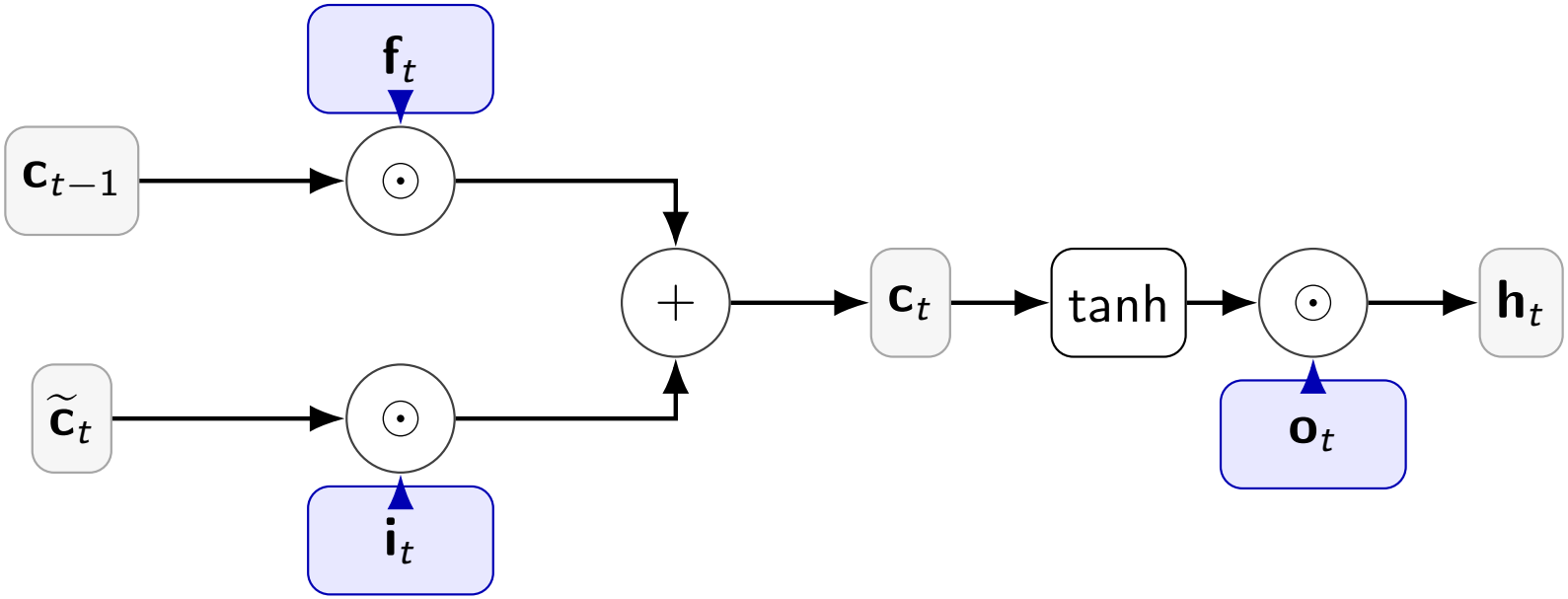
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- Gates *facilitate* long-range learning; they do not guarantee the network will discover the correct pattern.

LSTM: Inside One Time Step

Q: Where do the forget, input, and output gates act?

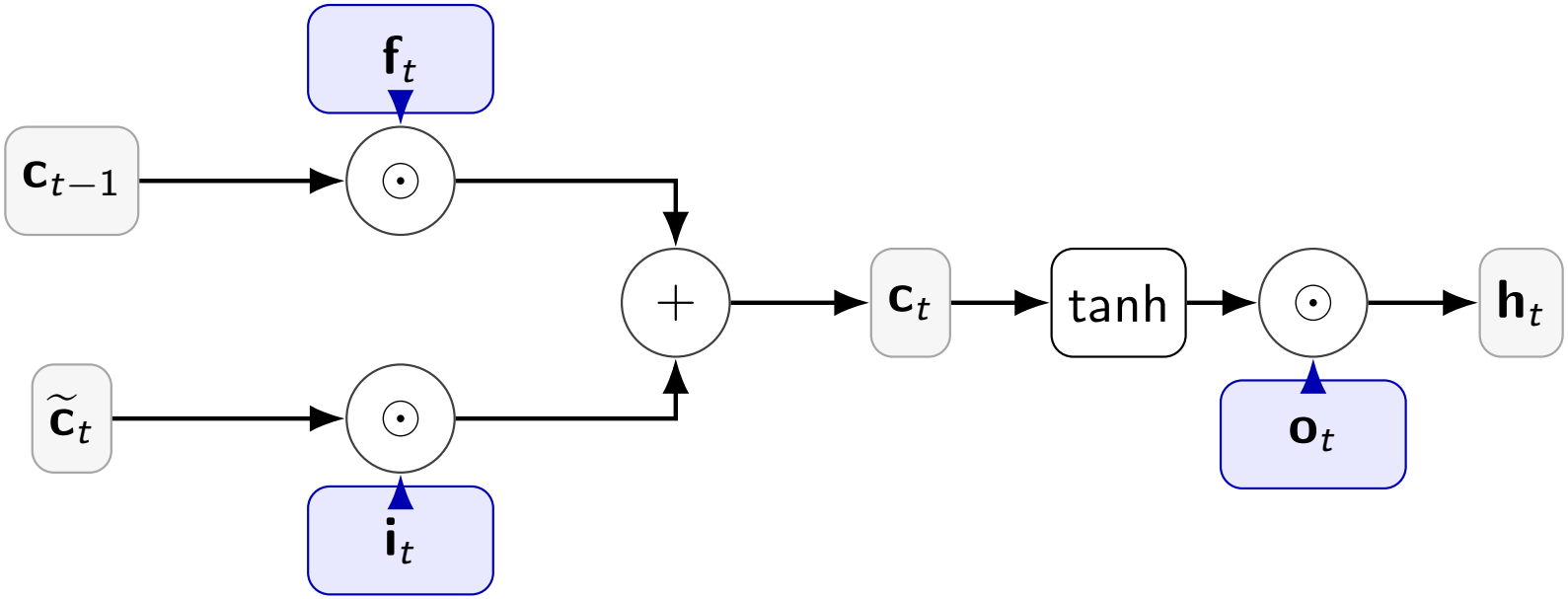
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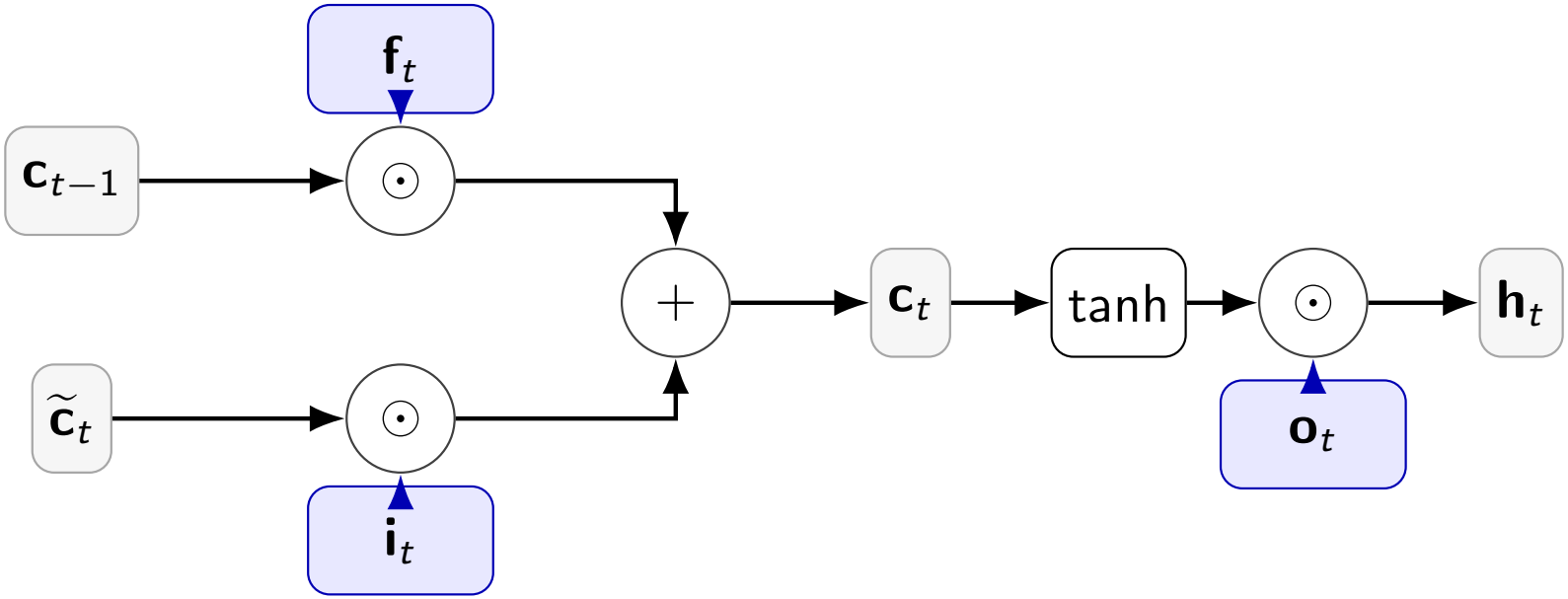
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- **Update cell memory:** $\mathbf{f}_t$ controls how much of $\mathbf{c}_{t-1}$ to retain; $\mathbf{i}_t$ controls how much candidate memory $\widetilde{\mathbf{c}}_t$ to add.
- **Produce hidden state:** $\mathbf{o}_t$ controls how much of the updated memory is exposed as $\mathbf{h}_t$. Both $\mathbf{c}_t$ and $\mathbf{h}_t$ pass to the next month.

The gate values and candidate memory are computed from $(\mathbf{u}_t, \mathbf{h}_{t-1})$ using learned parameters.

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- These equations determine the gate values; the next slide shows how the gates actually update memory.

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- g is an output layer that maps the hidden state to a sales forecast (or parameters of a predictive distribution).

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- **Tradeoff:** direct outputs avoid recursive error propagation; recursive decoding reuses a sequential prediction rule, but errors can accumulate.

From LSTMs to Transformers

Q: How can two architectures use distant observations differently?

From LSTMs to Transformers

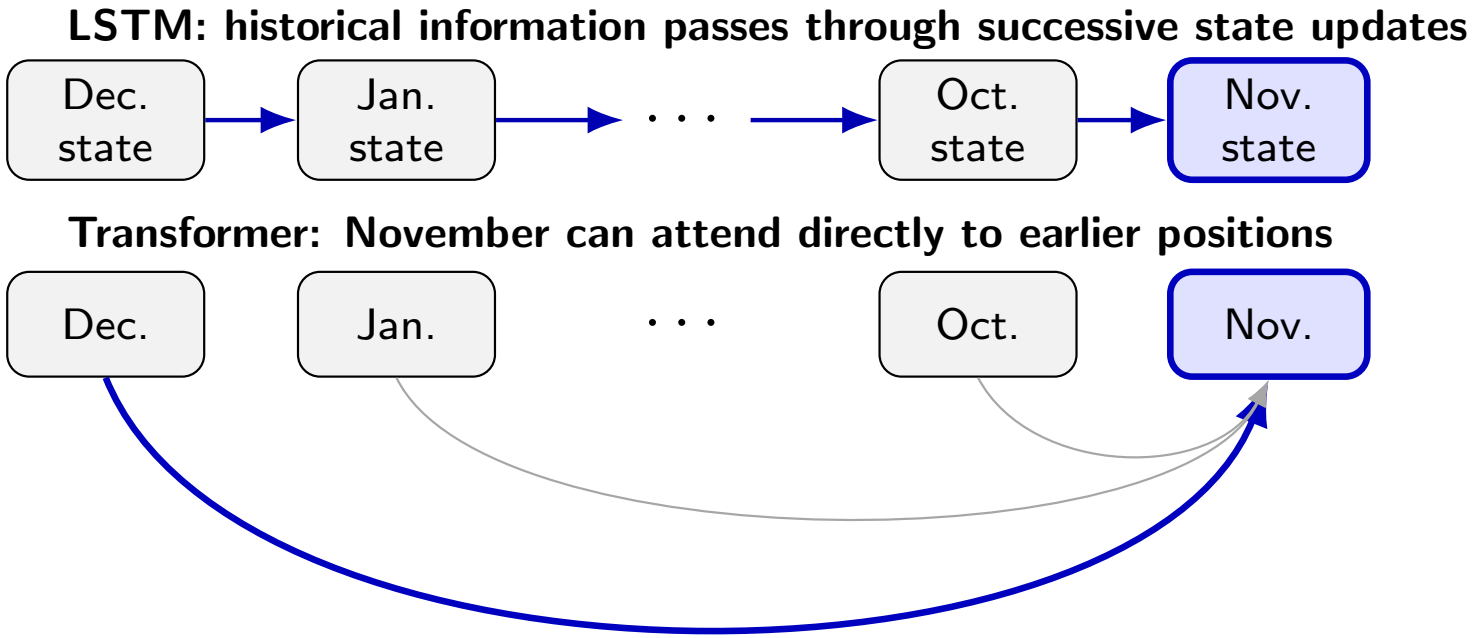
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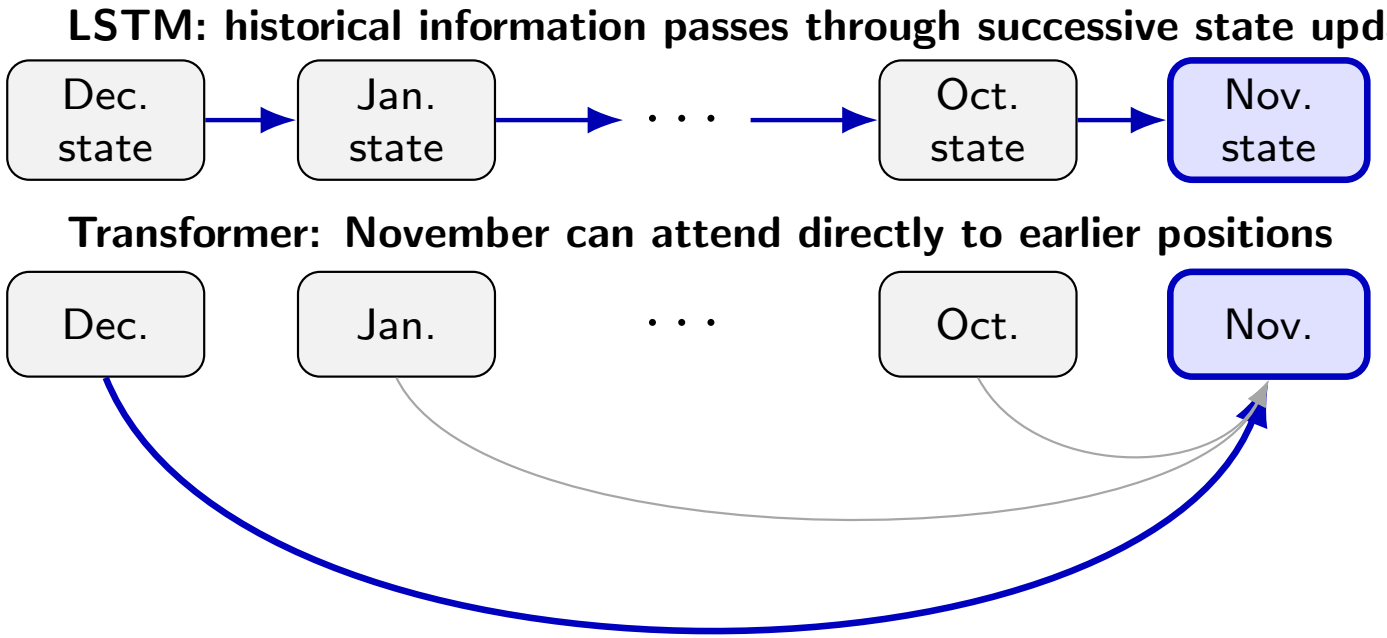
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- **Both can use last December's information.** An LSTM carries it forward through state updates; attention lets November's representation combine information from December directly.

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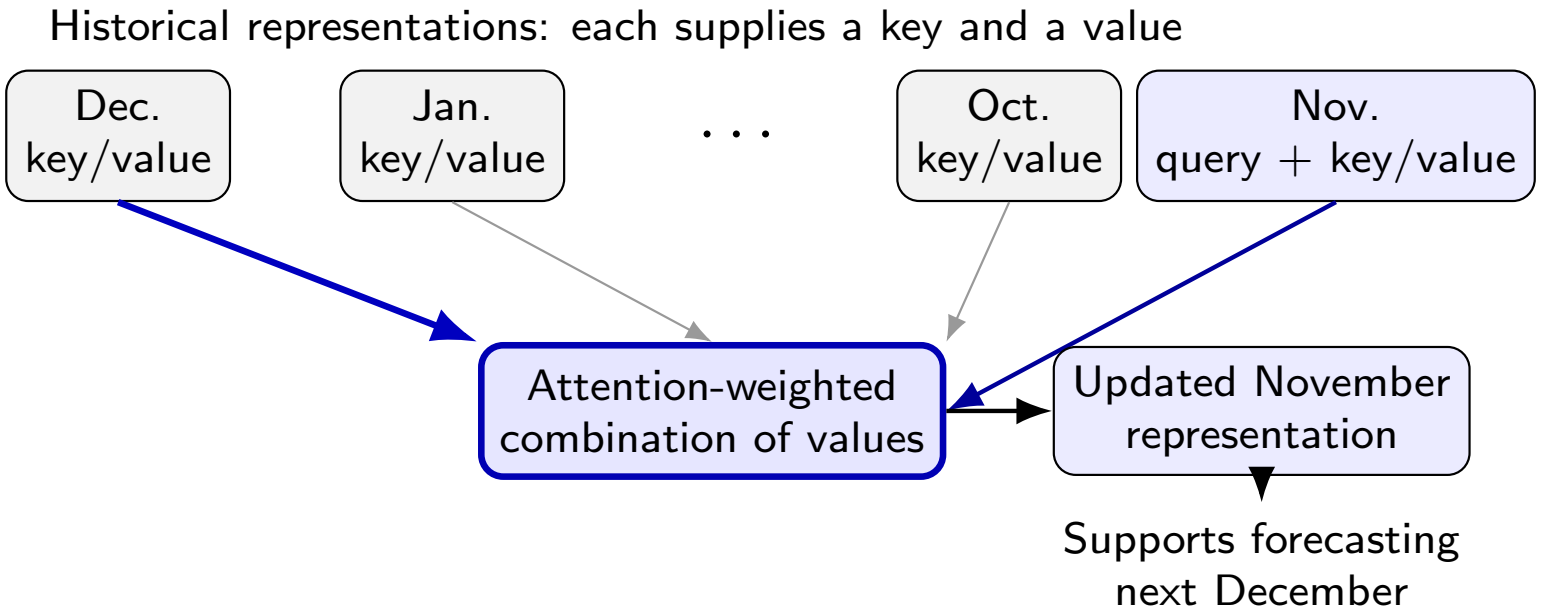
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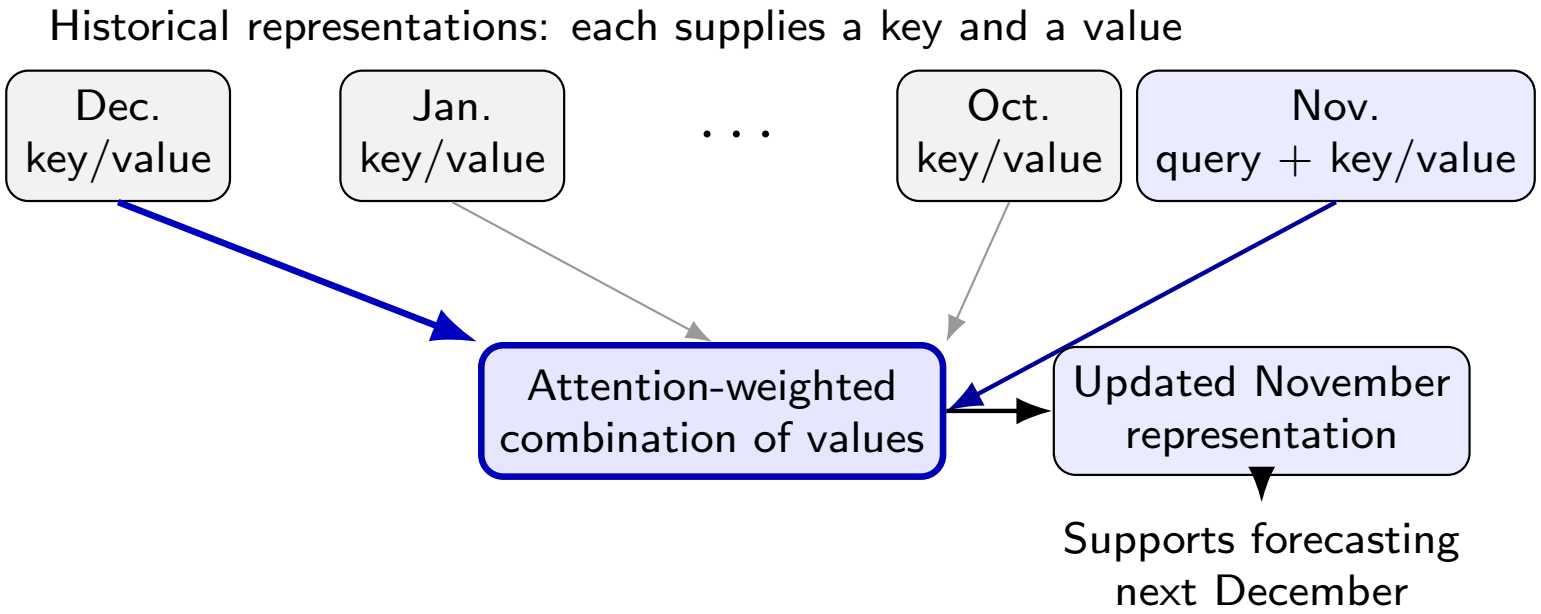
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- **Attention:** November's query is compared with historical keys to determine weights. Those weights combine historical values into an updated November representation, which can support future forecasts.

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- **Distinction:** Position encodes sequence order; calendar variables (e.g., month of year) can encode seasonality.

Transformer Architecture: Stacking Blocks

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input window

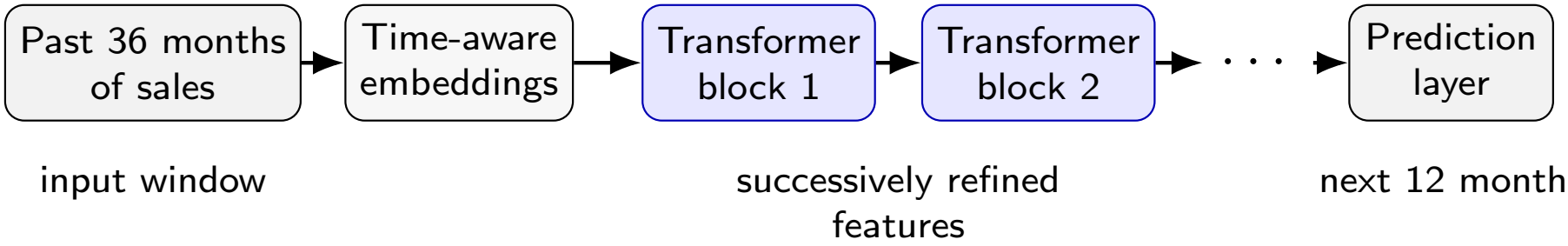
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next 12 months

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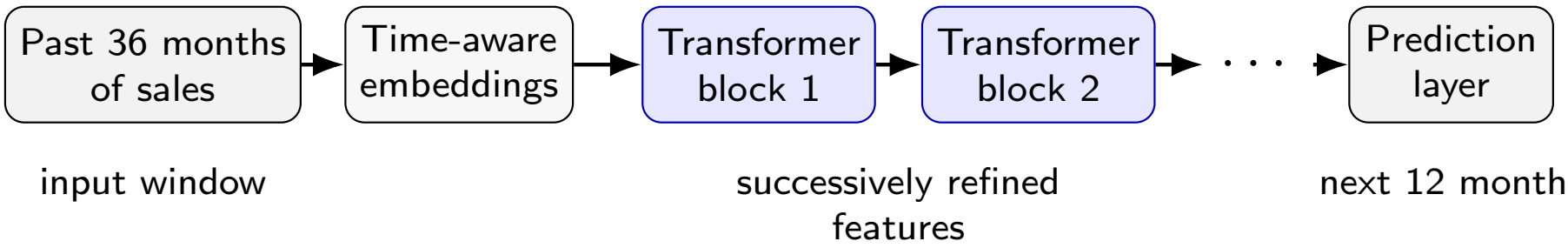


- **Each block performs learned feature engineering:** It takes the features constructed by the preceding block and transforms them, using attention to incorporate information across historical observations.

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- **Each block performs learned feature engineering:** It takes the features constructed by the preceding block and transforms them, using attention to incorporate information across historical observations.
- **Stacking blocks creates depth:** Successive blocks learn increasingly refined features. A final prediction layer converts these features into sales forecasts.

Inside a Transformer Block

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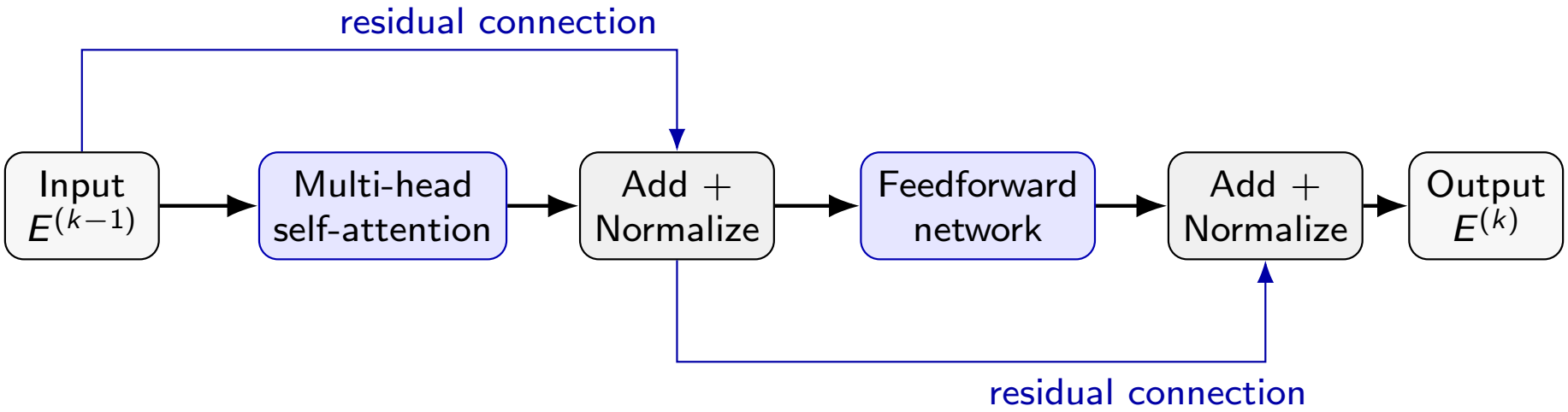
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A **block** is a subnetwork, not a single hidden layer. Unlike N-BEATS blocks, which contribute forecasts, transformer blocks produce updated learned features.

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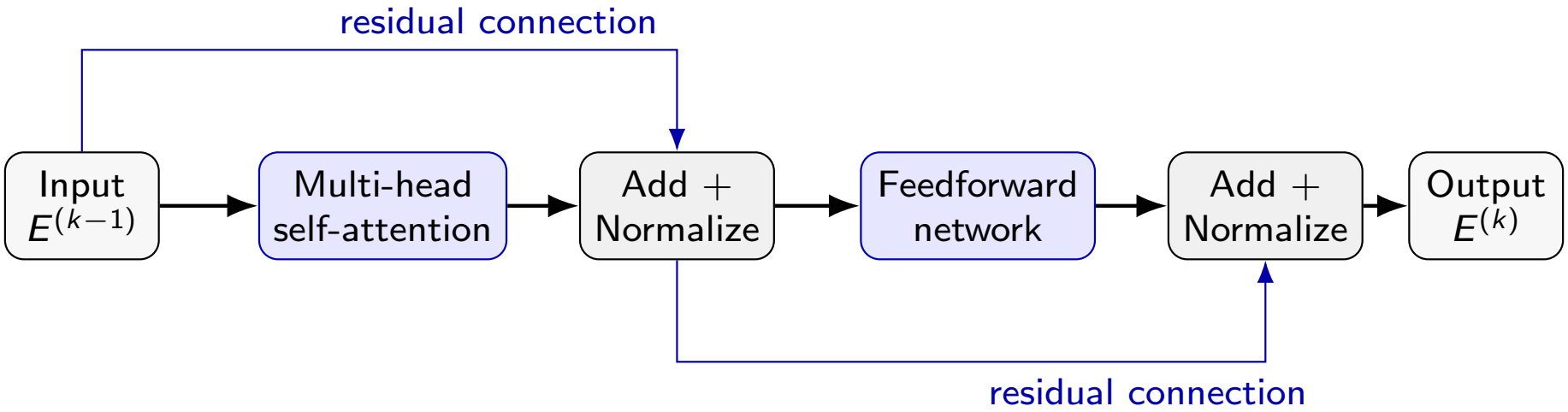
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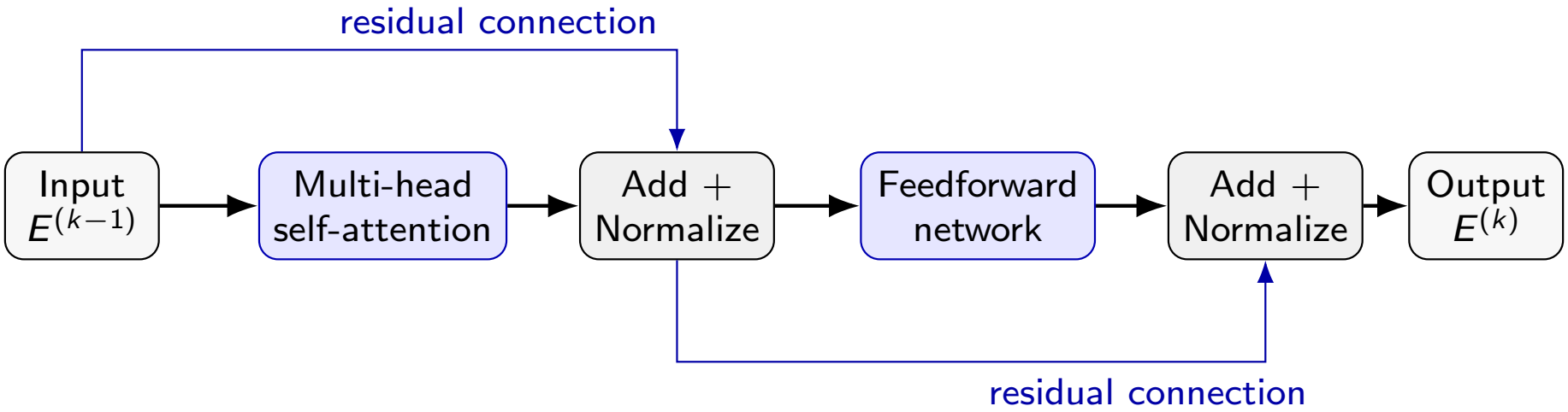


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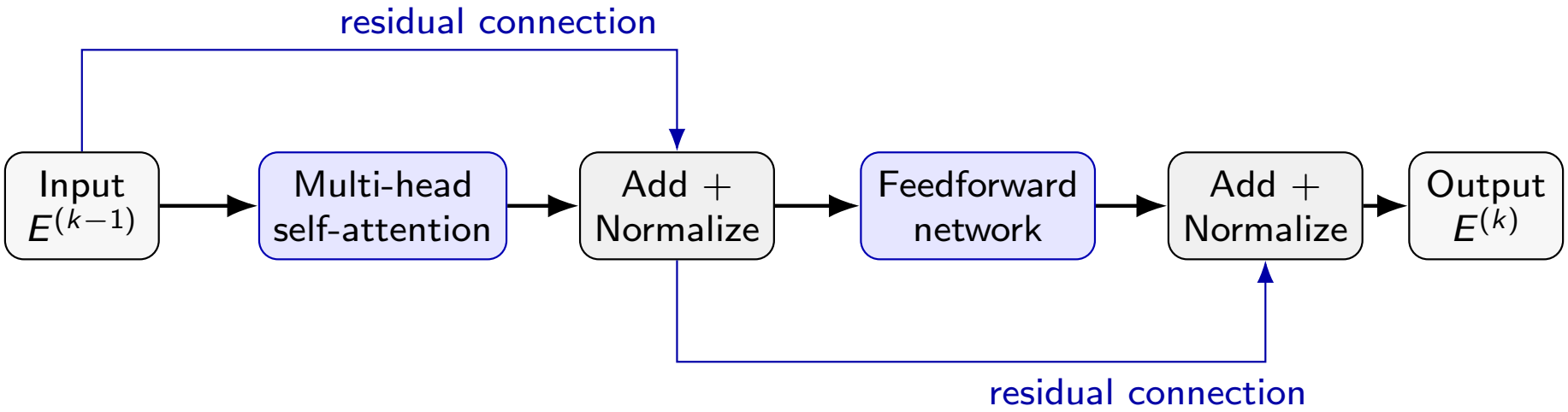


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- **Residual connections and normalization** support training when many blocks are stacked.

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- **Distinction:** Attention determines how the historical features are constructed. The output strategy determines how future predictions are generated.

Comparing Neural-Network Forecasting Architectures

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Feedforward NN	A fixed vector of lagged sales passes through nonlinear transformations.
N-BEATS / N-HiTS	A fixed window passes through specialized forecasting blocks; N-HiTS uses multiple resolutions.
RNN / LSTM	A hidden state evolves as each month's observations are processed; LSTM adds gated memory.
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Takeaway: These architectures differ in *how they process temporal information*. Network depth is a separate modeling choice.

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- **Empirical decision:** Compare against simple benchmarks using genuinely out-of-sample forecasts. No architecture consistently wins merely because it is newer or deeper.

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- **Next:** How can we exploit shared patterns across many SKUs and divisions without assuming every series behaves identically?

Outline

5 Forecast
- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
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- Paradigm Comparison

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- But pooling dissimilar products can also hurt: common parameters are not automatically appropriate for every SKU.

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- All SKU–month rows contribute to estimation of the same a and ϕ.
- Pooling may improve precision when histories are short and common dynamics are credible; it may introduce bias when they are not.

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- Its effective information depends on within-SKU dependence and cross-SKU correlation: $N \times T$ rows are not $N \times T$ independent observations.
- Alternatives: allow SKU intercepts a_i, division-specific slopes ϕ_d, partial pooling, or regularized hierarchical coefficients.

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- f_h can be linear, boosted trees, feedforward, recurrent, or transformer-based.
- **Nonlinearity** lets the response depend on the state; **SKU characteristics** help distinguish structurally different products.
- A flexible model cannot distinguish otherwise identical inputs unless relevant differences are supplied or represented.

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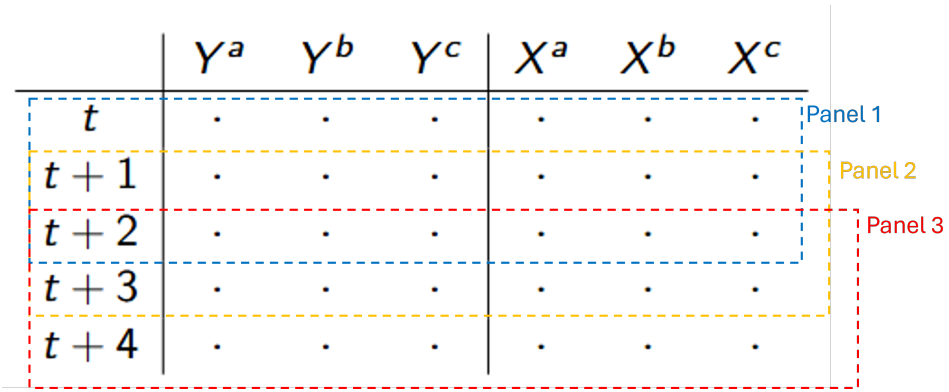
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- Sharing data across SKU histories is the *additional* source of information in global learning, not the existence of rolling windows alone.



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- A broader pool increases available examples but may also combine different sales dynamics and scales.
- Compare both using date-based, held-out forecast origins and metrics appropriate to the decision (e.g. SKU-level and division-level errors).
- Training only on some SKUs and evaluating on entirely unseen SKUs tests *cross-series transfer*; evaluating later months on previously seen SKUs tests *future forecasting*. These are different experiments.

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- Include SKU/division identifiers (or learned embeddings), calendar indicators, known prices, and other covariates when useful.
- Preserve the meaning of the target: we forecast *dollar sales* so that SKU values add to divisional and company revenue; modeling physical unit counts across unlike products poses a different aggregation problem.
- A global model learns shared patterns, but maintaining heterogeneity is a modeling design problem.

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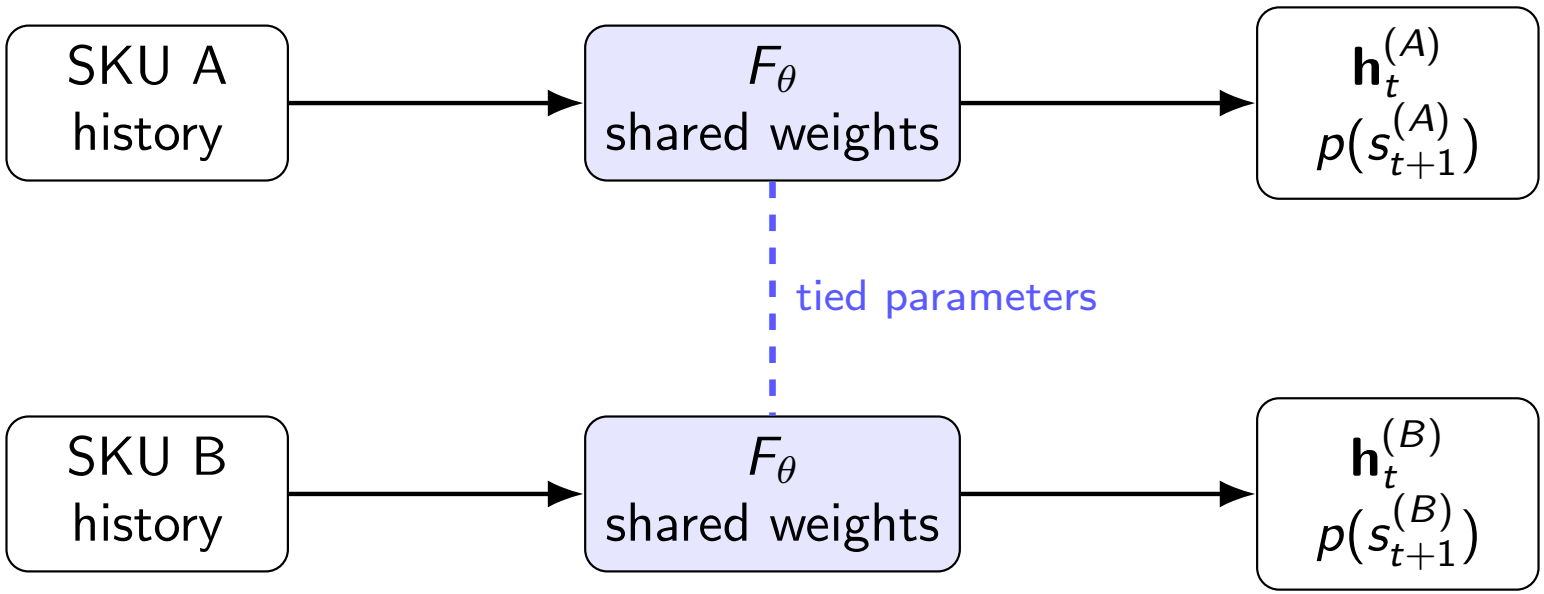
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- $\psi_t^{(i)}$ parametrizes a predictive distribution $p(s_t^{(i)} \mid \psi_t^{(i)})$.
- This separates *shared parameters* (θ) from each SKU's *time-varying state* ($\mathbf{h}_t^{(i)}$).

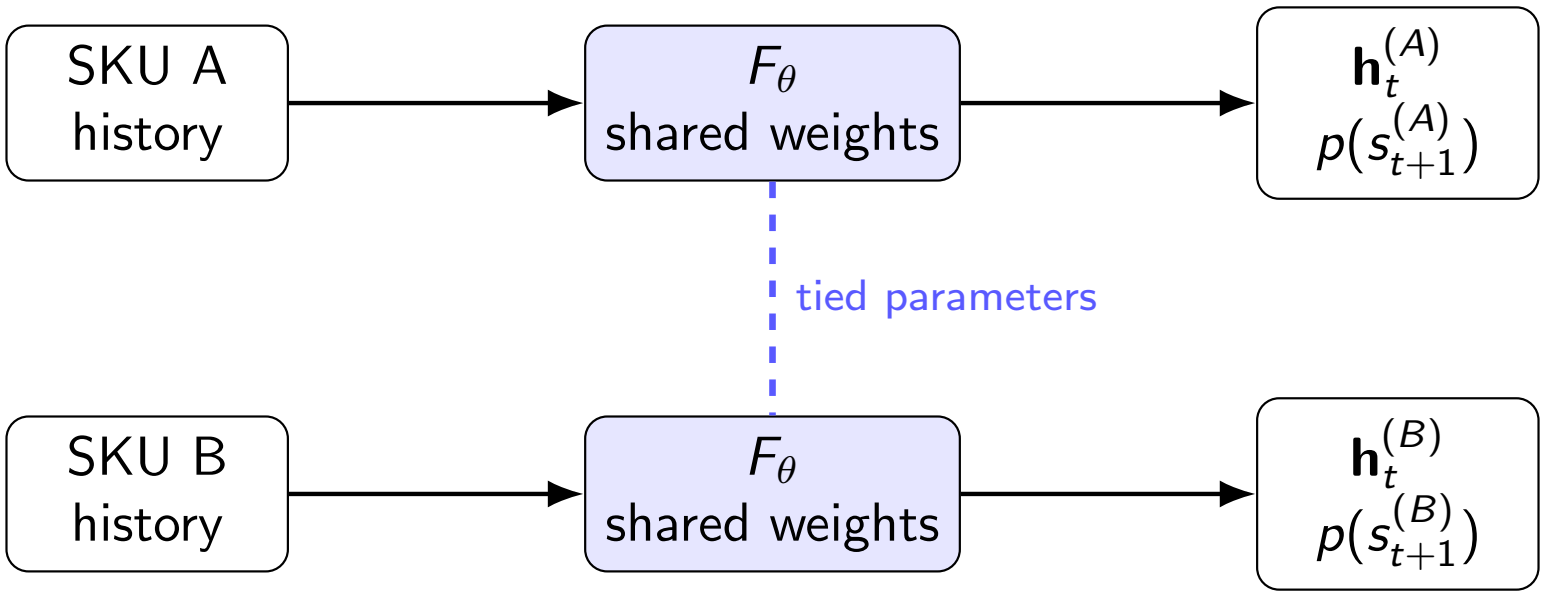
DeepAR: Shared Parameters, Different States

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- The network parameters are pooled; SKU-specific states reflect their distinct observed histories.
- Covariates and scale normalization can allow additional differences across products.

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- If instead modeling units sold as counts, a negative-binomial head can allow overdispersion.
- The likelihood must match the observed target, especially whether sales are counts, continuous revenue, intermittent, or bounded.

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- Model estimation may use sampled subsequences; the objective uses shared parameters while hidden states remain SKU-specific.

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2. Generate a draw $s_{t+1}^{(b)} \sim p(\cdot \mid \psi_{t+1})$.
3. Feed that draw and known future covariates back into the network; draw $s_{t+2}^{(b)}$, and continue to $t + H$.

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DeepAR: Simulating Multi-Month Paths

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- This is autoregressive *forecast generation*; it does not mean an ordinary linear AR model.
- Forecast paths reflect the assumed likelihood and trained model, and can be miscalibrated when assumptions or dynamics are wrong.

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- A global model may use any architecture: pooled regression, boosted trees, N-BEATS, DeepAR, or transformers/TFT.
- Local/global describes *training scope*; it says nothing by itself about whether forecasts are coherent across the company hierarchy.

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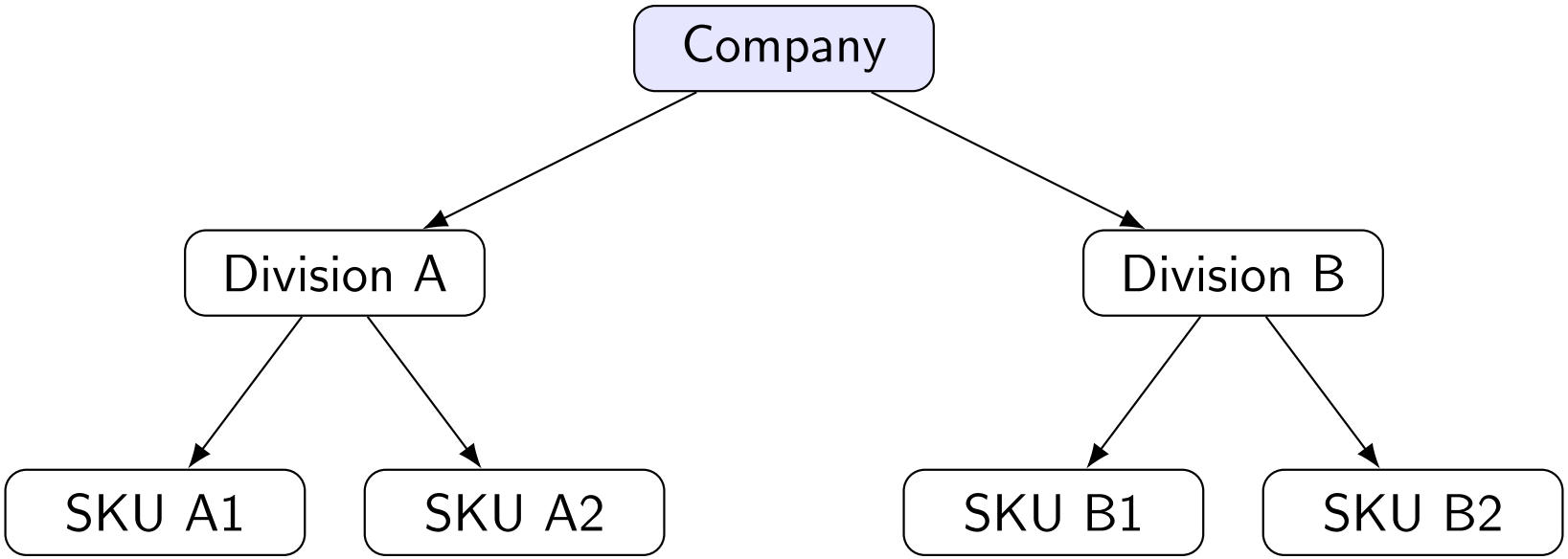
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- The issue is **coherence**: forecasts violate identities obeyed by the actual data.
- Even globally trained models do not generally enforce those identities automatically.

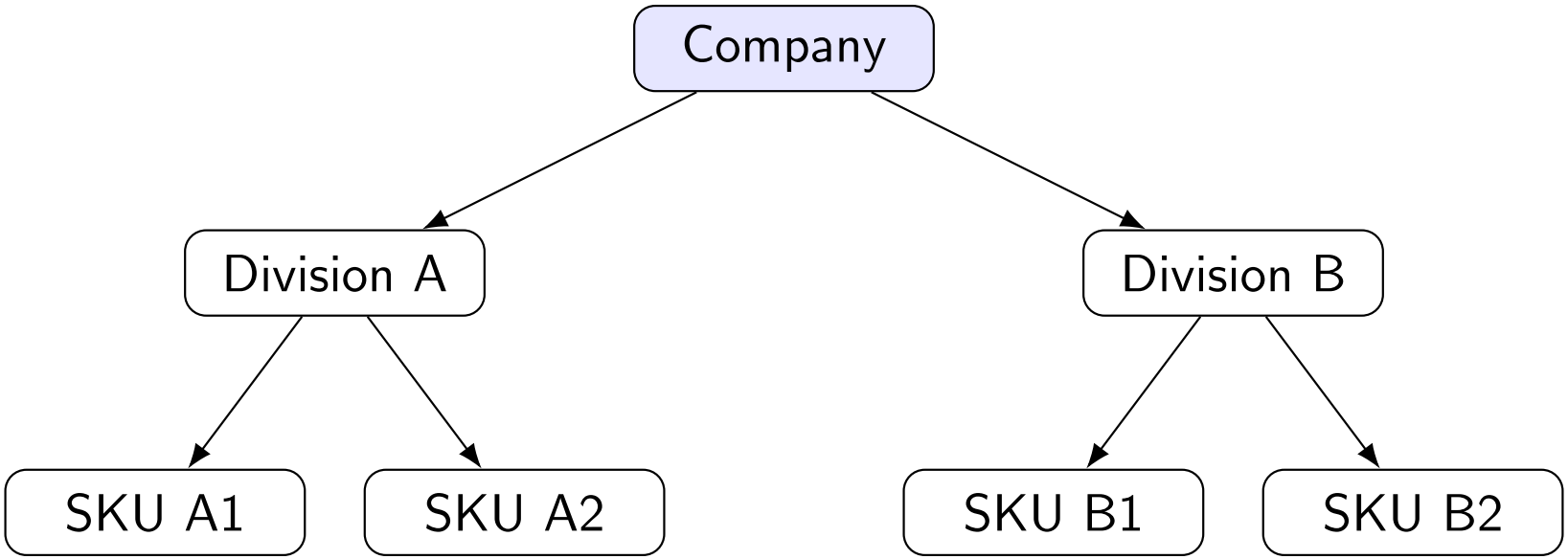
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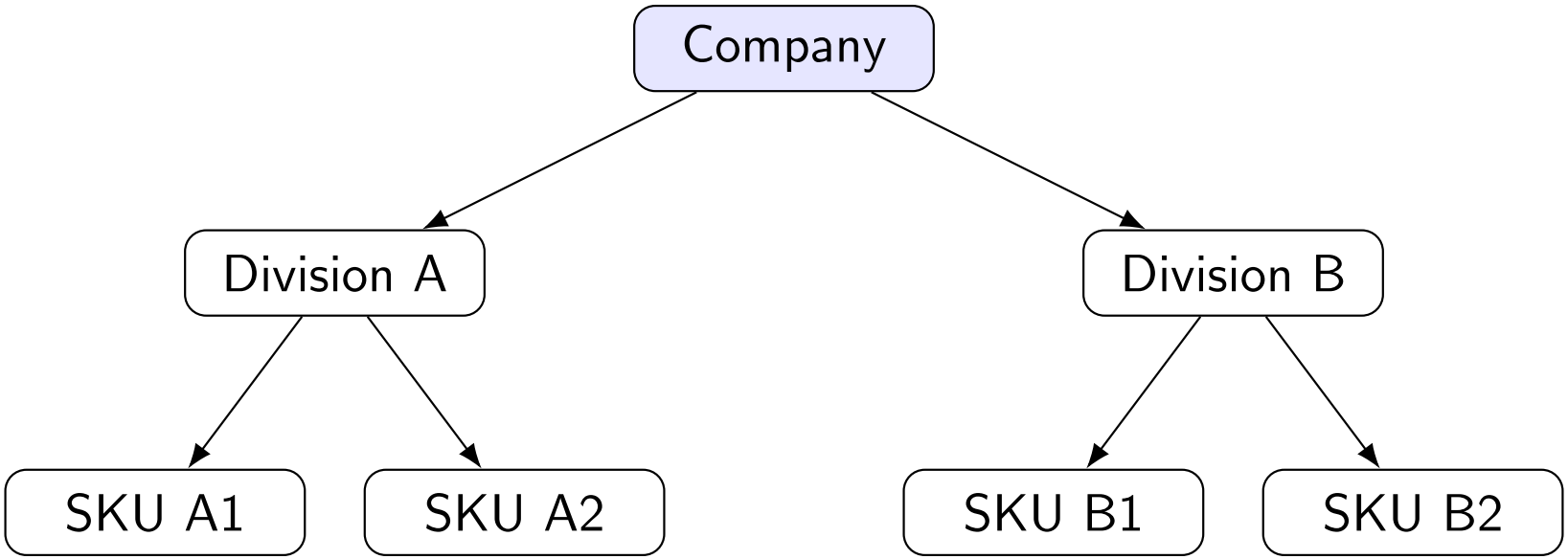
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- These are identities in *additive dollar sales*, not optional statistical relationships.

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- **Middle-out:** forecast divisions; sum up to company and allocate down to SKUs.
- Each enforces adding-up, but uses different information and imposes different assumptions.
- More general reconciliation can use base forecasts from *every* level rather than selecting one preferred level.

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- Historical allocations can be poor when SKU mix changes, products are launched, or demand shifts.

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	share	reconciled SKU sales
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- Check: $66 + 54 + 60 + 20 = 200$. Every level is coherent.

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$$\mathbf{y} = S\mathbf{b}, \qquad S = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

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- S is a fixed summing matrix. Any vector of the form $S\mathbf{b}$ is coherent.

General Forecast Reconciliation

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- Different reconciliation methods correspond to different choices of G_h.

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- **MinT** chooses a coherent linear reconciliation that minimizes the sum of reconciled forecast-error variances, under an unbiased-base-forecast framework.
- It uses a covariance matrix rather than assuming every level is equally reliable.

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- Practical issues: estimate W_h from legitimate forecast errors; use shrinkage/approximations if the covariance estimate is noisy or singular.
- Linear coherence does not automatically guarantee nonnegative forecasts or coherent predictive *distributions*.

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- Forecast accuracy and coherence should both be evaluated; coherence alone does not imply better forecast accuracy.

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- Pretraining does not guarantee transfer: unfamiliar products, intermittent sales, structural breaks, and specialized covariates may still be challenging.

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- Pretrained models differ in their input representations, architectures, forecast outputs, and covariate support.

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- We will develop **original Chronos** carefully as one example, then contrast alternative designs.

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5. Summarize simulated paths as means, medians, quantiles, and prediction intervals.

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- Handle all-zero histories using an explicit safeguard; do not divide by zero.
- The scale must depend only on the historical context at the forecast origin; including future target values would leak information.

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- Illustration: 0.8 maps to one bin, 1.0 to another, and 1.2 to another; the exact bins are a model design choice.
- Quantization loses precision and may clip extreme values, but permits a categorical prediction objective.
- Original Chronos adds special tokens where needed; the toy three-bin example is only intuition, not the actual vocabulary.

Original Chronos: Why a Transformer?

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- The transformer represents contextual relationships; it does not literally identify the structural data-generating process.

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- During training, known target tokens can condition later decoding steps; at inference they are replaced by generated tokens.
- This objective is *specific to the discretized generative example*; other foundation models may use continuous forecasts or quantile losses.

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- Sampling introduces Monte Carlo variation; increasing the sample count reduces that numerical variation but not model misspecification.

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- Both can forecast many SKUs; their *training provenance* is different.
- Evaluate on identical forecast origins, horizons, and targets; include seasonal-naive and other simple baselines.
- The model's default representation of time series may omit promotion plans or other company-specific features; check the particular model's covariate support.

Alternative Foundation Model Architectures

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Chronos-Bolt	Patch-based encoder with direct multi-step quantile forecasts; not the original token-by-token generation pipeline.
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These are *illustrative architecture families*; names, versions, and available input modalities evolve.

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- Options include adapting all weights, selected layers, or lightweight adapters where supported.
- Some models permit exogenous covariates, while others require separate mechanisms; check model-specific interfaces.
- Any fine-tuning and hyperparameter selection must use data preceding the held-out forecast origins.

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- Check seasonality, scale, intermittent/zero demand, forecast horizon, compute cost, and available covariates.
- Compare against seasonal naive, ETS/ARIMA, boosting, and global models using the *same* backtest design.
- Predictive distributions and hierarchical reconciliation can still require additional validation and processing.

Global vs Foundation: The Key Distinction

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	Company-trained global	**Pretrained foundation**
Training corpus	Company's many SKU histories	Broad pretraining corpus
Use on our SKUs	Estimate weights from our data	Reuse pretrained weights; optionally adapt
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Takeaway: the distinction is predominantly *where parameters were learned and how they are reused*, not whether a transformer is involved.

Outline

5 Forecast
- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

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- These intervals answer different questions. Forecast uncertainty typically includes future shocks and sometimes parameter uncertainty.

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- All methods require assumptions for trustworthy coverage; no procedure is universally assumption-free under arbitrary time dependence.

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- Under Gaussian shocks, a nominal 95% prediction interval is $\widehat{y}_{t+h|t} \pm 1.96\sigma_h$.
- Parameter estimation adds uncertainty; model-based intervals can be miscalibrated after structural changes.

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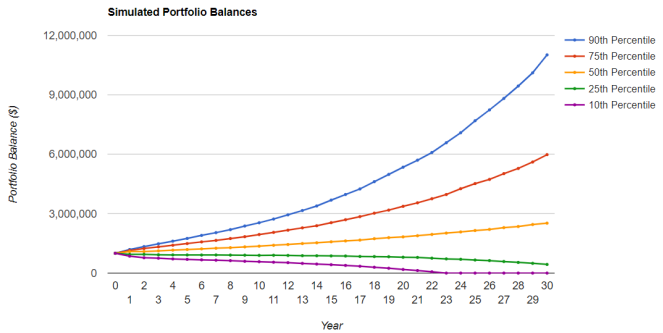
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- At each horizon, the empirical distribution across paths approximates the model's predictive distribution.
- The median and 5th/95th percentiles give a point summary and a central 90% **prediction interval**.
- Simulation also supports stress testing, nonlinear transformations, and aggregate outcomes.



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5. Repeat B times, setting and recording a pseudorandom seed for reproducibility.

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- Log prices are Gaussian at each horizon, hence prices are lognormal.
- This is an illustrative parametric simulation model, not a claim that actual asset returns are Gaussian or iid.

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- If independent log returns are Gaussian, their sum is Gaussian; the compounded *simple* return is generally not Gaussian.

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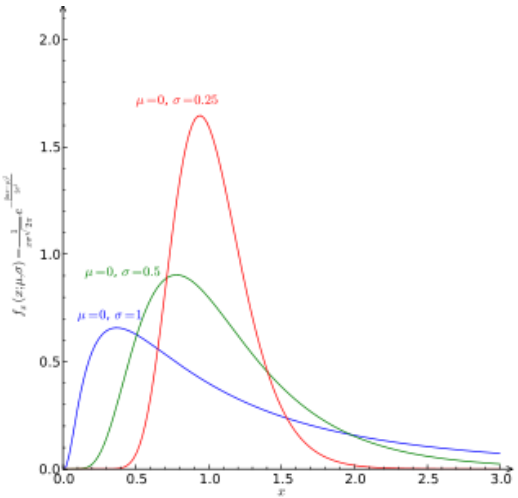
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- The support is positive, and the level distribution is right-skewed.
- This ensures positive simulated prices but does not remove tail risk or guarantee the lognormal model is empirically correct.



Bootstrap Methods: Two Different Goals

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- **Parameter inference:** resample data and re-estimate a statistic or model to approximate sampling variability (e.g. standard error of a coefficient).

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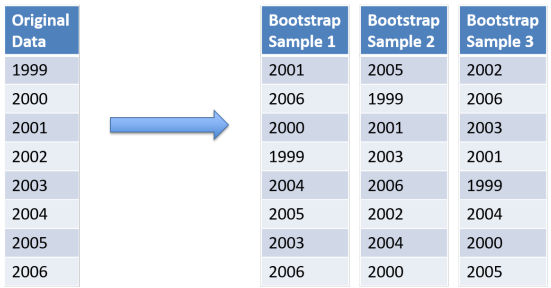
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- **Forecast uncertainty:** resample innovations/residuals or blocks, propagate the forecasting model, and form an empirical predictive distribution.
- These differ: a confidence interval for the *mean* is not the same as an interval for a *new observation*.
- The resampling design must reflect the dependence and forecast target.



Original Data
1999
2000
2001
2002
2003
2004
2005
2006

Bootstrap Sample 1	Bootstrap Sample 2	Bootstrap Sample 3
2001	2005	2002
2006	1999	2006
2000	2001	2003
1999	2003	2001
2004	2006	1999
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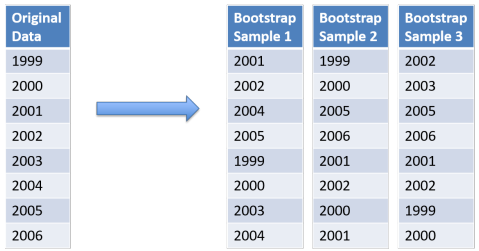
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- **Block bootstrap:** draw contiguous observed segments and concatenate them to preserve local temporal patterns.
- Examples include moving-block, circular-block, and stationary bootstraps.
- Block length trades retaining dependence against having enough distinct resampling units.
- A block bootstrap requires suitable assumptions (often weak dependence/stationarity); it is not a universal fix for trends, structural breaks, or nonstationarity.



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- To estimate forecast uncertainty, instead simulate future innovations and propagate them through the forecasting system.

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- Use a high empirical score quantile $q_{h,1-\alpha}$ to construct

$$\left[\widehat{s}_{T+h|T} - q_{h,1-\alpha},\ \widehat{s}_{T+h|T} + q_{h,1-\alpha} \right].$$

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- Then use $[\widehat{s}_{T+h|T} - q_h,\ \widehat{s}_{T+h|T} + q_h]$.
- The familiar finite-sample marginal coverage argument assumes calibration and test errors are exchangeable; dependent time-series scores generally require more care.

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- Practical adaptations include rolling calibration, horizon-specific errors, and dependence-aware procedures, but their guarantees need their own assumptions.
- Evaluate empirical coverage *and width* on later test origins; do not claim assumption-free time-series coverage.

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- Sharpness: among intervals with appropriate coverage, narrower is more informative.
- Quantile regression can estimate selected conditional quantiles directly; *pinball loss* evaluates them.
- Distributional scoring rules such as CRPS evaluate an entire predictive distribution, balancing calibration and sharpness.
- Always assess uncertainty by forecast horizon, period, and relevant SKU groups.

Outline

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- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
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- Continue in August, September, ... while the target month remains fixed.
- Write the forecast as $\widehat{s}_{T|t} = E[s_T \mid \mathcal{F}_t]$ with fixed target date T and expanding information sets $\mathcal{F}_t \subseteq \mathcal{F}_{t+1}$.

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- These operations can occur separately or together; none is equivalent to the other two.
- As the target approaches, the horizon $h = T - t$ also decreases.

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- These methods can all update coherently *with respect to their own specifications*; state-space modeling is one particularly explicit updating framework.

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- The Kalman filter updates estimates of hidden states as new sales observations arrive.

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- New data can change our estimate of the *current state* even if the estimated transition and observation matrices remain fixed.

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- v_t is the surprise relative to the model's prediction of newly observed sales.
- K_t controls how strongly the surprise revises the hidden-state estimate.
- This is *filtering*, not re-estimation of F or H.

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- Under correct linear-Gaussian assumptions, this gain gives the minimum mean-squared-error state update.

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- A small news shock may produce a small revision; a large or informative surprise may produce a substantial revision.
- Revisions are not necessarily *smooth* or monotone.

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- In this idealized sense, the *new information component* of the forecast revision should be unpredictable from yesterday's information.
- Systematically predictable revisions can indicate misspecification, insufficient information processing, or deliberate forecast smoothing.

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Operation	**What changes**	**Example**
Refresh features	New known sales, prices, calendar covariates	Boosted-tree forecast
Filter hidden state	Posterior estimate of current trend/seasonality	Kalman filter
Update recurrent state	Current hidden/cell state	LSTM / DeepAR
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Takeaway: forecast updating is universal; state space provides a particularly explicit decomposition of new data, uncertainty, and revisions.

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- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

A Different Forecasting Target: Volatility

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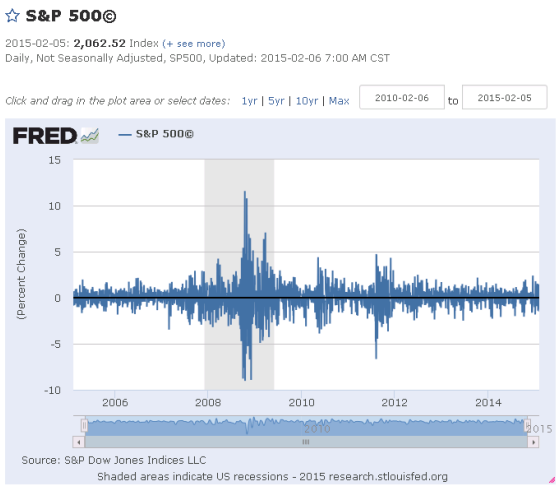
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- Volatility is the square root of variance; forecasting one is not identical to forecasting the other.



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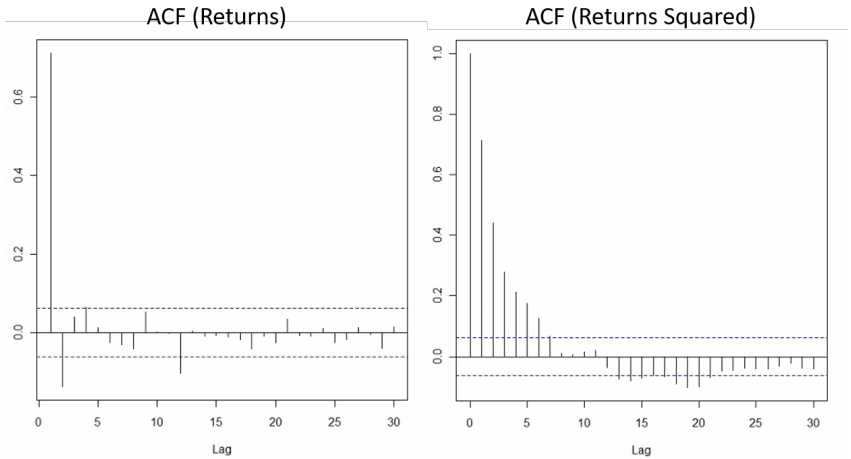
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- This motivates models for *conditional variance* rather than only conditional mean.



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- **Co-movement:** volatility can rise together across assets and markets.
- GARCH(1,1) targets persistence and clustering; extensions address asymmetry and cross-series dynamics.

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- O, H, L, C are open, high, low, close. Range estimators require assumptions (e.g. limited drift and no overnight jumps for the canonical formulas).

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- With 30-minute returns during a 6.5-hour regular US stock-market session, there are 13 intervals.
- Realized *variance* is an ex-post measure/proxy, not by itself a forecasting model.
- Microstructure noise, jumps, sampling frequency, and overnight returns can complicate the relationship to latent integrated variance.

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- But if turbulence clusters, a large recent shock may increase the next period's *conditional* variance.
- Two complementary strategies:
  - **GARCH:** model the latent conditional variance as a function of past shocks and variance.
  - **HAR-RV:** predict tomorrow's observed realized variance using realized volatility measures averaged across daily, weekly, and monthly scales.

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- ARCH $=$ *autoregressive conditional heteroskedasticity*; the variation is in conditional variance, not necessarily the unconditional variance.

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- Thus a changing *conditional* variance can coexist with a finite, constant *unconditional* variance.

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- In this *GARCH(p, q)* convention, p counts variance lags and q counts squared-shock lags. Check software conventions.
- Standard sufficient nonnegativity conditions: $\omega > 0$, $\alpha_i, \beta_j \geq 0$; $\sum_i \alpha_i + \sum_j \beta_j < 1$ is a familiar sufficient condition for finite long-run variance in the standard specification.

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- When $\alpha + \beta$ is near one, shocks to conditional variance decay slowly; this does not guarantee a stationary process in every more general specification.

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- With $\alpha + \beta < 1$, variance forecasts mean-revert toward $\bar{h} = \omega/(1 - \alpha - \beta)$.
- For a return prediction interval, forecast variance must be combined with distributional assumptions (e.g. conditional Gaussian or Student-t innovations).

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4. Rejecting suggests conditional heteroskedasticity remains in the residuals; it does not by itself select the ideal GARCH order.

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- HAR-RV (heterogeneous autoregressive realized volatility) models *observed realized variance* using averages over several timescales.
- Daily, trailing-week, and trailing-month information may represent different persistence horizons.
- This is a relatively simple regression-based forecasting model, rather than a latent conditional-variance recursion.
- It illustrates an important idea from earlier ML: carefully constructed lag summaries can capture rich temporal structure without a deep neural network.

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- The windows overlap; each coefficient reflects its predictor conditional on the other two.
- Here 5 and 22 are illustrative trading-day approximations to a week and a month.

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- β_d, β_w, and β_m capture the association of next-day realized variance with short-, medium-, and longer-window persistence.
- A linear HAR-RV forecast can be negative; practitioners may use log-RV or other positive transformations/model constraints.
- Re-estimation, prediction intervals, and backtests should respect chronological ordering and measurement conventions.

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Main inputs	Lagged squared shocks and conditional variance	Daily, weekly, monthly realized-variance averages
Data	Returns (often daily)	Intraday-derived realized-variance history
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Takeaway: complementary tools; compare on the same evaluation target and horizon.

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- **Stochastic volatility:** treat log variance as a separate latent stochastic process, often estimated with filtering or Bayesian methods.
- **Multivariate models:** represent co-movement and time-varying covariance across assets.

Extensions: Asymmetric and Multivariate Volatility

Q: What do these two workhorses leave out?

- **GJR-GARCH / EGARCH:** allow volatility reactions to depend on the sign as well as magnitude of shocks.
- **Stochastic volatility:** treat log variance as a separate latent stochastic process, often estimated with filtering or Bayesian methods.
- **Multivariate models:** represent co-movement and time-varying covariance across assets.
- **ML extensions:** predict realized variance with flexible lag features and covariates, or combine realized measures with parametric variance models.

Outline

5 Forecast
- Forecasting Framework
- Setup & Evaluation
- Classical Time Series Models
- Econometric Models
- Machine Learning Models
- Deep Learning
- Global / Panel Models
- Hierarchical
- Foundation Models
- Uncertainty and Prediction Intervals
- Forecast Updating
- Volatility Forecasting
- Paradigm Comparison

Forecasting Choices: Four Separate Dimensions

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Dimension	**Choice**	**Question answered**
Architecture	ARIMA/ADL, state space, tree ML, LSTM, transformer	How does the model represent dynamics?
Training scope	Local, pooled/global, pretrained foundation	Which histories inform estimation?
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Example: global DeepAR with autoregressive forecasts and uncertainty, followed by hierarchical reconciliation.

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One series, short history	Naive/seasonal naive, ETS, ARIMA
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Principle: start with a strong, appropriately chosen baseline; escalate complexity when out-of-sample evidence warrants it.

Forecasting: Model Selection Map (II)

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Many related series; limited history each	Pooled models, global boosting, DeepAR, global transformers
Pretraining potentially useful	Foundation models; check transfer against task-specific baselines
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Important: these are candidates, not a performance ranking. Use rolling-origin evaluation with a common target, horizon, and loss.

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- **Evaluation:** preserve chronology, forecast-origin information, benchmark comparisons, and the loss that matters for the business decision.